

Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity

T. Tony Cai¹, Dongwoo Kim¹, Yicheng Li², and Dongdong Xiang^{2*}

¹Department of Statistics and Data Science,
The Wharton School, University of Pennsylvania

²KLATASDS-MOE, School of Statistics,
East China Normal University

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Abstract

Transfer learning, which leverages auxiliary data to improve inference in a target domain, has become a powerful tool in machine learning. In this work, we study minimax and adaptive estimation of large bandable covariance matrices under a transfer learning framework.

We first establish the minimax rate of convergence under squared spectral norm loss and propose a rate-optimal estimator. Our analysis reveals phase transitions determined by the source-target disparity, the source sample size, the dimension, and the two smoothness indices. We then develop an adaptive estimator, a fully data-driven procedure that does not require the unknown smoothness and disparity parameters. In most regimes, it directly attains the minimax optimal rate without additional cost of adaptation.

Simulation studies support our theoretical results and demonstrate the practical effectiveness of the proposed adaptive procedure. We further illustrate its utility in a phoneme classification task, where incorporating related phonemes as auxiliary data significantly improves classification performance.

Keywords: Adaptive covariance estimation, Covariance matrix, High-dimensional statistics, Minimax rate of convergence, Transfer learning.

*The authors are listed in alphabetical order by surname.

1 Introduction

In high-dimensional data analysis, understanding covariance structure is fundamental. Covariance matrices are central to regression, discriminant analysis, clustering, PCA, and graphical modeling. However, as dimensionality increases, accurate estimation becomes increasingly challenging, motivating extensive research into structured covariance models; for comprehensive reviews, see Cai et al. (2016); Cai (2017).

Among structured models, bandable covariance matrices—those with entries decaying away from the diagonal—have received significant attention due to their relevance in applications such as finance, biology, phonetics, climatology, and engineering (He and Chen, 2016; Qiu and Chen, 2012; Bien et al., 2016; Li et al., 2017; Lee et al., 2022; Qiu and Chen, 2015; Yi and Zou, 2013). Several estimation methods have been developed to exploit this structure, with minimax optimality (Bickel and Levina, 2008; Cai et al., 2010; Cai and Yuan, 2012).

Concurrently, *transfer learning*—which leverages information from auxiliary datasets to improve inference in a target domain—has become a powerful tool in modern machine learning. It has shown success in domains such as natural language processing, bioinformatics, medical imaging, and recommendation systems (Han et al., 2021; Petegrosso et al., 2017; Maqsood et al., 2019; Hu et al., 2019). For comprehensive surveys, see Weiss et al. (2016); Zhuang et al. (2021). Recent theoretical work has extended transfer learning to various statistical tasks, including nonparametric models, GLMs, multi-armed bandits, and functional estimation (Cai and Wei, 2021; Cai and Pu, 2022; Li et al., 2023; Cai, Cai and Li, 2024; Cai, Kim and Pu, 2024).

When transfer learning and covariance matrix estimation intersect, they create a synergy capable of addressing more complex and intriguing statistical problems. It enables more efficient use of related but distinct datasets, particularly when target data is lim-

ited or prohibitively expensive to collect. This synergy has proven useful in applications such as GTEx analysis, EEG-based BCI, and text classification (Li et al., 2022; Rodrigues et al., 2019; Raina et al., 2006), where transfer learning methods can substantially improve accuracy of covariance estimation.

1.1 Formulation of the Problem

Our discussion begins with an illustrative example of covariance matrix estimation within the transfer learning context, based on phoneme sound analysis. Phonemes—the fundamental units of sound in language—are typically represented by periodogram vectors, which capture sound intensity across a wide range of frequencies from short audio recordings of the corresponding phoneme (Hamooni and Mueen, 2014).

Zooming in on a particular phoneme in English, say ZH as in “seizure”, we consider a target sample $\{X_1^{(t)}, \dots, X_{n_t}^{(t)}\}$ where each observation $X_i^{(t)} \in \mathbb{R}^d$ is independently and identically distributed. Within the context of our phoneme study, the dimensionality d corresponds to the number of frequencies chosen to extract periodogram vectors from the audio recordings. Our goal is to estimate the target covariance matrix $\Sigma^{(t)} \in \mathbb{R}^{d \times d}$ of the periodogram vectors for the ZH phoneme. This estimation problem naturally arises in applications such as phoneme classification tasks (Hastie et al., 1995; Bien et al., 2016).

A key characteristic of the periodogram vector is its inherent ordering from low to high frequencies, which induces a distinctive structure in the target covariance matrix $\Sigma^{(t)}$. Entries close to the diagonal reflect covariances between intensities at nearby frequencies and are expected to exhibit larger magnitudes. In contrast, entries farther from the diagonal—corresponding to covariances between intensities at widely separated frequencies—tend to decay toward zero. This structure is clearly visualized in Figure 5 in Section 4.

The described structure of the phoneme dataset aligns with a bandable structure, a concept formulated by Bickel and Levina (2008). We examine the away-from-diagonal

operator, denoted $\mathfrak{A}_d$, on the set of $d \times d$ matrices. Given a bandwidth $u > 0$, the operator is defined by:

$$\mathfrak{A}_d(B; u) := \left(b_{jk} \mathbb{1}(|j - k| > u) \right)_{j,k \in [d]} \quad \text{where } B = (b_{jk})_{j,k \in [d]} \in \mathbb{R}^{d \times d}.$$

Figure 2a in Section 2 provides a visual representation of the away-from-diagonal operator $\mathfrak{A}_d$. It retains elements beyond a certain distance from the main diagonal. The bandwidth $u > 0$ given in the operator specifies this threshold of distance. Under the bandable structure, it is assumed that the magnitude of the away-from-diagonal part diminishes geometrically as the bandwidth increases. We define the class $\mathcal{B}_\alpha(M)$ for bandable covariance matrices characterized by a decay rate $\alpha > 0$ and radius $M > 0$ as follows:

$$\mathcal{B}_\alpha(M) := \left\{ \Sigma \in \mathbb{S}_d^+ : \|\mathfrak{A}_d(\Sigma; u)\|_1 \leq M u^{-\alpha} \text{ for any } u > 0 \right\},$$

where $\mathbb{S}_d^+$ represents the collection of $d \times d$ symmetric and positive definite matrices and $\|\cdot\|_1$ indicates the matrix ℓ_1 -norm.

The conventional learning framework centers around finding an effective method to estimate the target covariance matrix based on the available target sample. However, we would encounter a significant challenge within the conventional learning framework, particularly when it comes to the ZH phoneme. Data points for this phoneme are typically derived from continuous speech recordings, such as the TIMIT dataset provided by the Linguistic Data Consortium (Garofolo et al., 1993). However, the ZH phoneme is the least frequently occurring sound in everyday English speech (Hamooni and Mueen, 2014; Hayden, 1950), making its recordings scarce and the collection of additional data prohibitively expensive.

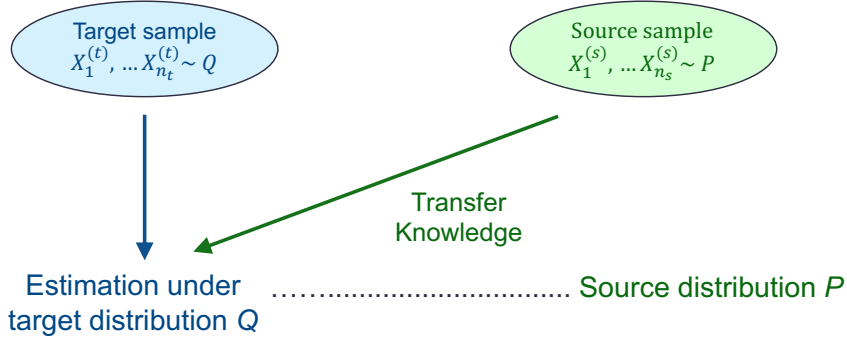


Figure 1: An illustration for transfer learning

One way to address this limitation is to extend the conventional learning framework through a transfer learning strategy. This approach allows us to utilize recordings of other phonemes that are different yet bear some similarities. For instance, the **S** phoneme, as in “see”, is categorized as an obstruent fricative in the hierarchy of English phonemes, like the **ZH** phoneme (Dekel et al., 2004; Lee and Hon, 1989). Moreover, the **S** phoneme is significantly more common, ranking as the fifth most frequent sound in English speech (Hamooni and Mueen, 2014; Hayden, 1950). Consequently, we can leverage the similarity and larger sample size of the **S** phoneme to enhance the covariance matrix estimation for the **ZH** phoneme.

From a statistical perspective, in addition to having an independently and identically distributed target sample $\{X_1^{(t)}, \dots, X_{n_t}^{(t)}\}$ from a target distribution Q , we have an auxiliary sample, $\{X_1^{(s)}, \dots, X_{n_s}^{(s)}\}$, comprising d -dimensional random vectors that are independently and identically distributed from a source distribution P . The source sample is independent of the target sample and is also characterized by its source covariance matrix $\Sigma^{(s)} \in \mathbb{R}^{d \times d}$. While our objective remains the same, namely to find an optimal estimation method for the target covariance matrix $\Sigma^{(t)}$, our methodology may benefit from leveraging the source sample alongside the target sample. Figure 1 illustrates the transfer learning strategy.

The effectiveness of transfer learning depends on a meaningful and quantifiable rela-

tionship between the source and target samples. In this paper, we assume that:

- Both the target and source covariance matrices are bandable. Specifically, for target parameters (α_t, M_t) and source parameters (α_s, M_s) , we assume $\Sigma^{(t)} \in \mathcal{B}_{\alpha_t}(M_t)$ and $\Sigma^{(s)} \in \mathcal{B}_{\alpha_s}(M_s)$.
- The disparity matrix $\Delta^{(s)} := \Sigma^{(t)} - \Sigma^{(s)}$, which measures the deviation between the target and source covariance matrices, is sparse in the matrix ℓ_1 -norm. For a given disparity threshold $\delta \geq 0$, we assume $\|\Delta^{(s)}\|_1 \leq \delta$.

More general disparity conditions and approximately bandable misspecification are discussed in the supplementary material.

Let $\mathcal{P}_{\alpha_t, \alpha_s, \delta}(M_t, M_s, \rho)$ denote the collection of joint distributions of the target and source samples satisfying these assumptions together with the uniform sub-Gaussian condition in Assumption 2.1 with parameter ρ for both samples. We focus on the bandable class as it serves as the canonical model for ordered variables and provides a transparent setting to study the interactions between structural decay, domain disparity, and relative sample sizes.

To establish minimax optimality, we evaluate estimators $\widehat{\Sigma}^{(t)}$ that utilize both target and source samples under the squared spectral-norm loss. The minimax risk is defined as:

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}(M_t, M_s, \rho)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2.$$

Our first main result characterizes the sharp rate of this risk.

1.2 Main Results and Our Contribution

Under the above model and loss, and for sufficiently large d , our first main result shows that the minimax risk satisfies the following rate whenever $\log d/n \rightarrow 0$ and $(\log d/n_t) \wedge \delta^2 \rightarrow 0$, the necessary and sufficient criterion for consistent estimation. By letting $n := n_t + n_s$ and

$\alpha_{\max} := \alpha_t \vee \alpha_s$, the minimax optimal rate is expressed as:

$$\begin{aligned} \Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d) := & \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \frac{\log d}{n} \\ & + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2 \right) + \left(\frac{\log d}{n_t} \wedge \delta^2 \right). \end{aligned} \quad (1)$$

That is, under the displayed consistency criterion, Equation (1) gives the order of the minimax risk for estimating the target covariance matrix in the transfer learning setting under the squared spectral-norm loss.

When there is no auxiliary source sample and no constraints on the disparity matrix, our setting simplifies to the conventional learning framework and the corresponding minimax rate $\Phi_{\alpha_t, \alpha_s, \infty}(n_t, 0, d)$ reduces to the known result in the conventional setting as detailed in Cai et al. (2010). Moreover, by comparing the minimax rate to the baseline performance, $\Phi_{\alpha_t, \alpha_s, \infty}(n_t, 0, d)$, we can identify the necessary and sufficient conditions under which the source sample and transfer learning are indeed effective. This identification reveals interesting phase transition phenomena.

The minimax optimal rate still leaves open the practical problem of constructing an estimator that does not depend on the unknown parameters α_t , α_s , or δ . The second major contribution of this paper is the *adaptive transfer block-tridiagonal estimator*, which does not require the smoothness or disparity parameters. In most of the regimes considered in this paper, this estimator directly attains the minimax optimal rate $\Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d)$, despite not knowing these parameters. This includes settings in which the source and target covariance matrices are sufficiently close, as well as broad regimes in which the source sample size and dimension grow regularly with the target sample size. The adaptive section describes these regimes and explains how the validation step governs the use of auxiliary data when transfer is less favorable.

We further demonstrate its practical effectiveness through both simulation studies and a real-data application. The simulation results validate the theoretical phase transition

phenomena: transfer learning improves estimation when the disparity is small, but offers relatively small benefit when the disparity is large. Additionally, we apply the adaptive estimator to a phoneme classification task using the TIMIT dataset. Incorporating auxiliary data from related phonemes substantially enhances classification accuracy in both QDA and LDA, with gains increasing as the auxiliary sample size grows, underscoring the practical utility of our proposed method.

1.3 Organization and Notation

The rest of the paper is organized as follows. We conclude this section by introducing the basic notation. Section 2 explores optimal transfer learning for covariance matrix estimation, presenting the blockwise tridiagonal estimator and establishing the minimax rate of convergence. Section 3 introduces the adaptive transfer estimator and discusses the regimes in which it attains Φ . Section 4 carries out simulation study and real-data analysis to evaluate the effectiveness of the proposed adaptive algorithm in various settings. Section 5 explores possible directions of future research. Finally, the proofs of the main theorems and technical results are presented in the supplementary material.

Throughout the paper, the primary asymptotic components are the sample sizes (n_t, n_s) , the dimensionality (d) and the disparity threshold (δ) , while all other variables are treated as constants. We conform to the conventional big-Oh(O), big-Omega(Ω) and big-Theta(Θ) notations. For conciseness, we occasionally substitute these notations with the symbols $\lesssim$, $\gtrsim$ and $\asymp$, respectively. Additionally, we adhere to the convention of little-oh(o) notation and simply write $a \ll b$ or $b \gg a$ to signify that $a = o(b)$.

In addition, let us introduce several notations pertinent to covariance matrices. We define the pooled covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$, which represents the weighted average of the population covariance matrices, $\Sigma^{(t)}$ and $\Sigma^{(s)}$. Concurrently, we define another disparity matrix $\Delta \in \mathbb{R}^{d \times d}$ as the deviation of the target covariance matrix $\Sigma^{(t)}$ from the pooled

covariance matrix Σ . By letting $n := n_t + n_s$, these are mathematically represented as:

$$\Sigma := \frac{n_t}{n}\Sigma^{(t)} + \frac{n_s}{n}\Sigma^{(s)}, \quad \text{and} \quad \Delta := \Sigma^{(t)} - \Sigma = \frac{n_s}{n}\Delta^{(s)}.$$

We proceed to define the sample counterparts of the covariance matrices. The target sample covariance matrix $\check{\Sigma}^{(t)}$ and the source sample covariance matrix $\check{\Sigma}^{(s)}$ are defined that for each group indicator $g \in \{t, s\}$,

$$\check{\Sigma}^{(g)} := \frac{1}{n_g} \sum_{i=1}^{n_g} (X_i^{(g)} - \bar{X}^{(g)})(X_i^{(g)} - \bar{X}^{(g)})^\top \quad \text{where } \bar{X}^{(g)} := \frac{1}{n_g} \sum_{i=1}^{n_g} X_i^{(g)}.$$

Subsequently, we define the pooled sample covariance matrix $\check{\Sigma}$ as a weighted combination of the target and source sample covariance matrices:

$$\check{\Sigma} := \frac{n_t}{n}\check{\Sigma}^{(t)} + \frac{n_s}{n}\check{\Sigma}^{(s)}.$$

Finally, it is apparent to define the sample disparity matrices $\check{\Delta}^{(s)}$ and $\check{\Delta}$ as:

$$\check{\Delta}^{(s)} := \check{\Sigma}^{(t)} - \check{\Sigma}^{(s)}, \quad \text{and} \quad \check{\Delta} := \check{\Sigma}^{(t)} - \check{\Sigma} = \frac{n_s}{n}\check{\Delta}^{(s)}.$$

2 Optimal Transfer Learning

In this section, our goal is to present an optimal method for estimating the target covariance matrix $\Sigma^{(t)}$ within the transfer learning framework. Recall that $\mathcal{P}_{\alpha_t, \alpha_s, \delta}(M_t, M_s, \rho)$ denotes the collection of joint distributions of the target and source samples such that $\Sigma^{(t)} \in \mathcal{B}_{\alpha_t}(M_t)$, $\Sigma^{(s)} \in \mathcal{B}_{\alpha_s}(M_s)$, $\|\Sigma^{(t)} - \Sigma^{(s)}\|_1 \leq \delta$, and both samples satisfy the following uniform sub-Gaussianity condition:

Assumption 2.1 (Uniform sub-Gaussianity). The target observations $X_1^{(t)}, \dots, X_{n_t}^{(t)}$ and the source observations $X_1^{(s)}, \dots, X_{n_s}^{(s)}$ are uniformly sub-Gaussian random vectors with a standard-deviation proxy $\rho > 0$. More specifically, we assume:

$$\left. \begin{aligned} &\|a^\top (X_1^{(t)} - \mathbb{E}X_1^{(t)})\|_{\psi_2} \\ &\|a^\top (X_1^{(s)} - \mathbb{E}X_1^{(s)})\|_{\psi_2} \end{aligned} \right\} \leq \rho, \quad \text{for any unit vector } a \in \mathbb{R}^d,$$

where we denote by $\|\cdot\|_{\psi_2}$ the sub-Gaussian norm or Orlicz ψ_2 -norm. As per Wainwright (2019) and Vershynin (2018), the sub-Gaussian norm of a random variable Z is defined by

$$\|Z\|_{\psi_2} := \inf\{r > 0 : \mathbb{E}e^{Z^2/r^2} \leq 2\}.$$

We are now ready to state the optimality theory for estimating the target covariance matrix under the transfer learning framework. The following theorem characterizes the minimax rate of convergence under the squared spectral-norm loss. In particular, it highlights how the interaction between the sample sizes (n_t, n_s) , the dimension (d) , and the disparity level (δ) governs the fundamental limits of transfer learning.

Theorem 2.2. *One can consistently estimate the target covariance matrix if and only if at least one of the following conditions holds:*

Assumption (a) $\log d \ll n_t$.

Assumption (b) $\log d \ll n$ and $\frac{\log d}{n_t} \wedge \delta^2 \ll 1$.

Under either condition, the optimal convergence rate under the squared spectral-norm loss is given by

$$\begin{aligned} \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\asymp \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \frac{\log d}{n} \\ &\quad + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2 \right) + \left(\frac{\log d}{n_t} \wedge \delta^2 \right). \end{aligned}$$

Remark 2.3. These results extend beyond the exact bandable model and the matrix ℓ_1 -disparity condition. Under weaker assumptions, we replace the ℓ_1 constraint with an upper-tail stability (UTS) condition and allow for approximately bandable misspecification. In this broader setting, the same upper bound holds, supplemented only by a term quantifying the misspecification level.

We maintain the exact ℓ_1 -based formulation in the main text to keep the phase-transition phenomena transparent. The more general statements are deferred to Subsection A.1 in

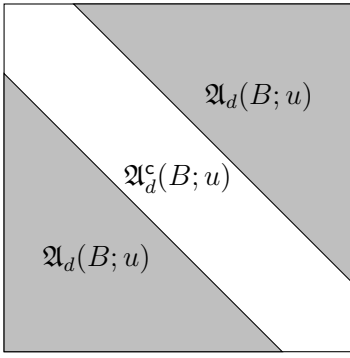
the supplementary material for pedagogical clarity, rather than any limitation in the theory itself. In the remainder of the paper, we retain the class $\mathcal{P}_{\alpha_t, \alpha_s, \delta}$ to streamline the presentation.

Due to space constraints, the proof of Theorem 2.2 is deferred to the supplementary material. We instead introduce the key technical ingredient for establishing the upper bound: the *blockwise tridiagonal operator*, denoted by $\mathfrak{T}_d$, which was previously employed in high-dimensional covariance estimation by Cai and Zhang (2016). This operator is defined for an integer bandwidth $u > 0$ and can be expressed as follows. Let $\lfloor x \rfloor$ denote the greatest integer not exceeding $x \in \mathbb{R}$.

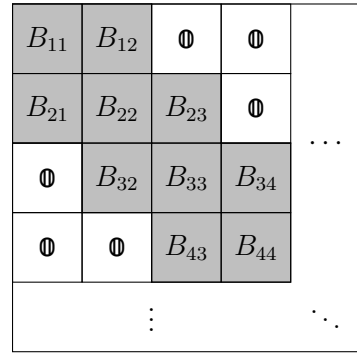
$$\mathfrak{T}_d(B; u) := \left(b_{jk} \mathbb{1}(|u_j - u_k| \leq 1) \right)_{j, k \in [d]} \quad \text{where} \quad \begin{cases} B = (b_{jk})_{j, k \in [d]} \in \mathbb{R}^{d \times d}, \\ u_j := 1 + \lfloor (j - 1)/u \rfloor, \quad (j \in [d]). \end{cases}$$

Intuitively, when the matrix B is partitioned into $u \times u$ blocks, the operator $\mathfrak{T}_d$ retains only the blockwise main diagonal and the immediate super- and sub-diagonal blocks, resulting in a block-tridiagonal structure. A visual illustration of this mechanism is provided in Figure 2b.

By applying the blockwise tridiagonal operator $\mathfrak{T}_d$ with a carefully chosen bandwidth, we can establish the upper bound in the minimax rate of Theorem 2.2. Since the bandable structure implies decay of the away-from-diagonal parts, it is natural to truncate the sample covariance matrices using $\mathfrak{T}_d$. Although full sample covariance matrices perform poorly in high-dimensional settings, the low-dimensional blocks retained by $\mathfrak{T}_d$ can yield accurate local estimates, provided that the bandwidth is selected appropriately. Detailed derivations are provided in the supplementary material.



(a) The output, $\mathfrak{A}_d(B; u)$, of the away-from-diagonal operator



(b) The output, $\mathfrak{T}_d(B; u)$, of the blockwise tridiagonal operator

Figure 2: A visual representation of important matrix operators

Given the extensive literature on covariance matrix estimation under the conventional setting, it is instructive to compare our minimax result with known rates. Notably, the transfer learning framework reduces to the conventional framework when $n_s = 0$ and $\delta = \infty$, as the source sample is absent and the source population provides no information about the target covariance structure. In this case, Cai et al. (2010) established that the minimax rate of convergence is given by

$$\Phi_{\alpha_t, \alpha_s, \infty}(n_t, 0, d) \asymp \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \right) + \frac{\log d}{n_t}. \quad (2)$$

The optimal convergence rate established in Theorem 2.2 also provides insight into the effectiveness of the transfer learning. A natural question is to identify the conditions under which transfer learning yields tangible benefits, and to quantify the extent of such improvements. Notably, the analysis reveals a phase transition phenomenon, where the utility of transfer learning undergoes a sharp change depending on the disparity level (δ) and the source sample size (n_s). The main findings are concisely summarized in Table 1.

The disparity threshold δ plays a crucial role in the phase transition since it serves as a key measure of similarity between the target and source covariance matrices. Let us define

two critical points $\delta_1^* \lesssim \delta_2^*$ by

$$\begin{aligned}\delta_1^* &:= \left(n^{-\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \sqrt{\frac{d}{n}} \right) + \sqrt{\frac{\log d}{n}} \\ \delta_2^* &:= \left(n_t^{-\alpha_t/(2\alpha_t+1)} \wedge \sqrt{\frac{d}{n_t}} \right) + \sqrt{\frac{\log d}{n_t}}.\end{aligned}$$

Depending on the level of the disparity threshold δ , we naturally classify our model into three distinct categories: the minimal, the moderate and the strong disparity models as outlined in Table 1.

Recall that $\Phi^{\text{LE}} := \Phi_{\alpha_t, \alpha_s, \infty}(n_t, 0, d)$ accords with the minimax rate of convergence in the conventional learning scenario. It is noteworthy that the minimax rate Φ must be no slower than Φ^{LE} , as it is possible to construct an estimator that relies solely on the target sample. Hence, the rate Φ^{LE} serves as a baseline and can be regarded as the least effective rate of convergence within the transfer learning framework.

Our model experiences a notable transition at the critical point $\delta \asymp \delta_2^*$, which dictates the potential benefits of transfer learning compared to conventional learning. Specifically, under the strong disparity model, characterized by $\delta \gtrsim \delta_2^*$, the minimax rate of convergence aligns exactly with Φ^{LE} . Despite the availability of the source sample, the disparity threshold δ is too substantial to facilitate the effective transfer of knowledge. This also highlights a safety property at the minimax level: as the disparity increases into the strong-disparity regime, the transfer benchmark degrades back to the target-only minimax rate Φ^{LE} , rather than becoming worse than the conventional baseline. Conversely, in the context of the minimal or moderate disparity model, characterized by $\delta \ll \delta_2^*$, incorporating the source sample presents an opportunity to enhance the accuracy of estimating the target covariance matrix.

Given that the disparity threshold is sufficiently small as in the minimal or moderate disparity models, the size of the source sample becomes a key factor in this enhancement. We explore the necessary and sufficient conditions for the enhancement in two separate

Table 1: Does transfer learning offer benefits? If so, does increasing the source sample size offer benefits?

		Minimal disparity model $(\delta \ll \delta_1^*)$	Moderate disparity model $(\delta \gtrsim \delta_1^* \text{ and } \delta \ll \delta_2^*)$	Strong disparity model $(\delta \gtrsim \delta_2^*)$
Limited source sampling regime $(n_s \lesssim n_t)$	without auto- smoothing benefit	No		
	with auto- smoothing benefit	No		
Ample source sampling regime $(n_s \gg n_t)$		Yes / Yes	Yes / No	

scenarios: the ample and the limited source sampling regimes as presented in Table 1. Additionally, a comparative analysis of these regimes reveals another phase transition phenomenon.

Within the ample source sampling regime where $n_s \gg n_t$ holds, it is readily verifiable that the minimax rate of convergence diminishes more rapidly than the baseline rate Φ^{LE} . In other words, a substantial size of the source sample is conducive to effective transfer learning, aligning with our expectations under the minimal or moderate disparity model. On the contrary, under the limited source sampling regime, defined by $n_s \lesssim n_t$, the effectiveness of the source sample may be compromised. Typically, the minimax rate of convergence aligns with the baseline rate Φ^{LE} . Nevertheless, the transfer learning can still benefit from the scarce source sample if the following extra conditions are satisfied:

- The source covariance matrix exhibits an expedited decay, as indicated by $\alpha_s > \alpha_t$.
- The dimensionality is neither too small nor large, as $n_t^{1/(2\alpha_s+1)} \ll d \ll \exp(n_t^{1/(2\alpha_t+1)})$.

These conditions give rise to the *auto-smoothing* effect, which allows the target covariance matrix to be analyzed as if it possesses the source decay rate α_s rather than the target decay rate α_t . If the source covariance matrix exhibits an expedited decay, the auto-smoothing now facilitates the more accurate estimation for the target covariance matrix. Auto-smoothing is akin to the concept of transferable smoothness described by Cai and Pu (2022) in the context of transfer learning for nonparametric regression. The dimensionality condition is equally critical, ensuring that the improved decay rate effectively aids in estimation. Therefore, those extra conditions are both essential and adequate to gain benefit from the auto-smoothing and to ensure effective transfer learning under the limited source sampling regime as outlined in Table 1.

Moving forward, our analysis now centers on regimes and models where transfer learning proves advantageous. Pertaining to the magnitude of this advantage, a final phase

transition is observed at the critical point, $\delta \asymp \delta_1^*$. Within the moderate disparity model, satisfying $\delta \gtrsim \delta_1^*$, the minimax rate of convergence is simply established as δ^2 . Although the established rate indicates an enhancement over the baseline rate Φ^{LE} , the expansion of the source sample size fails to accelerate the convergence rate as long as other conditions remain the same. On the other hand, within the minimal disparity model, characterized by $\delta \ll \delta_1^*$, an increase in the source sample size yields additional benefits. It can be demonstrated that the minimax rate of convergence precisely matches the most efficient rate Φ^{ME} , defined as follows. By leveraging a model with a negligible disparity threshold, we can extract greater benefits from the larger source sample, leading to enhanced performance of estimation.

$$\Phi^{\text{ME}} := \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \frac{\log d}{n}.$$

Let us further examine the case where the minimax rate coincides with the most effective rate Φ^{ME} . It is immediate that $\Phi^{\text{ME}} = \Phi_{\alpha_{\max}, \alpha_{\max}, 0}(n_t, n_s, d)$ holds, suggesting that the target and the source samples could be considered as originating from the same population. This representation is underscored by the auto-smoothing effect, which arises from the condition $\alpha_t = \alpha_s = \alpha_{\max}$. As we have discussed before, this effect equalizes the decay rates of the target and source covariance matrices, ensuring both exhibit the fastest decay rate, $\alpha_{\max}$. Not only that, another condition $\delta = 0$ indicates that the source and target covariance matrices are identical, effectively recasting the transfer learning framework into the conventional learning framework with augmented sample size, $n = n_t + n_s$. While the source population may differ from the target population, to estimate the target covariance matrix, the minimax rate of convergence guarantees that they are fundamentally equivalent.

3 Adaptive Transfer Learning

The minimax estimator in Section 2 uses the unknown smoothness indices α_t and α_s , as well as the unknown disparity level δ . We now construct a single data-driven estimator that does not use any of these parameters. The construction first fits an adaptive bandable covariance estimator to a single collection of observations, then compares its target-only and pooled instances through an independent one-dimensional validation statistic.

The benchmark is the ordinary classwise minimax rate $\Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d)$ in Equation (1). For later reference, let

$$p_m(a) := m^{-2a/(2a+1)} \wedge \frac{d}{m}, \quad P_m(a) := p_m(a) + \frac{\log d}{m}, \quad \alpha_{\max} := \alpha_t \vee \alpha_s.$$

Then

$$\Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d) = P_n(\alpha_{\max}) + (p_{n_t}(\alpha_t) \wedge \delta^2) + \left(\frac{\log d}{n_t} \wedge \delta^2 \right), \quad n = n_t + n_s. \quad (3)$$

Our main result gives an additive oracle bound relative to this benchmark. Within the consistency regime of Theorem 2.2, it yields simultaneous minimax adaptivity up to a logarithmic factor, with exact matching when transfer is exact or sufficiently close to exact.

3.1 Adaptive transfer block-tridiagonal estimator

For convenience, we assume that $\mathbb{E}X_i^{(g)} = 0$ throughout this subsection for $g \in \{t, s\}$. For the general case, we may first apply the pairing symmetrization

$$W_i^{(g)} = \frac{X_{2i}^{(g)} - X_{2i-1}^{(g)}}{\sqrt{2}}, \quad 1 \leq i \leq \left\lfloor \frac{N_g}{2} \right\rfloor, \quad g \in \{t, s\},$$

to the original target and source samples, where N_g denotes the original sample size in group g . The paired observations are independent and satisfy $\mathbb{E}W_i^{(g)} = 0$, $\text{Cov}(W_i^{(g)}) = \Sigma^{(g)}$, $\sup_{\|u\|_2=1} \|u^\top W_i^{(g)}\|_{\psi_2} \leq C\rho$. In practice, one may instead center the samples by subtracting their empirical means, and the same theoretical results extend to this setting. Here and

below, let $L = C_L \rho^2$ be a fixed constant chosen so that the spectra of $\Sigma^{(t)}$ and $\Sigma^{(s)}$ lie in $[0, L]$.

A single-source adaptive estimator We first give a general single-source adaptive estimator as the building block of our adaptive transfer estimator. It treats the input data as a single sample, regardless of whether they come from the target sample or the pooled sample. The main idea is to construct a dyadic family of block-tridiagonal candidates, measure the cumulative spectral change from each candidate to its coarser scales, and select the bandwidth that minimizes this change together with a scale-dependent stochastic penalty. For a symmetric matrix B , we let $\text{Clip}_L(B)$ denote the spectral clipping of B to $[0, L]$, obtained by truncating its eigenvalues in the spectral decomposition.

Algorithm 1 FitAdaptive

Require: A collection $\mathcal{D}$ of r independent observations; fixed constants λ and L .

- 1: Set $\hat{\Sigma}_{\mathcal{D}} = r^{-1} \sum_{X \in \mathcal{D}} X X^\top$.
- 2: Set $u_- = d \wedge \lceil \log(ed) \rceil$, $u_+ = u_- \vee (d \wedge \lfloor r / \log(er) \rfloor)$, and $\mathcal{U}_r = \{2^k u_- : k \geq 0, 2^k u_- \leq u_+\}$.
- 3: **for** $u \in \mathcal{U}_r$ **do**
- 4: Set $\ell_u = \max\{\ell \geq 0 : 2^\ell u \leq u_+\}$ and

$$F_{\mathcal{D}}(u) = \sum_{\ell=1}^{\ell_u} \|\mathfrak{T}_d(\hat{\Sigma}_{\mathcal{D}}; 2^\ell u) - \mathfrak{T}_d(\hat{\Sigma}_{\mathcal{D}}; 2^{\ell-1} u)\|_2 + \lambda \rho^2 \sqrt{\frac{u + \log(ed)}{r}}.$$

- 5: Set $\hat{u} = \arg \min_{u \in \mathcal{U}_r} F_{\mathcal{D}}(u)$.
 - 6: **return** $\text{Clip}_L(\mathfrak{T}_d(\hat{\Sigma}_{\mathcal{D}}; \hat{u}))$.
-

Nested dyadic block bands

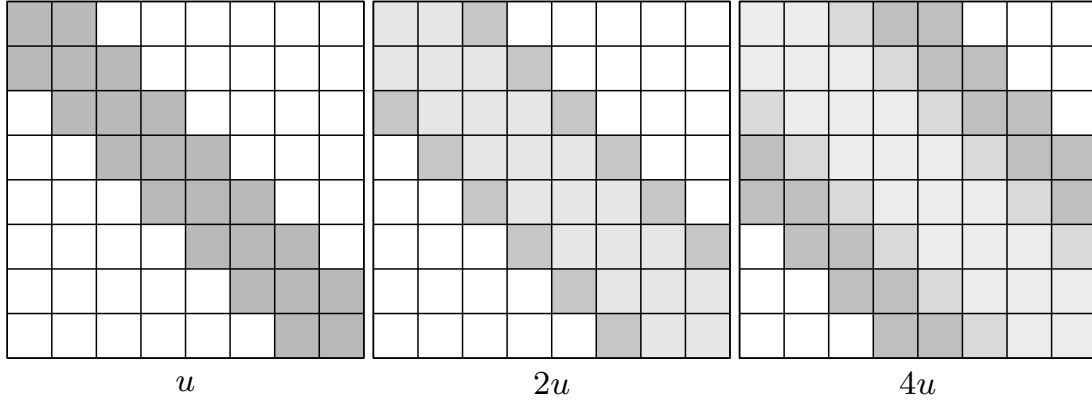


Figure 3: Nested dyadic block bands at scales u , $2u$, and $4u$.

Adaptive transfer estimator via validation For the adaptive transfer estimator, we first compute two candidates from the target samples only and the pooled samples, using the adaptive estimator in Algorithm 1. Then, we use a separate validation set from the target samples to choose between the two candidates. We use the estimator based on the pooled samples unless it deviates too much from the estimator based only on the target samples, in which case we fall back to the latter. The challenge is that directly validating two covariance matrices in high dimension with the operator norm incurs a dimension-dependent error, which can overwhelm the difference between the candidates. To avoid this cost, we first find a unit direction along which the two candidates differ most and then validate their projected quadratic forms in that direction using the held-out target samples. The validation problem is therefore scalar, so its statistical and computational cost is small. Moreover, when the two candidates are close in operator norm, either choice has essentially the same risk. The one-sided rule falls back to the estimator based only on the target samples when the validation data confirm that the candidate discrepancy is large enough to matter. This allows the estimator to retain the pooled candidate whenever transfer

is useful while protecting against substantial discrepancy between source and target, and hence to attain the minimax transfer rate without knowing the disparity level in advance.

Algorithm 2 Adaptive transfer block-tridiagonal estimator

Require: Independent target and source samples; a validation size $m_f \leq n_t/2$; fixed constants

λ , L , and C_τ .

1: Partition the target indices into disjoint $\mathcal{I}_t^{(0)}$ and $\mathcal{I}_t^{(f)}$, with $|\mathcal{I}_t^{(f)}| = m_f$ and $|\mathcal{I}_t^{(0)}| = n_t - m_f$.

2: Set $\mathcal{D}_t = \{X_i^{(t)} : i \in \mathcal{I}_t^{(0)}\}$, $\mathcal{D}_f = \{X_i^{(t)} : i \in \mathcal{I}_t^{(f)}\}$, and $\mathcal{D}_p = \mathcal{D}_t \cup \{X_i^{(s)} : 1 \leq i \leq n_s\}$.

3: Set $A_{\text{tar}} = \text{FitAdaptive}(\mathcal{D}_t)$ and $A_{\text{pool}} = \text{FitAdaptive}(\mathcal{D}_p)$.

4: Choose a unit v with $|v^\top (A_{\text{tar}} - A_{\text{pool}})v| = \|A_{\text{tar}} - A_{\text{pool}}\|_2$.

5: Set $\hat{q} = \Pi_{[0,L]}(|\mathcal{D}_f|^{-1} \sum_{X \in \mathcal{D}_f} (v^\top X)^2)$, $a = v^\top A_{\text{tar}}v$, and $b = v^\top A_{\text{pool}}v$.

6: Set $\tau = C_\tau \rho^2(\sqrt{\log(en)/m_f} + \log(en)/m_f)$.

7: **if** $|a - \hat{q}| + 4\tau \leq |b - \hat{q}|$ **then**

8: **return** $\hat{\Sigma}_{\text{ad}}^{(t)} = A_{\text{tar}}$.

9: **else**

10: **return** $\hat{\Sigma}_{\text{ad}}^{(t)} = A_{\text{pool}}$.

3.2 Adaptive Upper Bound

We first provide the risk guarantees of the two candidate estimators over the target and pooled samples. These bounds separate the estimation error of the adaptive covariance fit from the discrepancy introduced by pooling.

Proposition 3.1 (Risk bounds for the two candidates). *Suppose Assumption 2.1 holds and choose λ and C_L sufficiently large. Then the two calls to **FitAdaptive** satisfy*

$$\mathbb{E}\|A_{\text{tar}} - \Sigma^{(t)}\|_2^2 \lesssim P_{n_t}(\alpha_t), \quad (4)$$

$$\mathbb{E}\|A_{\text{pool}} - \Sigma^{(t)}\|_2^2 \lesssim P_n(\alpha_{\max}) + \delta^2. \quad (5)$$

The target candidate therefore achieves $P_{n_t}(\alpha_t)$, the optimal rate for estimating $\Sigma^{(t)}$ from the target sample alone. The pooled candidate achieves $P_n(\alpha_{\max}) + \delta^2$, which is the

rate obtained by treating the target and source observations as a single pooled sample, up to the squared discrepancy between their covariance matrices. These are exactly the two candidate rates identified in Section 2.

With these candidate bounds in hand, the remaining task is to show that the directional validation rule selects a candidate with the appropriate risk. The following theorem establishes this oracle behavior without requiring prior knowledge of either smoothness index or the disparity level. Define the general selector remainder

$$\Gamma_{m_f, n} := \rho^4 \left(\frac{\log(en)}{m_f} + \frac{\log^2(en)}{m_f^2} \right). \quad (6)$$

Theorem 3.2. *Suppose Assumption 2.1 holds and choose λ, C_L, C_τ sufficiently large. For every fixed $\alpha_t, \alpha_s > 0$ and every $\delta \geq 0$, the estimator in Algorithm 2 satisfies*

$$\sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}_{\text{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim \Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d) + (\delta^2 \wedge \Gamma_{m_f, n}). \quad (7)$$

The implicit constant is uniform when (α_t, α_s) ranges over a fixed compact subset of $(0, \infty)^2$.

Theorem 3.2 shows that adaptively selecting between the two candidates only suffers from an additive cost $\delta^2 \wedge \Gamma_{m_f, n}$. In many cases, the additive cost can be absorbed into the optimal rate $\Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d)$, which we recall is defined in Equation (1).

Corollary 3.3. *Under the conditions of Theorem 3.2, suppose either of the following holds:*

- $\delta^2 \lesssim p_{n_t}(\alpha_t) \vee \frac{\log d}{n_t}$;
- for fixed $\alpha_t, \alpha_s > 0$, $m_f = \lfloor n_t / \log(en_t) \rfloor$, $n_s \asymp n_t^\kappa$, and $d \asymp n_t^\xi$, with $\kappa \geq 0$ and $\xi > 0$.

Then

$$\sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}_{\text{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim \Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d). \quad (8)$$

Our theory shows that adaptation can still be achieved without an additional leading-order cost under transfer learning. In the two particular settings in Corollary 3.3, the

adaptivity is achieved without additional cost. First, a small disparity is enough to make the adaptivity cost disappear: δ^2 only needs to be no larger than the statistical scale for estimating from target data alone, namely $p_{n_t}(\alpha_t) \vee (\log d)/n_t$, which includes exact transfer as a special case. Second, the adaptivity cost also disappears under a deliberately mild polynomial growth condition. It allows both the source sample size and the dimension to grow polynomially with the target sample size, with $n_s \asymp n_t^\kappa$, $d \asymp n_t^\xi$, and a validation size of order $n_t/\log n_t$. It imposes no small disparity condition and no ordering condition between α_t and α_s . Outside these two regimes, Theorem 3.2 isolates the price of transfer adaptation in the additive term $\delta^2 \wedge \Gamma_{m_f, n}$, rather than a multiplicative factor. This term is generally small when the validation size m_f is chosen proportional to n_t .

The low cost of adaptivity comes from our validation strategy, which reduces the comparison of the two covariance candidates to a scalar validation problem. We choose a direction that exposes the largest discrepancy between the candidates and compare their projected quadratic forms along this direction, rather than estimating an operator norm over the full dimension. The resulting concentration cost depends on the validation size and the confidence level, but does not introduce an additional factor that grows with d . This separation is important in high dimensions, where a direct operator norm comparison could obscure the difference between the candidates.

The procedure also has a natural safety interpretation. When the pooled candidate is compatible with the target covariance, the validation step retains it and makes use of the larger effective sample and the smoother covariance structure available after pooling. When the discrepancy is too large for the pooled candidate to be reliable, the one-sided rule can fall back to the candidate built from the target data alone. The same estimator therefore adapts simultaneously to the two smoothness indices, the source sample size, and the unknown transfer quality, while keeping the candidate built from target data alone available as a safety benchmark.

In this sense, the phase transitions described in Section 2 characterize when transfer is beneficial rather than how the estimator must be tuned. In practice, our adaptive procedure does not need to classify the problem into a strong, moderate, or minimal disparity regime, or into an ample or limited source regime, before fitting the estimator. Those regimes emerge only in the risk interpretation of the same adaptive procedure.

4 Numerical Experiments

This section evaluates the numerical performance of the adaptive transfer estimator, as depicted in Theorem 3.2. Subsection 4.1 examines its behavior under varying source sample sizes (n_s) and disparity levels (δ), confirming consistency with our theoretical findings. Subsection 4.2 demonstrates its practical utility in a phoneme classification task. Supplementary Material also contains additional experiments that examine robustness beyond the baseline setting, including more general source-target disparity structures, misspecification of exact bandability, and non-Gaussian sampling.

4.1 Simulation Study

We consider the target and source covariance matrices as follows:

$$\begin{aligned} \Sigma^{(t)} &:= (\sigma_{jk}^{(t)})_{j,k \in [d]} & \text{where } \sigma_{jk}^{(t)} &:= \begin{cases} 1 & \text{if } j = k, \\ C_t |j - k|^{-3} \xi_{jk}^{(t)} & \text{if } j \neq k, \end{cases} \\ \Sigma^{(s)} &:= (\sigma_{jk}^{(s)})_{j,k \in [d]} & \text{where } \sigma_{jk}^{(s)} &:= \begin{cases} 1 & \text{if } j = k, \\ \sigma_{jk}^{(t)} + C_s |j - k|^{-2} \xi_{jk}^{(s)} & \text{if } j \neq k. \end{cases} \end{aligned}$$

In this context, $\xi_{jk}^{(t)}$ and $\xi_{jk}^{(s)}$ ($1 \leq j < k \leq d$) represent independent Rademacher variables with $\xi_{jk}^{(t)} = \xi_{kj}^{(t)}$ and $\xi_{jk}^{(s)} = \xi_{kj}^{(s)}$ ($j, k \in [d]$). The constant C_t is set at 0.1, while the other constant $C_s > 0$ is adjusted in accordance with the disparity threshold $\delta := \|\Sigma^{(t)} - \Sigma^{(s)}\|_1$.

Notably, the target covariance matrix $\Sigma^{(t)}$ exhibits a target decay rate of $\alpha_t = 2$, whereas the source covariance matrix $\Sigma^{(s)}$ presents a source decay rate of $\alpha_s = 1$. This configuration merits careful attention because the condition $\alpha_s < \alpha_t$ makes the source covariance less regular than the target covariance and introduces additional challenges in estimating the target covariance matrix.

The target sample size n_t and the dimensionality d remain fixed at $(n_t, d) = (50, 50)$, reflecting their non-critical role in the effectiveness of transfer learning. Our investigation instead includes various combinations of the source sample size n_s and the disparity threshold δ . The source sample size is assessed at $n_s \in \{50, 100, 200, 400, 800\}$, and the disparity threshold is evaluated at $\delta \in \{0, 0.1, 0.4, 0.7, 1, 1.3\}$. Importantly, we introduce the scenario $(n_s, \delta) = (0, \infty)$ for comparative analysis. This scenario represents the conventional learning paradigm and thus establishes the baseline performance for our simulation study.

For each specified configuration, we repeat 1,000 simulations to generate target and source samples from mean-zero Gaussian distributions. Upon each simulated dataset, we employ the adaptive transfer estimator, as described in Theorem 3.2, and calculate its squared spectral-norm loss. The comprehensive results are visually represented in Figure 4, which unveils several intriguing patterns and corroborates our theoretical findings.

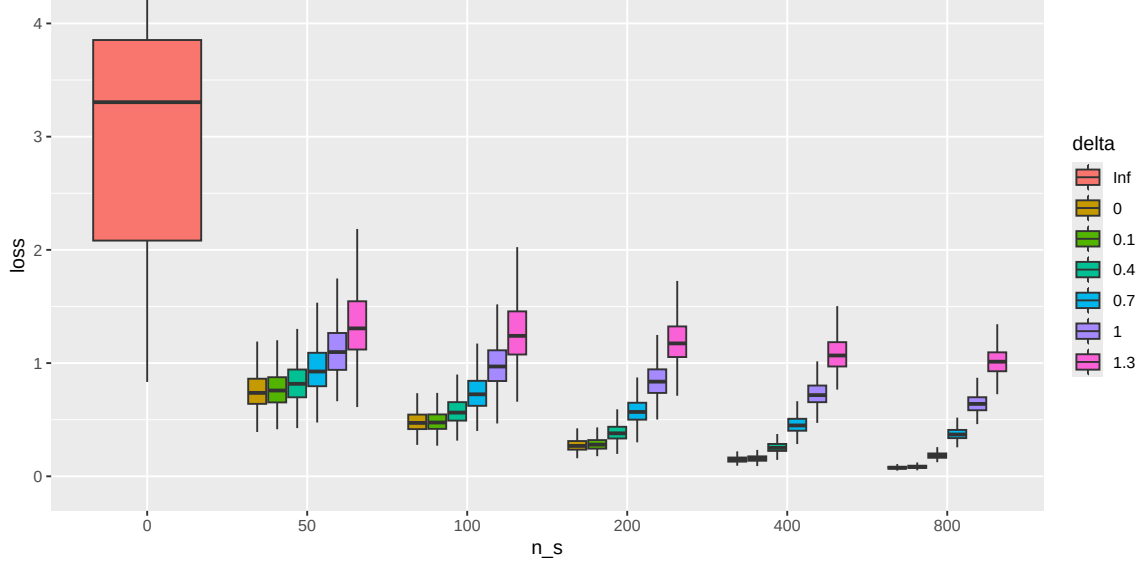


Figure 4: The results of numerical experiments under $\alpha_t = 2$ and $\alpha_s = 1$

First, when $\delta = 1.3$, transfer learning still improves over the baseline scenario, where $(n_s, \delta) = (0, \infty)$, although the gain is much smaller than in the low-disparity configurations. For small and moderate source sample sizes, the substantial discrepancy between the target and source covariance matrices limits the useful information to borrow. As n_s becomes larger, the loss falls further below the target-only baseline.

The configuration $\delta = 1.3$ represents a high-disparity setting in which borrowing information from the source is more challenging. Nevertheless, as depicted in Figure 4, the performance of the adaptive transfer estimator under $\delta = 1.3$ does not markedly deteriorate compared to the target-only baseline $(n_s, \delta) = (0, \infty)$. Thus, under the displayed settings, the adaptive transfer estimator remains competitive with the target-only baseline even when the source-target discrepancy is large.

Conversely, when δ is set below 1.3, Figure 4 exhibits an evident improvement relative to the baseline scenario, where $(n_s, \delta) = (0, \infty)$. This improvement amplifies as the source sample size n_s expands or as the disparity threshold δ diminishes. Remarkably, the model's performance at $\delta = 0.1$ is nearly indistinguishable from that at $\delta = 0$, suggesting that the

source sample generated with $\delta = 0.1$ is as potent as one with $\delta = 0$. These empirical findings are consistent with our theoretical assertions and validate in practice the effectiveness of the adaptive transfer estimator.

4.2 Application to Discriminant Analysis of Phoneme Dataset

In this section, we apply the adaptive transfer estimator to a real-world dataset. Since the true covariance matrix is generally unknown in practice, directly assessing the accuracy of our estimate is infeasible. However, any method that relies on covariance estimation is expected to benefit from a more accurate covariance estimator when a similar but distinct source sample is available. To evaluate the practical utility of our estimator, we employ discriminant analysis in a classification problem, using classification accuracy as an indirect measure of performance. Specifically, we anticipate higher accuracy in discriminant analysis when incorporating a source sample and applying our covariance estimator, as knowledge transfer should lead to a more accurate estimate of the covariance matrix.

We consider a binary classification problem as described in Hastie et al. (2009). The dataset is derived from the TIMIT database, a widely used resource for speech recognition research, provided by the Linguistic Data Consortium (Garofolo et al., 1993). The processed phoneme benchmark considered here follows the version used in Hastie et al. (1995). It contains short sound recordings of male speakers pronouncing one of two similar-sounding phonemes. The objective is to develop a classifier for automatically labeling these sounds. The predictor vectors, x , are log-periodograms, which represent the log-intensity of the sound recordings across $d = 256$ frequencies. Each log-periodogram is labeled with its corresponding phoneme, y .

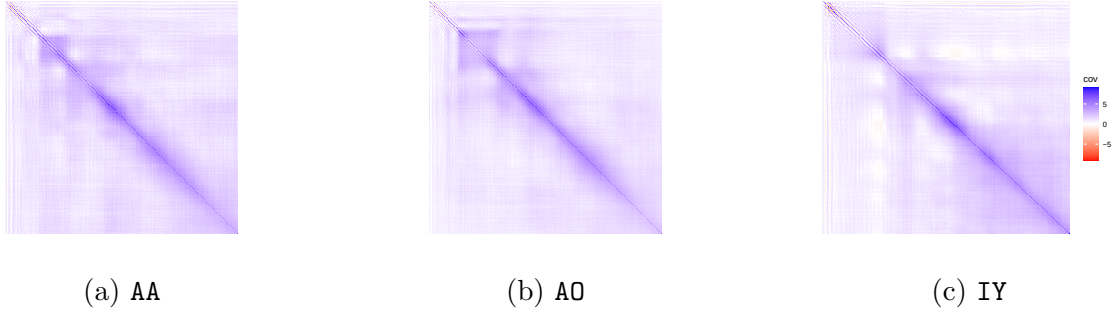


Figure 5: Sample Covariance Matrices of Three Phoneme Sounds

Our analysis focuses on three phonemes: **AA** as the vowel in “dark”, **AO** as the first vowel in “water” and **IY** as the vowel in “she”. Specifically, we develop a classifier to distinguish between **AA** and **AO** phonemes. They sound very similar and are difficult to differentiate, even for human listeners, given the short duration of the recordings. We randomly split the data into training and test sets, with the training set containing **AA** samples of $n_{\text{AA}} = 69$ and **AO** samples of $n_{\text{AO}} = 102$.

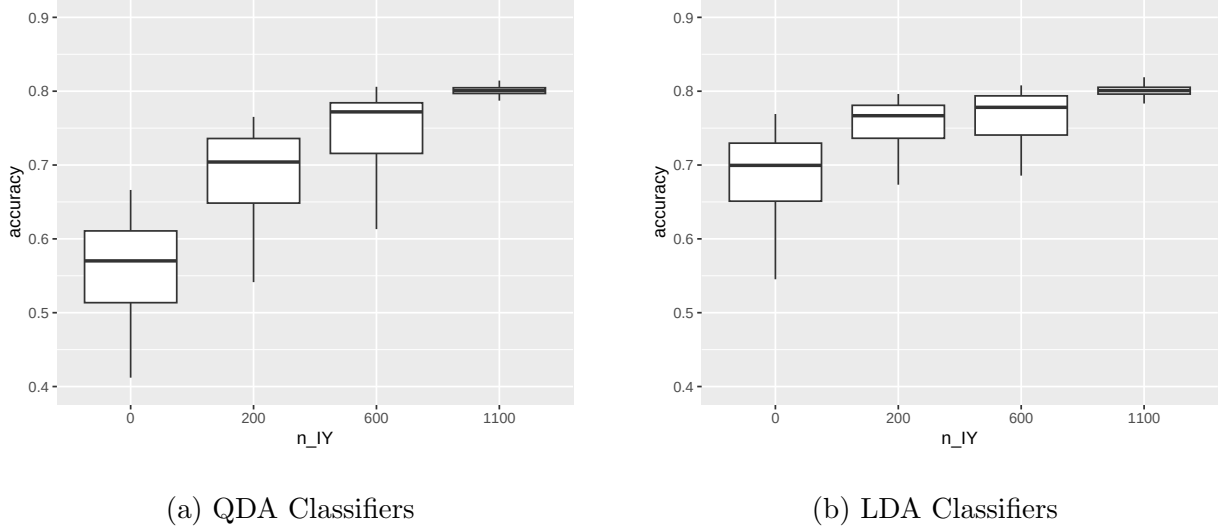


Figure 6: Accuracy of Discriminant Analysis of Phoneme Dataset

Note that the training set is relatively small compared to the dimension $d = 256$, which hinders both the accurate estimation of the covariance matrix and the performance of discriminant analysis. One way to mitigate the effects of limited sample size is to integrate

similar but distinct sounds. In particular, the IY sound is classified as a sonorant vowel, like both AA and AO (Dekel et al., 2004; Lee and Hon, 1989). Figure 5 supports the use of the adaptive transfer estimator, as all covariance matrices exhibit a bandable structure, and IY shows some degree of similarity with both AA and AO.

With 1,163 auxiliary IY samples available, we utilize subsets of these to enhance covariance matrix estimation for Quadratic Discriminant Analysis (QDA) and Linear Discriminant Analysis (LDA). Specifically, we randomly sample IY sounds at sizes $n_{\text{IY}} \in \{0, 200, 600, 1100\}$. The case $n_{\text{IY}} = 0$ represents the target-only version of the adaptive transfer estimator, without auxiliary data.

For each random split, LDA utilizes the pooled sample covariance matrix of the AA and AO training data, while QDA employs class-specific sample covariance matrices. When $n_{\text{IY}} > 0$, we compute the sample covariance of the IY subsample and provide it as the source input to the adaptive transfer estimator—applied once for the pooled covariance in LDA and separately for the two class-specific matrices in QDA. Class means and priors are consistently estimated from the AA/AO training data alone. The resulting covariance estimates are then plugged into the standard LDA or QDA decision rules.

Theoretical support for this approach is provided in Proposition E.1 of Supplementary Material E, which links operator-norm covariance error to excess classification risk in binary Gaussian LDA. The classification accuracy of these models is presented in Figure 6. Across the 100 random splits, the mean accuracies at $n_{\text{IY}} \in \{0, 200, 600, 1100\}$ are $(0.562, 0.683, 0.741, 0.801)$ for QDA and $(0.677, 0.743, 0.752, 0.801)$ for LDA, respectively. Thus, on this grid, accuracy increases monotonically with the auxiliary sample size; at $n_{\text{IY}} = 1100$, the gains over the target-only fit are 0.239 for QDA and 0.124 for LDA.

5 Discussions

This paper explores minimax and adaptive estimation of high-dimensional bandable covariance matrices within a transfer learning framework. We establish the minimax rate of convergence under squared spectral-norm loss and introduce a blockwise tridiagonal estimator that achieves optimality. This minimax optimal rate, denoted by Φ , reveals phase transitions that characterize when source samples improve target-domain estimation.

For unknown smoothness and disparity parameters, we develop the adaptive transfer block-tridiagonal estimator. Its risk satisfies the global additive oracle bound $\Phi + (\delta^2 \wedge \Gamma_{m_f, n})$, where the selector remainder depends on the validation size and only logarithmically on the total sample size. The remainder is absorbed by Φ under near transfer, including exact transfer, and under the stated polynomial growth regime for the source sample size and dimension.

The numerical results provide complementary evidence for the adaptive transfer estimator. Across the reported source-sample sizes and disparity levels, transfer gains are larger for smaller disparity and larger source sample sizes, whereas at high disparity the adaptive estimator remains close to the target-only baseline. Application to the phoneme data further demonstrates the usefulness of our adaptive estimator.

Our study focuses on estimation under the squared spectral-norm loss. While conventional bandable covariance estimation has been investigated under other measures—such as the matrix ℓ_1 -norm, Frobenius norm, and general Bregman divergences—extending these to the transfer learning framework represents a compelling direction for future work. Similarly, extending these results to other structures, such as sparse and spiked covariance matrices, or to the estimation of bandable precision matrices, remains an open and promising area of research.

Finally, we identify robustness to heavy-tailed or contaminated observations as a critical

future direction. Although simulations in the supplementary material suggest the proposed procedure remains empirically competitive beyond Gaussian settings, a rigorous extension would require robust estimators tailored to the transfer setting. Potential avenues include matrix-truncation and spectrum-truncation approaches (Minsker and Wei, 2018; Ke et al., 2019), or contamination-robust procedures under Huber’s model (Chen et al., 2018; Avella-Medina et al., 2018). Retaining sharp upper bounds and adaptive guarantees in these robust-transfer settings remains a substantial methodological challenge.

These topics are left for future investigation. We anticipate that the framework, insights, and theoretical results presented here will serve as foundational resources for subsequent studies in this domain.

SUPPLEMENTARY MATERIAL

Title: Supplement to “Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity.”

This supplementary material provides the complete proofs of the main theorems and technical results presented in the paper titled “Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity.” The contents are organized as follows. Supplementary Material A introduces fundamental notation, develops weaker-assumption and misspecified-model extensions through Proposition A.1 and Theorems A.3 and A.4, and contains the proofs of Theorems A.5, 2.2 and 3.2. Supplementary Material B and Supplementary Material C provide proofs of the technical results for optimality and adaptive aggregation, respectively. Supplementary Material D contains additional numerical experiments for the adaptive transfer estimator. Supplementary Material E gives a preliminary classification guarantee for plug-in LDA.

Acknowledgments

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Abstract

This supplementary material provides the complete proofs of the main theorems and technical results presented in the paper titled “Transfer Learning for Covariance Matrix Estimation: Optimality and Adaptivity.” The contents are organized as follows. Supplementary Material A introduces fundamental notation, develops weaker-assumption and misspecified-model extensions through Proposition A.1 and Theorems A.3 and A.4, and contains the proofs of Theorems A.5, 2.2 and 3.2. Supplementary Material B and Supplementary Material C provide proofs of the technical results for optimality and adaptive aggregation, respectively. Supplementary Material D contains additional numerical experiments for the adaptive transfer block-tridiagonal estimator. Supplementary Material E gives a preliminary classification guarantee for plug-in LDA.

A Proof of Main Theorems

Before looking into the technical specifics, it is essential to establish some fundamental notations that will be consistently employed throughout this paper. First of all, let us revisit the core operators introduced earlier in the main text: the blockwise tridiagonal operator $\mathfrak{T}_d$ and the superbanding operator $\mathfrak{S}_d$. For any given bandwidths $u > 0$ and $u_2 > u_1 > 0$, we shall inspect the output matrices, $\mathfrak{T}_d(B; u)$ and $\mathfrak{S}_d(B; u_1, u_2)$ ($B \in \mathbb{R}^{d \times d}$), in the blockwise standpoint.

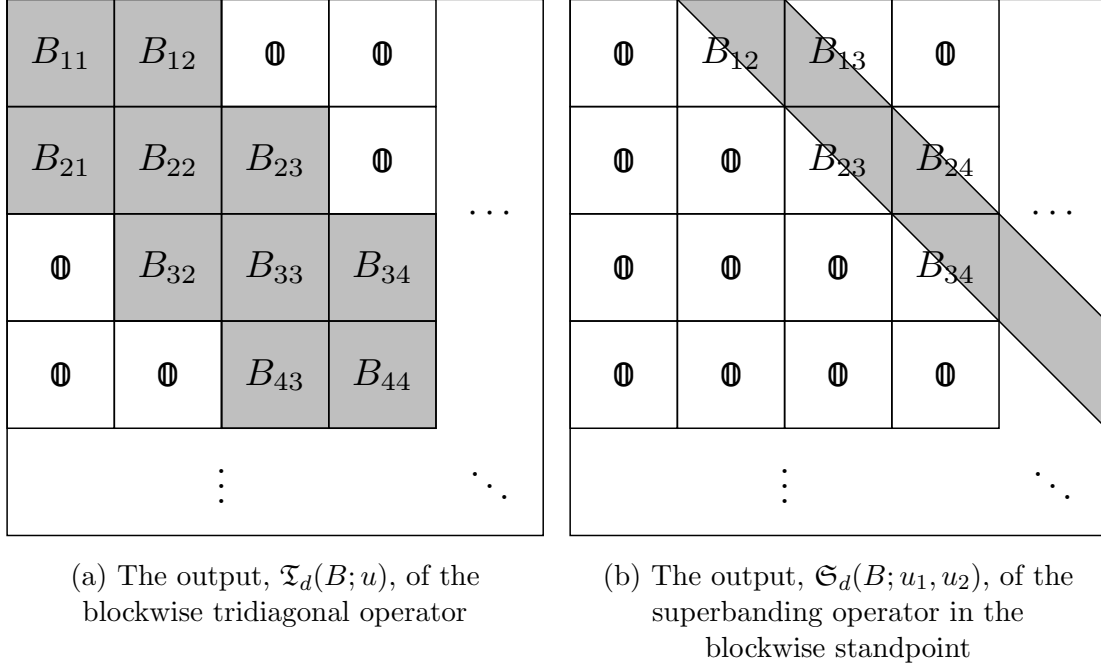


Figure 7: A visual representation of important operators

As previously established in the main text and illustrated in Figure 7a, the matrix $\mathfrak{T}_d(B; u)$ is identifiable as a tridiagonal matrix from a blockwise viewpoint. For ease of reference, let $\mathcal{T}_d(B; u)$ denote the fixed collection of coordinate submatrices at every main-, first super-, and first sub-diagonal block position retained by $\mathfrak{T}_d(\cdot; u)$. The collection includes a retained block even when that block happens numerically to be zero. Its index set therefore depends only on (d, u) , not on the realization of B , and each member has both dimensions at most u .

On the other hand, the matrix $\mathfrak{S}_d(B; u_1, u_2)$ can be recognized as a double superdiagonal matrix from a blockwise viewpoint. As depicted in Figure 7b, the first superdiagonal of the block matrix $\mathfrak{S}_d(B; u_1, u_2)$ contains lower triangular sub-matrices, while the second superdiagonal consists of upper triangular sub-matrices. Most nonzero blocks are $(u_2 - u_1) \times (u_2 - u_1)$ submatrices, although the terminal blocks on the two superdiagonals may have different sizes depending on $u_2 - u_1$ and d . For later convenience, $\mathcal{S}_d(B; u_1, u_2)$ will denote the collection of all non-zero sub-matrices in the superbanding matrix $\mathfrak{S}_d(B; u_1, u_2)$. Every submatrix in $\mathcal{S}_d(B; u_1, u_2)$ has dimensions at most $(u_2 - u_1) \times (u_2 - u_1)$.

Moving forward, we will talk about some general notations. Following the works of Vershynin (2018) and Wainwright (2019), we define the sub-exponential norm, or Orlicz

ψ_1 -norm, of a random variable Z as:

$$\|Z\|_{\psi_1} := \inf\{r > 0 : \mathbb{E}e^{|Z|/r} \leq 2\}.$$

In the context of bandable covariance matrices, we define two quantities: the minimum radius $M_{\min} := M_t \wedge M_s$ and the maximum radius $M_{\max} := M_t \vee M_s$. The Hadamard product is represented by the operator $\circ$, and the Frobenius norm by $\|\cdot\|_F$. Given a symmetric matrix $B \in \mathbb{R}^{d \times d}$, we write its ordered eigenvalues as

$$\gamma_{\max}(B) = \gamma_1(B) \geq \gamma_2(B) \geq \cdots \geq \gamma_d(B) = \gamma_{\min}(B).$$

For any vectors $v, w \in \mathbb{R}^d$, we denote their Hamming distance as $d_H(v, w)$. Plus, we consider the mutual information $I(Z; W)$, which quantifies the dependence between the random variables Z and W . Lastly, we express the Kullback-Leibler divergence as $D_{\text{KL}}(\mathbb{Q}_1 \parallel \mathbb{Q}_0)$ and the total variation as $\|\mathbb{Q}_0 - \mathbb{Q}_1\|_{\text{TV}}$, both of which capture the dissimilarity between the probability distributions $\mathbb{Q}_0$ and $\mathbb{Q}_1$.

A.1 Weaker Assumptions and Misspecified Model

For the upper bound proofs below, we isolate a generic upper-tail stability (UTS) condition that can be reused for both the source-target disparity matrix and the misspecification remainder terms. For any matrix $B = (b_{jk})_{j,k \in [d]} \in \mathbb{R}^{d \times d}$ and any bandwidth $u > 0$, let us define its upper-tail operator as

$$\mathfrak{A}_d^+(B; u) := \left(b_{jk} \mathbb{1}(k - j > u) \right)_{j,k \in [d]},$$

where we further adopt the convention $\mathfrak{A}_d^+(B; 0) := B$.

We say that a matrix $A \in \mathbb{R}^{d \times d}$ is *upper-tail stable at level $r \geq 0$* if

$$\|\mathfrak{A}_d^+(A; u)\|_2 \leq r, \quad \text{for every bandwidth } u \geq 0. \quad (9)$$

In particular, if A is symmetric, then for every bandwidth $u > 0$,

$$\mathfrak{A}_d(A; u) = \mathfrak{A}_d^+(A; u) + \left(\mathfrak{A}_d^+(A; u) \right)^\top, \quad \|\mathfrak{A}_d(A; u)\|_2 \leq 2\|\mathfrak{A}_d^+(A; u)\|_2. \quad (10)$$

We further record a robustness extension of the upper bound under tail-stable misspecification. Throughout the supplement, let $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$ denote the collection of joint distributions for the target and source samples that satisfy:

- the uniformly sub-Gaussian assumption Assumption 2.1;
- target and source covariance matrices $\Sigma^{(t)}$ and $\Sigma^{(s)}$ are approximately bandable: they admit decompositions

$$\Sigma^{(t)} = \Sigma_0^{(t)} + R^{(t)}, \quad \Sigma^{(s)} = \Sigma_0^{(s)} + R^{(s)}, \quad (11)$$

where

$$\Sigma_0^{(t)} \in \mathcal{B}_{\alpha_t}(M_t), \quad \Sigma_0^{(s)} \in \mathcal{B}_{\alpha_s}(M_s),$$

and the symmetric remainder matrices $R^{(t)}$ and $R^{(s)}$ are upper-tail stable at level η .

- the actual source-target disparity matrix $\Delta^{(s)} := \Sigma^{(t)} - \Sigma^{(s)}$ is upper-tail stable at level δ , namely

$$\|\mathfrak{A}_d^+(\Delta^{(s)}; u)\|_2 \leq \delta, \quad \text{for every bandwidth } u \geq 0. \quad (12)$$

The exact bandable model $\mathcal{P}_{\alpha_t, \alpha_s, \delta}$ from the main text is contained in $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$ when $R^{(t)} = R^{(s)} = \mathbb{0}$, equivalently $\eta = 0$. Moreover, the original disparity condition $\|\Delta^{(s)}\|_1 \leq \delta$ implies Equation (12) by Proposition A.1. Moreover, the upper-tail stability requirement is mild and can be verified under several familiar sufficient conditions. This extension allows us to establish the robustness of the proposed estimator against general disparity and misspecification of the model.

In detail, the next proposition shows that upper-tail stability itself is a mild requirement:

Proposition A.1. *Let $A \in \mathbb{R}^{d \times d}$ be symmetric and $r \geq 0$. Then A is upper-tail stable at level r if one of the following conditions is satisfied:*

- (1) $\|A\|_1 \leq r$.
- (2) $\|A_{\text{abs}}\|_2 \leq r$, where $A_{\text{abs}} := (|a_{jk}|)_{j,k \in [d]}$.
- (3) $\|A\|_{\mathcal{S}_p} \leq r$ for some $p \in [1, 2]$, where $\|\cdot\|_{\mathcal{S}_p}$ denotes the Schatten p -norm.
- (4) $d \geq 2$ and $\|A\|_2 \leq c(\log d)^{-1}r$ for a sufficiently small absolute constant $c > 0$.

Proof. Fix any bandwidth $u \geq 0$.

If condition (1) holds, then for $u > 0$,

$$\|\mathfrak{A}_d^+(A; u)\|_2 \leq (\|\mathfrak{A}_d^+(A; u)\|_1 \|\mathfrak{A}_d^+(A; u)\|_\infty)^{1/2} \leq (\|A\|_1 \|A\|_\infty)^{1/2} = \|A\|_1 \leq r,$$

while the case $u = 0$ follows immediately from $\|A\|_2 \leq \|A\|_1$.

If condition (2) holds, then for any unit vectors $x, y \in \mathbb{R}^d$, where $|x|$ and $|y|$ are taken entrywise,

$$|y^\top \mathfrak{A}_d^+(A; u)x| \leq |y|^\top \mathfrak{A}_d^+(A_{\text{abs}}; u)|x| \leq |y|^\top A_{\text{abs}}|x| \leq \|A_{\text{abs}}\|_2.$$

Taking the supremum over all unit vectors x, y yields $\|\mathfrak{A}_d^+(A; u)\|_2 \leq r$. The same argument with $u = 0$ gives $\|A\|_2 \leq \|A_{\text{abs}}\|_2 \leq r$.

If condition (3) holds, then

$$\|\mathfrak{A}_d^+(A; u)\|_2 \leq \|\mathfrak{A}_d^+(A; u)\|_F \leq \|A\|_F \leq \|A\|_{\mathcal{S}_p} \leq r,$$

where the third inequality follows from $1 \leq p \leq 2$. The case $u = 0$ follows from $\|A\|_2 \leq \|A\|_{\mathcal{S}_p}$.

Finally, if condition (4) holds, then the standard operator-norm bound for triangular projection (Kwapień and Pełczyński, 1970) implies that there exists an absolute constant $C > 0$ such that

$$\|\mathfrak{A}_d^+(A; u)\|_2 \leq C(\log d)\|A\|_2 \leq r$$

for every $u > 0$, provided $c \leq C^{-1}$. Since $d \geq 2$, the same assumption also gives $\|A\|_2 \leq r$ when $u = 0$. $\square$

Remark A.2. The upper-tail stability condition is strictly weaker than the matrix ℓ_1 -condition used in the exact model. Indeed, fix any $d \geq 3$ and $r > 0$, and define $b \in \mathbb{R}^d$ by

$$b_1 = 0, \quad b_2 = \cdots = b_d = \frac{r}{\sqrt{d-1}},$$

together with the symmetric hub matrix

$$A := e_1 b^\top + b e_1^\top.$$

Equivalently, A takes the explicit hub form

$$A = \frac{r}{\sqrt{d-1}} \begin{pmatrix} 0 & 1 & 1 & \cdots & 1 \\ 1 & 0 & 0 & \cdots & 0 \\ 1 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & 0 \end{pmatrix},$$

so all non-zero entries are concentrated on the first row and the first column. For every bandwidth $u > 0$, we have $\mathfrak{A}_d^+(A; u) = e_1 (b^{(u)})^\top$, where $b_k^{(u)} := b_k \mathbf{1}(k > 1 + u)$. Hence,

$$\|\mathfrak{A}_d^+(A; u)\|_2 = \|b^{(u)}\|_2 \leq \|b\|_2 = r.$$

Since the only non-zero eigenvalues of A are $\pm \|b\|_2$, the convention $\mathfrak{A}_d^+(A; 0) = A$ yields $\|\mathfrak{A}_d^+(A; 0)\|_2 = \|A\|_2 = r$. Therefore, A is upper-tail stable at level r . On the other hand,

$$\|A\|_1 = \max_{k \in [d]} \sum_{j=1}^d |a_{jk}| = \sum_{k=2}^d |b_k| = r\sqrt{d-1} > r.$$

Thus, the upper-tail stability requirement is strictly weaker than the matrix ℓ_1 -condition in this case, particularly when d is large.

Theorem A.3 (Block tridiagonal estimator). *Consider blockwise tridiagonal estimator $\widehat{\Sigma}^{(t)}$ for the target covariance matrix $\Sigma^{(t)}$ as follows:*

$$\widehat{\Sigma}^{(t)} := \begin{cases} \mathfrak{T}_d(\check{\Sigma}^{(t)}; u^*) & \text{if } n_t \delta^2 > (n_t^{1/(2\alpha_t+1)} \wedge d) \vee \log d, \\ \mathfrak{T}_d(\check{\Sigma}; v^*) & \text{if } n_t \delta^2 \leq (n_t^{1/(2\alpha_t+1)} \wedge d) \vee \log d, \end{cases}$$

where the optimal bandwidths $u^*, v^* > 0$ are chosen such that:

$$u^* \asymp n_t^{1/(2\alpha_t+1)} \wedge d \quad \text{and} \quad v^* \asymp n^{1/(2\alpha_{\max}+1)} \wedge d.$$

The maximal risk of the blockwise tridiagonal estimator $\widehat{\Sigma}^{(t)}$ now satisfies:

$$\sup_{\mathbb{P} \in \widetilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim \Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d) + \eta^2,$$

as long as either of the following assumptions holds:

Assumption (a).* $\log d \leq c_1 n_t$ for a small enough constant $c_1 > 0$.

Assumption (b^{}).* $\log d \leq c_2 n$ and $\delta \leq c_2$ for a small enough constant $c_2 > 0$.

Theorem A.4 (Adaptive estimator). *Let $n = n_t + n_s$. For every fixed $\alpha_t, \alpha_s > 0$ and every $\delta, \eta \geq 0$, the estimator in Subsection C.1 satisfies*

$$\sup_{\mathbb{P} \in \tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E} \|\hat{\Sigma}_{\text{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim \Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d) + \eta^2 + (\delta^2 \wedge \Gamma_{m_f, n}), \quad (13)$$

where

$$\Gamma_{m_f, n} := \rho^4 \left(\frac{\log(en)}{m_f} + \frac{\log^2(en)}{m_f^2} \right). \quad (14)$$

The implicit constant is uniform when (α_t, α_s) ranges over a fixed compact subset of $(0, \infty)^2$. No smallness assumption on η is required.

A.2 Proof of Theorem 2.2

The upper bound has already been established in Theorem A.3. Notice that Assumption (a^{*}) and Assumption (b^{*}) are strictly weaker than Assumption (a) and Assumption (b), respectively. In particular, Theorem A.3 specializes to the exact bandable model $\mathcal{P}_{\alpha_t, \alpha_s, \delta}$ by taking $\eta = 0$, and therefore guarantees consistent estimation of the target covariance matrix under the transfer learning framework whenever either Assumption (a) or Assumption (b) holds.

We next present the corresponding lower bound, which characterizes the fundamental difficulty of the estimation problem.

Theorem A.5 (Minimax lower bound). *Suppose that d is sufficiently large. Given any estimator $\hat{\Sigma}^{(t)}$ of the target covariance matrix $\Sigma^{(t)}$, we have*

$$\begin{aligned} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\hat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\gtrsim \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \left(\frac{\log d}{n} \wedge 1 \right) \\ &\quad + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2 \right) + \left(\frac{\log d}{n_t} \wedge \delta^2 \wedge 1 \right). \end{aligned}$$

According to Theorem A.5, consistent estimation requires $\log d/n \rightarrow 0$ and $(\log d/n_t) \wedge \delta^2 \rightarrow 0$. Under this exact criterion, the lower bound in Theorem A.5 matches the specialization of the upper bound in Theorem A.3 to the exact bandable model, that is, to $\eta = 0$. Assumptions (a) and (b) in Theorem 2.2 are two common sufficient regimes for the criterion. This establishes the minimax rate of convergence, $\Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d)$, as desired. We do not claim a matching misspecified minimax lower bound here.

A.3 Auxiliary Risk Bounds for the Upper Bound

Proposition A.6. *Under the misspecified model $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$, we can identify a sufficiently small constant $c_t > 0$ ensuring that for any bandwidth $u > 0$ and any dimension $d \in \mathbb{Z}^+$ satisfying the condition $u \vee \log d \leq c_t n_t$,*

$$\mathbb{E} \|\mathfrak{T}_d(\tilde{\Sigma}^{(t)}; u) - \Sigma^{(t)}\|_2^2 \lesssim \frac{u + \log d}{n_t} + (u^{-2\alpha_{\max}} + (u^{-2\alpha_t} \wedge \delta^2) + \eta^2) \mathbf{1}(u < d).$$

Proposition A.7. *Under the misspecified model $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$, we can identify a sufficiently small constant $c_p > 0$ ensuring that for any bandwidth $v > 0$ and any dimension $d \in \mathbb{Z}^+$ satisfying the condition $v \vee \log d \leq c_p n$,*

$$\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}; v) - \Sigma\|_2^2 \lesssim \frac{v + \log d}{n} + (v^{-2\alpha_{\max}} + (v^{-2\alpha_{\min}} \wedge \delta^2) + \eta^2) \mathbb{1}(v < d).$$

A.4 Proof of Theorem A.3

We prove Theorem A.3 by considering the same two regimes.

Case I: The disparity threshold δ is sufficiently large as $n_t \delta^2 > (n_t^{1/(2\alpha_t+1)} \wedge d) \vee \log d$

Under this regime, $\hat{\Sigma}^{(t)} = \mathfrak{T}_d(\check{\Sigma}^{(t)}; u^*)$. To apply Proposition A.6, it suffices to verify the preliminary condition $u^* \vee \log d \leq c_t n_t$. Since $u^* \asymp n_t^{1/(2\alpha_t+1)} \wedge d$, we already have $u^* \lesssim n_t$. For the logarithmic term, Assumption (a*) directly gives $\log d \leq c_1 n_t$. Under Assumption (b*), we additionally have $\delta \leq c_2$, and the current regime yields

$$\log d \leq (n_t^{1/(2\alpha_t+1)} \wedge d) \vee \log d \leq n_t \delta^2 \leq c_2^2 n_t.$$

Hence $\log d \leq (c_1 \vee c_2^2) n_t$ whenever either Assumption (a*) or Assumption (b*) is satisfied. Therefore the preliminary condition in Proposition A.6 is ensured after shrinking the small constants $c_1, c_2 > 0$ if necessary and absorbing the implied constant in u^* into c_t . Using Proposition A.6, we obtain

$$\begin{aligned} \mathbb{E} \|\hat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &= \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)}; u^*) - \Sigma^{(t)}\|_2^2 \\ &\lesssim \frac{u^* + \log d}{n_t} + ((u^*)^{-2\alpha_{\max}} + ((u^*)^{-2\alpha_t} \wedge \delta^2) + \eta^2) \mathbb{1}(u^* < d) \end{aligned}$$

Here $u^* \asymp n_t^{1/(2\alpha_t+1)} \wedge d$, so $u^*/n_t \asymp (u^*)^{-2\alpha_t}$ whenever $u^* < d$, while the indicator vanishes when $u^* = d$. Moreover, the current regime gives

$$\frac{u^*}{n_t} \vee \frac{\log d}{n_t} \leq \delta^2.$$

Hence both u^*/n_t and $(\log d)/n_t$ are absorbed by the two δ^2 -clipped target terms in the claimed rate. Also, $(u^*)^{-2\alpha_{\max}} \leq (u^*)^{-2\alpha_t}$ when $u^* < d$. Therefore

$$\begin{aligned} \mathbb{E} \|\hat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\lesssim \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \frac{\log d}{n} \\ &\quad + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2 \right) + \left(\frac{\log d}{n_t} \wedge \delta^2 \right) + \eta^2. \end{aligned}$$

Case II: The disparity threshold δ is sufficiently small as $n_t \delta^2 \leq (n_t^{1/(2\alpha_t+1)} \wedge d) \vee \log d$

Under this regime, $\hat{\Sigma}^{(t)} = \mathfrak{T}_d(\check{\Sigma}; v^*)$. To apply Proposition A.7, it suffices to verify the preliminary condition $v^* \vee \log d \leq c_p n$. Since $v^* \asymp n^{1/(2\alpha_{\max}+1)} \wedge d$, we have $v^* \lesssim n$. For the logarithmic term, Assumption (a*) implies

$$\log d \leq c_1 n_t \leq c_1 n,$$

whereas Assumption (b*) directly gives $\log d \leq c_2 n$. Hence $\log d \leq (c_1 \vee c_2)n$ whenever either Assumption (a*) or Assumption (b*) is satisfied. Therefore the preliminary condition in Proposition A.7 is ensured after shrinking the small constants $c_1, c_2 > 0$ if necessary and absorbing the implied constant in v^* into c_p . By the upper-tail condition Equation (12) with $u = 0$ and the identity $\Delta = \frac{n_s}{n} \Delta^{(s)}$, we have $\|\Delta\|_2 \leq \delta$. Using Proposition A.7, we obtain

$$\begin{aligned} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\leq 2\mathbb{E}\|\mathfrak{T}_d(\check{\Sigma}; v^*) - \Sigma\|_2^2 + 2\|\Delta\|_2^2 \\ &\lesssim \frac{v^* + \log d}{n} + ((v^*)^{-2\alpha_{\max}} + ((v^*)^{-2\alpha_{\min}} \wedge \delta^2) + \eta^2)\mathbb{1}(v^* < d) + \delta^2 \end{aligned}$$

Since $v^* \asymp n^{1/(2\alpha_{\max}+1)} \wedge d$, the two pooled terms v^*/n and $(v^*)^{-2\alpha_{\max}}$ are both of order $n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge d/n$. Moreover, $((v^*)^{-2\alpha_{\min}} \wedge \delta^2) \leq \delta^2$, and the current regime yields

$$\delta^2 \leq \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \right) \vee \frac{\log d}{n_t} \leq \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \right) + \frac{\log d}{n_t}.$$

Therefore both the extra $\|\Delta\|_2^2$ term and the pooled bias term $((v^*)^{-2\alpha_{\min}} \wedge \delta^2)$ are absorbed by the two δ^2 -clipped target terms in the claimed rate. Hence

$$\begin{aligned} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\lesssim \left(n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n} \right) + \frac{\log d}{n} \\ &\quad + \left(n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2 \right) + \left(\frac{\log d}{n_t} \wedge \delta^2 \right) + \eta^2. \end{aligned}$$

This completes the proof of Theorem A.3. $\square$

A.5 Proof of Theorem A.5

It suffices to prove the following four lower bounds. We establish them one by one in the subsequent subsections and combine them at the end.

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n}, \quad (15)$$

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim \frac{\log d}{n} \wedge 1, \quad (16)$$

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2, \quad (17)$$

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E}\|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim \frac{\log d}{n_t} \wedge \delta^2 \wedge 1. \quad (18)$$

Under the convention in Assumption 2.1, if $Z \sim N(0, \sigma^2)$, then

$$\mathbb{E} \exp(Z^2/\rho^2) = \left(1 - \frac{2\sigma^2}{\rho^2} \right)^{-1/2} \leq 2 \quad \text{whenever} \quad \sigma^2 \leq \frac{3\rho^2}{8}.$$

Thus, the Gaussian constructions below satisfy Assumption 2.1 whenever their covariance matrix Σ obeys

$$\gamma_{\max}(\Sigma) \leq \frac{3\rho^2}{8}.$$

Proof of lower bound in Equation (15)

Suppose that we are given a dimension $r \in \mathbb{Z}^+$ and a tolerance $\varepsilon > 0$ which will be specified later. For any $\theta \in \{0, 1\}^r$ on the r -dimensional hypercube with $2r \leq d$, we define a symmetric matrix $\Sigma_\theta \in \mathbb{R}^{d \times d}$ such that:

$$\Sigma_\theta := \frac{5}{16}\rho^2 I_d + \varepsilon \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \quad \text{where } \mathbf{1}_r := (1, 1, \dots, 1)^\top \in \mathbb{R}^r.$$

Consider a probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ ($\theta \in \{0, 1\}^r$) indexed by the r -dimensional hypercube. Under $\mathbb{Q}_\theta$, we assume the following target and source samples. It is worth noting that both target and source samples share the same covariance matrix Σ_θ .

$$X_1^{(t)}, \dots, X_{n_t}^{(t)}, X_1^{(s)}, \dots, X_{n_s}^{(s)} \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, \Sigma_\theta).$$

We need to check the validity of covariance matrix Σ_θ as well as other modeling assumptions to confirm $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ for any given $\theta \in \{0, 1\}^r$. Observe that Σ_θ is positive definite and a valid covariance matrix if $r\varepsilon \leq \rho^2/16$:

$$\gamma_{\min}(\Sigma_\theta) = \frac{5}{16}\rho^2 + \varepsilon \gamma_{\min} \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \geq \frac{5}{16}\rho^2 - \varepsilon \left\| \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \right\|_1 \geq \frac{5}{16}\rho^2 - r\varepsilon \geq \frac{1}{4}\rho^2.$$

Not only that, the probability measure $\mathbb{Q}_\theta$ should meet the uniformly sub-Gaussian condition in Assumption 2.1. It is worth noting that $a^\top X_1^{(t)}$ follows a Gaussian distribution with variance $a^\top \Sigma_\theta a$ for any vector $a \in \mathbb{R}^d$. The same restriction also guarantees the uniform sub-Gaussianity assumption:

$$\gamma_{\max}(\Sigma_\theta) = \frac{5}{16}\rho^2 + \varepsilon \gamma_{\max} \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \leq \frac{5}{16}\rho^2 + \varepsilon \left\| \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \right\|_1 \leq \frac{5}{16}\rho^2 + r\varepsilon \leq \frac{3}{8}\rho^2.$$

The covariance matrix of the measure $\mathbb{Q}_\theta$ needs to satisfy the bandable structure. It suffices to argue $\Sigma_\theta \in \mathcal{B}_{\alpha_t}(M_t) \cap \mathcal{B}_{\alpha_s}(M_s)$ since the target and source covariance matrices are equal under $\mathbb{Q}_\theta$. This condition is simply ensured by the condition $r^{\alpha_{\max}+1}\varepsilon \leq 2^{-\alpha_{\max}}M_{\min}$. For $0 < u < 1$, the same condition gives, for $g \in \{t, s\}$,

$$\|\mathfrak{A}(\Sigma_\theta; u)\|_1 \leq r\varepsilon \leq M_{\min}(2r)^{-\alpha_{\max}} \leq M_{\min} \leq M_g u^{-\alpha_g}.$$

For $1 \leq u \leq 2r$, we instead have

$$\|\mathfrak{A}(\Sigma_\theta; u)\|_1 \leq M_{\min} u^{-\alpha_{\max}} \leq (M_t u^{-\alpha_t}) \wedge (M_s u^{-\alpha_s}),$$

whereas $\mathfrak{A}(\Sigma_\theta; u) = 0$ for $u > 2r$. Finally, the disparity condition is always satisfied; we have assumed no disparity between the target and source covariance matrices. To sum up, we have constructed the valid probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ ($\theta \in \{0, 1\}^r$) as long as the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ are chosen to be:

$$2r \leq d, \quad r\varepsilon \leq \frac{\rho^2}{16}, \quad \text{and} \quad r^{\alpha_{\max}+1}\varepsilon \leq 2^{-\alpha_{\max}}M_{\min}. \quad (19)$$

We will next construct a sub-model $\mathcal{Q}(r, \varepsilon) := \{\mathbb{Q}_\theta : \theta \in \Theta\}$ for some $\Theta \subset \{0, 1\}^r$. The parameter space Θ could be sufficiently separated in terms of Hamming distance while leaving enough number of parameters in Θ . This is a well-known result as Varshamov-Gilbert bound (see, e.g. Gilbert (1952) or Tsybakov (2009)), but we provide the full statement for completeness of the proof.

Lemma A.8 (Varshamov-Gilbert bound). *For any $r \geq 8$, there exists a subset Θ of the hypercube $\{0, 1\}^r$ such that*

$$|\Theta| \geq 2^{r/8} \quad \text{and} \quad d_H(\theta^{[1]}, \theta^{[2]}) \geq \frac{r}{8} \quad \text{for any distinct } \theta^{[1]}, \theta^{[2]} \in \Theta.$$

We shall define the sub-model $\mathcal{Q}(r, \varepsilon)$ based on the parameter space $\Theta \subset \{0, 1\}^r$ shown in Lemma A.8. It follows that the target covariance matrices in $\mathcal{Q}(r, \varepsilon)$ are $(r\varepsilon/8)$ -separated in terms of matrix ℓ_2 -norm: for any distinct $\theta^{[1]} = (\theta_1^{[1]}, \dots, \theta_r^{[1]}) \in \Theta$ and $\theta^{[2]} = (\theta_1^{[2]}, \dots, \theta_r^{[2]}) \in \Theta$,

$$\begin{aligned} \|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2 &\geq \frac{\|(\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}})z\|_2}{\|z\|_2} \geq \varepsilon d_H(\theta^{[1]}, \theta^{[2]}) \geq \frac{r\varepsilon}{8}, \\ \text{where } \begin{cases} z_0 := (\mathbb{1}(\theta_1^{[1]} \neq \theta_1^{[2]}), \dots, \mathbb{1}(\theta_r^{[1]} \neq \theta_r^{[2]})) \in \mathbb{R}^r, \\ z := (\mathbb{0}_r, z_0, \mathbb{0}_{d-2r}) \in \mathbb{R}^d. \end{cases} \end{aligned}$$

It is thus immediate from Fano's lemma (see, e.g. Yu (1997)) that

$$\begin{aligned} \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \\ &\geq \left(\frac{r\varepsilon}{16}\right)^2 \left(1 - \frac{\log 2 + I(X; \vartheta)}{\log |\Theta|}\right), \end{aligned} \quad (20)$$

where a random variable ϑ follows the uniform distribution on the parameter space Θ and a random vector $X := (X_1^{(t)}, \dots, X_{n_t}^{(t)}, X_1^{(s)}, \dots, X_{n_s}^{(s)})$ given ϑ follows the distribution $\mathbb{Q}_\vartheta$. When we denote by $\overline{\mathbb{Q}}$ the uniform mixture of distributions in $\mathcal{Q}(r, \varepsilon)$, the definition of mutual information and the convexity of Kullback-Leibler divergence imply that:

$$I(X; \vartheta) = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} D_{\text{KL}}(\mathbb{Q}_\theta \| \overline{\mathbb{Q}}) \leq \frac{1}{|\Theta|^2} \sum_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \| \mathbb{Q}_{\theta^{[2]}}). \quad (21)$$

The following lemma gives a useful tool for controlling the Kullback-Leibler divergence in the last term.

Lemma A.9. *Given any symmetric and positive definite matrices $\Sigma_1, \Sigma_2 \in \mathbb{R}^{p \times p}$ such that $2\|\Sigma_1 - \Sigma_2\|_2 \leq \gamma_{\min}(\Sigma_2)$, we have*

$$D_{\text{KL}}(N(\mathbb{0}, \Sigma_1) \| N(\mathbb{0}, \Sigma_2)) \leq \frac{\|\Sigma_1 - \Sigma_2\|_F^2}{2\gamma_{\min}^2(\Sigma_2)}.$$

Under the assumption in Equation (19), we have shown that $\gamma_{\min}(\Sigma_\theta) \geq \rho^2/4$, so Σ_θ is positive definite for every $\theta \in \Theta$. Not only that, for any distinct $\theta^{[1]}, \theta^{[2]} \in \Theta$, we may easily check:

$$\frac{2\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2}{\gamma_{\min}(\Sigma_{\theta^{[2]}})} \leq \frac{2r\varepsilon}{\rho^2/4} \leq \frac{1}{2}.$$

As a result, Lemma A.9 immediately yields that:

$$\begin{aligned} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) &= n D_{\text{KL}}(N(\mathbf{0}, \Sigma_{\theta^{[1]}}) \parallel N(\mathbf{0}, \Sigma_{\theta^{[2]}})) \\ &\leq \frac{n \|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_F^2}{2\gamma_{\min}^2(\Sigma_{\theta^{[2]}})} = \frac{nr\varepsilon^2 d_{\text{H}}(\theta^{[1]}, \theta^{[2]})}{\gamma_{\min}^2(\Sigma_{\theta^{[2]}})} \leq \frac{16}{\rho^4} nr^2 \varepsilon^2. \end{aligned}$$

Continuing from Equation (21), the mutual information is now controlled as:

$$I(X; \vartheta) \leq \max_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) \leq \frac{16}{\rho^4} nr^2 \varepsilon^2.$$

If we select the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ such that both $512nr\varepsilon^2 \leq \rho^4 \log 2$ and $r \geq 32$ hold, it follows from Equation (20) that

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \frac{r^2 \varepsilon^2}{512}. \quad (22)$$

Write $a = \alpha_{\max}$, and choose

$$\begin{aligned} r^* &= (\lfloor n^{1/(2a+1)} \rfloor \vee 32) \wedge \left\lfloor \frac{d}{2} \right\rfloor, \\ q^* &= \frac{\rho^2}{16} \wedge (2^{-a} M_{\min}(r^*)^{-a}) \wedge \left(\rho^2 \sqrt{\frac{r^* \log 2}{512n}} \right), \quad \varepsilon^* = \frac{q^*}{r^*}. \end{aligned}$$

Because d is sufficiently large, $r^* \geq 32$ and $2r^* \leq d$. The definition of q^* verifies every condition in Equation (19) and the Fano condition $512nr^*(\varepsilon^*)^2 \leq \rho^4 \log 2$. Moreover, with constants allowed to depend on the fixed parameters ρ , $M_{\min}$, and a ,

$$(q^*)^2 \asymp 1 \wedge (r^*)^{-2a} \wedge \frac{r^*}{n} \asymp n^{-2a/(2a+1)} \wedge \frac{d}{n}.$$

The desired result is therefore immediate from the last argument in Equation (22).

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim (r^* \varepsilon^*)^2 = (q^*)^2 \gtrsim n^{-2\alpha_{\max}/(2\alpha_{\max}+1)} \wedge \frac{d}{n}.$$

Proof of lower bound in Equation (16)

Suppose that we are given a dimension $r \in \mathbb{Z}^+$ and a tolerance $\varepsilon > 0$ which will be specified later. For any given $\theta \in \{1, \dots, r\}$ with $r \leq d$, we define a symmetric matrix $\Sigma_\theta \in \mathbb{R}^{d \times d}$ such that:

$$\Sigma_\theta := \frac{5}{16} \rho^2 I_d + \varepsilon \text{diag}(e_\theta),$$

where $e_\theta \in \mathbb{R}^d$ denotes the θ^{th} standard basis vector of the Euclidean space $\mathbb{R}^d$, whose components are all zero except the θ^{th} entry, which equals one. We would consider a probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ indexed by $\theta \in \{1, \dots, r\}$. Under $\mathbb{Q}_\theta$, we assume the following target and source samples. It is worth noting that both target and source samples share the same covariance matrix Σ_θ .

$$X_1^{(t)}, \dots, X_{n_t}^{(t)}, X_1^{(s)}, \dots, X_{n_s}^{(s)} \stackrel{i.i.d.}{\sim} N_d(\mathbf{0}, \Sigma_\theta).$$

For any given $\theta \in \{1, \dots, r\}$, we know that Σ_θ is a diagonal matrix with positive diagonal entries and thus, a valid covariance matrix. It is left to check other modeling assumptions to confirm $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$. We will check whether the probability measure $\mathbb{Q}_\theta$ meets the uniformly sub-Gaussian condition in Assumption 2.1. It is worth noticing that $a^\top X_1^{(t)}$ follows a Gaussian distribution with variance $a^\top \Sigma_\theta a$ for any vector $a \in \mathbb{R}^d$. Consequently, the condition $\varepsilon \leq \rho^2/16$ implies

$$\gamma_{\max}(\Sigma_\theta) = \frac{5}{16}\rho^2 + \varepsilon \leq \frac{3}{8}\rho^2,$$

and hence guarantees the uniform sub-Gaussianity assumption. On the other hand, the diagonal covariance matrix Σ_θ belongs to the bandable collection of any decay rate and any radius. Moreover, the disparity condition is always satisfied; we have assumed no disparity between the target and source covariance matrices. To summarize, we have constructed the valid probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ for every $\theta \in \{1, \dots, r\}$ as long as the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ are chosen to be:

$$r \leq d \quad \text{and} \quad \varepsilon \leq \frac{\rho^2}{16}. \quad (23)$$

Consider a sub-model $\mathcal{Q}(r, \varepsilon) := \{\mathbb{Q}_\theta : \theta \in \Theta\}$ with $\Theta := \{1, \dots, r\}$. We may easily check that the target covariance matrices in $\mathcal{Q}(r, \varepsilon)$ are ε -separated in terms of matrix ℓ_2 -norm: for any distinct $\theta^{[1]}, \theta^{[2]} \in \Theta$,

$$\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2 = \varepsilon \|\text{diag}(e_{\theta^{[1]}}) - \text{diag}(e_{\theta^{[2]}})\|_2 = \varepsilon.$$

It is thus immediate from Fano's lemma (see, e.g. Yu (1997)) that

$$\begin{aligned} \inf_{\hat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\hat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\geq \inf_{\hat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\hat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \\ &\geq \frac{\varepsilon^2}{4} \left(1 - \frac{\log 2 + I(X; \vartheta)}{\log |\Theta|} \right), \end{aligned} \quad (24)$$

where a random variable ϑ follows the uniform distribution on the parameter space Θ and a random vector $X := (X_1^{(t)}, \dots, X_{n_t}^{(t)}, X_1^{(s)}, \dots, X_{n_s}^{(s)})$ given ϑ follows the distribution $\mathbb{Q}_\theta$. When we denote by $\bar{\mathbb{Q}}$ the uniform mixture of distributions in $\mathcal{Q}(r, \varepsilon)$, the definition of mutual information and the convexity of Kullback-Leibler divergence implies that:

$$I(X; \vartheta) = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} D_{\text{KL}}(\mathbb{Q}_\theta \| \bar{\mathbb{Q}}) \leq \frac{1}{|\Theta|^2} \sum_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \| \mathbb{Q}_{\theta^{[2]}}). \quad (25)$$

Notice that Σ_θ is positive definite for any $\theta \in \Theta$. Not only that, for any distinct $\theta^{[1]}, \theta^{[2]} \in \Theta$, the assumption in Equation (23) implies that:

$$\frac{2\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2}{\gamma_{\min}(\Sigma_{\theta^{[2]}})} = \frac{2\varepsilon}{5\rho^2/16} \leq \frac{2}{5}.$$

As a result, Lemma A.9 immediately yields that:

$$\begin{aligned} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \| \mathbb{Q}_{\theta^{[2]}}) &= n D_{\text{KL}}(N(\mathbf{0}, \Sigma_{\theta^{[1]}}) \| N(\mathbf{0}, \Sigma_{\theta^{[2]}})) \\ &\leq \frac{n \|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_F^2}{2\gamma_{\min}^2(\Sigma_{\theta^{[2]}})} = \frac{256}{25\rho^4} n \varepsilon^2 \leq \frac{16}{\rho^4} n \varepsilon^2. \end{aligned}$$

Continuing from Equation (25), the mutual information is now controlled as:

$$I(X; \vartheta) \leq \max_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) \leq \frac{16}{\rho^4} n \varepsilon^2.$$

If we select the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ such that both $64n\varepsilon^2 \leq \rho^4 \log r$ and $r \geq 16$ hold, it follows from Equation (24) that

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \frac{\varepsilon^2}{8}. \quad (26)$$

To this end, choose

$$r^* = d \quad \text{and} \quad \varepsilon^* = \frac{\rho^2}{16} \left(\sqrt{\frac{\log d}{n}} \wedge 1 \right).$$

Since d is sufficiently large, $r^* \geq 16$, while $\varepsilon^* \leq \rho^2/16$ and $64n(\varepsilon^*)^2 \leq \rho^4 \log r^*$. The desired result is therefore immediate from the last argument in Equation (26).

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim (\varepsilon^*)^2 \gtrsim \frac{\log d}{n} \wedge 1.$$

Proof of lower bound in Equation (17)

If $\delta = 0$, the right-hand side of Equation (17) is zero and the claim is immediate. Hence, we assume $\delta > 0$ below. Suppose that we are given a dimension $r \in \mathbb{Z}^+$ and a tolerance $\varepsilon > 0$ which will be specified later. For any $\theta \in \{0, 1\}^r$ on the r -dimensional hypercube with $2r \leq d$, we define a symmetric matrix $\Sigma_\theta \in \mathbb{R}^{d \times d}$ such that:

$$\Sigma_\theta := \frac{5}{16} \rho^2 I_d + \varepsilon \begin{pmatrix} \mathbb{0} & \theta \mathbf{1}_r^\top & \mathbb{0} \\ \mathbf{1}_r \theta^\top & \mathbb{0} & \mathbb{0} \\ \mathbb{0} & \mathbb{0} & \mathbb{0} \end{pmatrix} \quad \text{where } \mathbf{1}_r := (1, 1, \dots, 1)^\top \in \mathbb{R}^r.$$

Consider a probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ ($\theta \in \{0, 1\}^r$) indexed by the r -dimensional hypercube. Under $\mathbb{Q}_\theta$, we assume the following target and source samples:

$$X_1^{(t)}, \dots, X_{n_t}^{(t)} \stackrel{i.i.d.}{\sim} N_d(\mathbb{0}, \Sigma_\theta) \quad \text{and} \quad X_1^{(s)}, \dots, X_{n_s}^{(s)} \stackrel{i.i.d.}{\sim} N_d(\mathbb{0}, \Sigma_\theta).$$

We need to check the validity of covariance matrix Σ_θ as well as other modeling assumptions to confirm $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ for any given $\theta \in \{0, 1\}^r$. Observe that Σ_θ is positive definite and a valid covariance matrix if $r\varepsilon \leq \rho^2/16$:

$$\gamma_{\min}(\Sigma_\theta) = \frac{5}{16} \rho^2 + \varepsilon \gamma_{\min} \begin{pmatrix} \mathbb{0} & \theta \mathbf{1}_r^\top & \mathbb{0} \\ \mathbf{1}_r \theta^\top & \mathbb{0} & \mathbb{0} \\ \mathbb{0} & \mathbb{0} & \mathbb{0} \end{pmatrix} \geq \frac{5}{16} \rho^2 - \varepsilon \left\| \begin{pmatrix} \mathbb{0} & \theta \mathbf{1}_r^\top & \mathbb{0} \\ \mathbf{1}_r \theta^\top & \mathbb{0} & \mathbb{0} \\ \mathbb{0} & \mathbb{0} & \mathbb{0} \end{pmatrix} \right\|_1 \geq \frac{5}{16} \rho^2 - r\varepsilon \geq \frac{1}{4} \rho^2.$$

Not only that, the probability measure $\mathbb{Q}_\theta$ should meet the uniformly sub-Gaussian condition in Assumption 2.1. Notice that $a^\top X_1^{(t)}$ and $a^\top X_1^{(s)}$ follow Gaussian distributions with

variances $a^\top \Sigma_\theta a$ and $a^\top \Sigma_0 a$, respectively, for any vector $a \in \mathbb{R}^d$. The same restriction also guarantees the uniform sub-Gaussianity assumption:

$$\gamma_{\max}(\Sigma_\theta) = \frac{5}{16}\rho^2 + \varepsilon\gamma_{\max} \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \leq \frac{5}{16}\rho^2 + \varepsilon \left\| \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \right\|_1 \leq \frac{5}{16}\rho^2 + r\varepsilon \leq \frac{3}{8}\rho^2.$$

The covariance matrix of the measure $\mathbb{Q}_\theta$ needs to satisfy the bandable structure. It suffices to argue $\Sigma_\theta \in \mathcal{B}_{\alpha_t}(M_t)$ since the source covariance matrix is a diagonal matrix that conforms to the bandable structure of any rate and any radius. This condition is simply ensured by the condition $r^{\alpha_t+1}\varepsilon \leq 2^{-\alpha_t}M_t$. For any bandwidth $u \in (0, 2r]$, we have

$$\|\mathfrak{A}(\Sigma_\theta; u)\|_1 \leq r\varepsilon \leq M_t(2r)^{-\alpha_t} \leq M_t u^{-\alpha_t}.$$

For $u > 2r$, the away-from-diagonal matrix $\mathfrak{A}(\Sigma_\theta; u)$ is zero. Finally, the disparity condition must be satisfied under the measure $\mathbb{Q}_\theta$. The condition $r\varepsilon \leq \delta$ is indeed sufficient for it:

$$\|\Sigma_\theta - \Sigma_0\|_1 = \varepsilon \left\| \begin{pmatrix} \mathbf{0} & \theta \mathbf{1}_r^\top & \mathbf{0} \\ \mathbf{1}_r \theta^\top & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{pmatrix} \right\|_1 \leq r\varepsilon \leq \delta.$$

To summarize, we have constructed the valid probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ ($\theta \in \{0, 1\}^r$) as long as the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ are chosen to be:

$$2r \leq d, \quad r\varepsilon \leq \delta \wedge \frac{\rho^2}{16}, \quad \text{and} \quad r^{\alpha_t+1}\varepsilon \leq 2^{-\alpha_t}M_t. \quad (27)$$

Let us define the sub-model $\mathcal{Q}(r, \varepsilon)$ based on the parameter space $\Theta \subset \{0, 1\}^r$ constructed in Lemma A.8. It follows that the target covariance matrices in $\mathcal{Q}(r, \varepsilon)$ are $(r\varepsilon/8)$ -separated in terms of matrix ℓ_2 -norm: for any distinct $\theta^{[1]} = (\theta_1^{[1]}, \dots, \theta_r^{[1]}) \in \Theta$ and $\theta^{[2]} = (\theta_1^{[2]}, \dots, \theta_r^{[2]}) \in \Theta$,

$$\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2 \geq \frac{\|(\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}})z\|_2}{\|z\|_2} \geq \varepsilon d_H(\theta^{[1]}, \theta^{[2]}) \geq \frac{r\varepsilon}{8},$$

where $\begin{cases} z_0 := (\mathbf{1}(\theta_1^{[1]} \neq \theta_1^{[2]}), \dots, \mathbf{1}(\theta_r^{[1]} \neq \theta_r^{[2]})) \in \mathbb{R}^r, \\ z := (\mathbf{0}_r, z_0, \mathbf{0}_{d-2r}) \in \mathbb{R}^d. \end{cases}$

It is thus immediate from Fano's lemma (see, e.g. Yu (1997)) that

$$\begin{aligned} \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \\ &\geq \left(\frac{r\varepsilon}{16}\right)^2 \left(1 - \frac{\log 2 + I(X; \vartheta)}{\log |\Theta|}\right), \end{aligned} \quad (28)$$

where a random variable ϑ follows the uniform distribution on the parameter space Θ and a random vector $X := (X_1^{(t)}, \dots, X_{n_t}^{(t)}, X_1^{(s)}, \dots, X_{n_s}^{(s)})$ given ϑ follows the distribution $\mathbb{Q}_\theta$. When we denote by $\overline{\mathbb{Q}}$ the uniform mixture of distributions in $\mathcal{Q}(r, \varepsilon)$, the definition of mutual information and the convexity of Kullback-Leibler divergence implies that:

$$I(X; \vartheta) = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} D_{\text{KL}}(\mathbb{Q}_\theta \| \overline{\mathbb{Q}}) \leq \frac{1}{|\Theta|^2} \sum_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \| \mathbb{Q}_{\theta^{[2]}}). \quad (29)$$

Under the assumption in Equation (27), we have shown that $\gamma_{\min}(\Sigma_\theta) \geq \rho^2/4$, so Σ_θ is positive definite for every $\theta \in \Theta$. Not only that, for any distinct $\theta^{[1]}, \theta^{[2]} \in \Theta$, we may easily check:

$$\frac{2\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2}{\gamma_{\min}(\Sigma_{\theta^{[2]}})} \leq \frac{2r\varepsilon}{\rho^2/4} \leq \frac{1}{2}.$$

As a result, Lemma A.9 immediately yields that:

$$\begin{aligned} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) &= n_t D_{\text{KL}}(N(\mathbf{0}, \Sigma_{\theta^{[1]}}) \parallel N(\mathbf{0}, \Sigma_{\theta^{[2]}})) \\ &\leq \frac{n_t \|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_F^2}{2\gamma_{\min}^2(\Sigma_{\theta^{[2]}})} = \frac{n_t r \varepsilon^2 d_{\text{H}}(\theta^{[1]}, \theta^{[2]})}{\gamma_{\min}^2(\Sigma_{\theta^{[2]}})} \leq \frac{16}{\rho^4} n_t r^2 \varepsilon^2. \end{aligned}$$

Continuing from Equation (29), the mutual information is now controlled as:

$$I(X; \vartheta) \leq \max_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) \leq \frac{16}{\rho^4} n_t r^2 \varepsilon^2.$$

If we choose the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ such that both $512n_t r \varepsilon^2 \leq \rho^4 \log 2$ and $r \geq 32$ hold, it follows from Equation (28) that

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \frac{r^2 \varepsilon^2}{512}. \quad (30)$$

Write $a = \alpha_t$, and choose

$$\begin{aligned} r^* &= \left(\left\lfloor n_t^{1/(2a+1)} \vee \delta^{-1/a} \right\rfloor \vee 32 \right) \wedge \left\lfloor \frac{d}{2} \right\rfloor, \\ q^* &= \delta \wedge \frac{\rho^2}{16} \wedge (2^{-a} M_t (r^*)^{-a}) \wedge \left(\rho^2 \sqrt{\frac{r^* \log 2}{512 n_t}} \right), \quad \varepsilon^* = \frac{q^*}{r^*}. \end{aligned}$$

Because d is sufficiently large, $r^* \geq 32$ and $2r^* \leq d$. The definition of q^* verifies every condition in Equation (27) and the Fano condition $512n_t r^* (\varepsilon^*)^2 \leq \rho^4 \log 2$. Moreover, with constants allowed to depend on the fixed parameters ρ , M_t , and a ,

$$(q^*)^2 \asymp \delta^2 \wedge 1 \wedge (r^*)^{-2a} \wedge \frac{r^*}{n_t} \asymp n_t^{-2a/(2a+1)} \wedge \frac{d}{n_t} \wedge \delta^2.$$

The desired result is therefore immediate from the last argument in Equation (30).

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim (r^* \varepsilon^*)^2 = (q^*)^2 \gtrsim n_t^{-2\alpha_t/(2\alpha_t+1)} \wedge \frac{d}{n_t} \wedge \delta^2.$$

Proof of lower bound in Equation (18)

If $\delta = 0$, the right-hand side of Equation (18) is zero and the claim is immediate. Hence, we assume $\delta > 0$ below. Suppose that we are given a dimension $r \in \mathbb{Z}^+$ and a tolerance $\varepsilon > 0$ which will be specified later. For any given $\theta \in \{1, \dots, r\}$ with $r \leq d$, we define a symmetric matrix $\Sigma_\theta \in \mathbb{R}^{d \times d}$ such that:

$$\Sigma_\theta := \frac{5}{16} \rho^2 I_d + \varepsilon \text{diag}(e_\theta),$$

where $e_\theta \in \mathbb{R}^d$ denotes the θ^{th} standard basis vector of the Euclidean space $\mathbb{R}^d$, whose components are all zero except the θ^{th} entry, which equals one. We would consider a probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ indexed by $\theta \in \{1, \dots, r\}$. Under $\mathbb{Q}_\theta$, we assume the following target and source samples: by letting $\Sigma_0 := (5\rho^2/16)I_d$,

$$X_1^{(t)}, \dots, X_{n_t}^{(t)} \stackrel{i.i.d.}{\sim} N_d(\mathbb{0}, \Sigma_\theta) \quad \text{and} \quad X_1^{(s)}, \dots, X_{n_s}^{(s)} \stackrel{i.i.d.}{\sim} N_d(\mathbb{0}, \Sigma_0).$$

For any given $\theta \in \{1, \dots, r\}$, we know that Σ_θ is a diagonal matrix with positive diagonal entries and thus, a valid covariance matrix. It is left to check other modeling assumptions to confirm $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$. We will check whether the probability measure $\mathbb{Q}_\theta$ meets the uniformly sub-Gaussian condition in Assumption 2.1. It is worth noticing that $a^\top X_1^{(t)}$ and $a^\top X_1^{(s)}$ follow Gaussian distributions with variances $a^\top \Sigma_\theta a$ and $a^\top \Sigma_0 a$, respectively, for any vector $a \in \mathbb{R}^d$. Consequently, the condition $\varepsilon \leq \rho^2/16$ implies

$$\gamma_{\max}(\Sigma_\theta) \leq \frac{5}{16}\rho^2 + \varepsilon \leq \frac{3}{8}\rho^2, \quad \text{for any } \theta \in \{0, 1, \dots, r\},$$

and hence guarantees the uniform sub-Gaussianity assumption. On the other hand, the diagonal covariance matrix Σ_θ ($\theta \in \{0, 1, \dots, r\}$) belongs to the bandable collection of any decay rate and any radius. Finally, the disparity condition must be always satisfied under $\mathbb{Q}_\theta$ for any ($\theta \in \{1, \dots, r\}$). The condition $\varepsilon \leq \delta$ is indeed sufficient for it:

$$\|\Sigma_\theta - \Sigma_0\|_1 = \varepsilon \|\text{diag}(e_\theta)\|_1 = \varepsilon \leq \delta.$$

To summarize, we have constructed the valid probability measure $\mathbb{Q}_\theta \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}$ for every $\theta \in \{1, \dots, r\}$ as long as the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ are chosen to be:

$$r \leq d \quad \text{and} \quad \varepsilon \leq \delta \wedge \frac{\rho^2}{16}. \quad (31)$$

Consider a sub-model $\mathcal{Q}(r, \varepsilon) := \{\mathbb{Q}_\theta : \theta \in \Theta\}$ with $\Theta := \{1, \dots, r\}$. We may easily check that the target covariance matrices in $\mathcal{Q}(r, \varepsilon)$ are ε -separated in terms of matrix ℓ_2 -norm: for any distinct $\theta^{[1]}, \theta^{[2]} \in \Theta$,

$$\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2 = \varepsilon \|\text{diag}(e_{\theta^{[1]}}) - \text{diag}(e_{\theta^{[2]}})\|_2 = \varepsilon.$$

It is thus immediate from Fano's lemma (see, e.g. Yu (1997)) that

$$\begin{aligned} \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_s, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 &\geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \\ &\geq \frac{\varepsilon^2}{4} \left(1 - \frac{\log 2 + I(X; \vartheta)}{\log |\Theta|} \right), \end{aligned} \quad (32)$$

where a random variable ϑ follows the uniform distribution on the parameter space Θ and a random vector $X := (X_1^{(t)}, \dots, X_{n_t}^{(t)}, X_1^{(s)}, \dots, X_{n_s}^{(s)})$ given ϑ follows the distribution $\mathbb{Q}_\vartheta$. When we denote by $\overline{\mathbb{Q}}$ the uniform mixture of distributions in $\mathcal{Q}(r, \varepsilon)$, the definition of mutual information and the convexity of Kullback-Leibler divergence implies that:

$$I(X; \vartheta) = \frac{1}{|\Theta|} \sum_{\theta \in \Theta} D_{\text{KL}}(\mathbb{Q}_\theta \| \overline{\mathbb{Q}}) \leq \frac{1}{|\Theta|^2} \sum_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \| \mathbb{Q}_{\theta^{[2]}}). \quad (33)$$

Notice that Σ_θ is positive definite for any $\theta \in \Theta$. Not only that, for any distinct $\theta^{[1]}, \theta^{[2]} \in \Theta$, the assumption in Equation (31) implies that:

$$\frac{2\|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_2}{\gamma_{\min}(\Sigma_{\theta^{[2]}})} = \frac{2\varepsilon}{5\rho^2/16} \leq \frac{2}{5}.$$

As a result, Lemma A.9 immediately yields that:

$$\begin{aligned} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) &= n_t D_{\text{KL}}(N(\mathbb{0}, \Sigma_{\theta^{[1]}}) \parallel N(\mathbb{0}, \Sigma_{\theta^{[2]}})) \\ &\leq \frac{n_t \|\Sigma_{\theta^{[1]}} - \Sigma_{\theta^{[2]}}\|_F^2}{2\gamma_{\min}^2(\Sigma_{\theta^{[2]}})} = \frac{256}{25\rho^4} n_t \varepsilon^2 \leq \frac{16}{\rho^4} n_t \varepsilon^2. \end{aligned}$$

Continuing from Equation (33), the mutual information is now controlled as:

$$I(X; \vartheta) \leq \max_{\theta^{[1]}, \theta^{[2]} \in \Theta} D_{\text{KL}}(\mathbb{Q}_{\theta^{[1]}} \parallel \mathbb{Q}_{\theta^{[2]}}) \leq \frac{16}{\rho^4} n_t \varepsilon^2.$$

If we choose the dimension $r \in \mathbb{Z}^+$ and tolerance $\varepsilon > 0$ such that both $64n_t \varepsilon^2 \leq \rho^4 \log r$ and $r \geq 16$ hold, it follows from Equation (32) that

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_S, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{Q}(r, \varepsilon)} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \geq \frac{\varepsilon^2}{8}. \quad (34)$$

To this end, choose

$$r^* = d \quad \text{and} \quad \varepsilon^* = \delta \wedge \frac{\rho^2}{16} \wedge \left(\frac{\rho^2}{16} \sqrt{\frac{\log d}{n_t}} \right).$$

Since d is sufficiently large, $r^* \geq 16$, while $\varepsilon^* \leq \delta \wedge (\rho^2/16)$ and $64n_t(\varepsilon^*)^2 \leq \rho^4 \log r^*$. The desired result is therefore immediate from the last argument in Equation (34).

$$\inf_{\widehat{\Sigma}^{(t)}} \sup_{\mathbb{P} \in \mathcal{P}_{\alpha_t, \alpha_S, \delta}} \mathbb{E} \|\widehat{\Sigma}^{(t)} - \Sigma^{(t)}\|_2^2 \gtrsim (\varepsilon^*)^2 \gtrsim \frac{\log d}{n_t} \wedge \delta^2 \wedge 1.$$

Finally, let R^* denote the minimax risk in Theorem A.5, and let $a_1, \dots, a_4$ denote the four terms on its right-hand side. The four bounds above give $R^* \geq c_j a_j$ for constants $c_j > 0$, $j = 1, \dots, 4$. Therefore,

$$R^* \geq \max_{1 \leq j \leq 4} c_j a_j \geq \frac{\min_{1 \leq j \leq 4} c_j}{4} \sum_{j=1}^4 a_j,$$

which proves the theorem. $\square$

A.6 Proof of Theorems A.4 and 3.2

It suffices to prove the misspecified theorem, since the exact model follows by setting $\eta = 0$. The two fitted matrices are constructed from sample second moments and spectrally clipped as specified in Subsection C.1; hence they are measurable with respect to data independent of the final target validation fold.

The candidate risk bounds in Propositions C.5 and C.6 give

$$\sup_{\mathbb{P} \in \tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E} \|A_{\text{tar}} - \Sigma^{(t)}\|_2^2 \lesssim P_{n_t}(\alpha_t) + \eta^2, \quad (35)$$

$$\sup_{\mathbb{P} \in \tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E} \|A_{\text{pool}} - \Sigma^{(t)}\|_2^2 \lesssim P_n(\alpha_{\max}) + \delta^2 + \eta^2, \quad (36)$$

The final estimator uses the one-sided selector in Lemma C.7, with A_{pool} as its default and threshold based on $x = c \log(en)$. The two bounds from that lemma, together with spectral clipping on the exceptional event, yield

$$\sup_{\mathbb{P}} \mathbb{E} \|\widehat{\Sigma}_{\text{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim P_{n_t}(\alpha_t) + \eta^2 + \Gamma_{m_f, n}, \quad (37)$$

$$\sup_{\mathbb{P}} \mathbb{E} \|\widehat{\Sigma}_{\text{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \lesssim P_n(\alpha_{\max}) + \delta^2 + \eta^2. \quad (38)$$

Here $\sup_{\mathbb{P}}$ denotes the supremum over $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$. The one-sided selector uses only the joint measurability of the candidates with respect to data outside the final target fold. The exceptional-event contribution is $O(\rho^4(en)^{-2})$, which is absorbed by $P_n(\alpha_{\max})$. Write

$$r = p_{n_t}(\alpha_t), \quad \ell = \frac{\log d}{n_t}, \quad u = P_n(\alpha_{\max}), \quad z = \delta^2.$$

Taking the smaller of Equations (37) and (38) and using

$$\min(r + \ell + C\Gamma_{m_f, n}, u + z) \leq u + \min(r + \ell + C\Gamma_{m_f, n}, z) \leq u + (r \wedge z) + (\ell \wedge z) + C(\Gamma_{m_f, n} \wedge z)$$

proves Equation (13). Setting $\eta = 0$ proves the exact main-text theorem. $\square$

B Proof of Technical Results in Optimality

B.1 Proof of Proposition A.6

For any given bandwidth $u > 0$, let us initiate by decomposing the ℓ_2 -loss into variance and bias terms:

$$\begin{aligned} & \|\mathfrak{T}_d(\check{\Sigma}^{(t)}; u) - \Sigma^{(t)}\|_2 \\ & \leq \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2 + \|\Sigma^{(t)} - \mathfrak{T}_d(\Sigma^{(t)}; u)\|_2 \\ & \lesssim \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2 + (u^{-\alpha_{\max}} + (u^{-\alpha_t} \wedge \delta) + \eta)\mathbf{1}(u < d). \end{aligned} \quad (39)$$

where the last inequality follows from Lemma B.1.

Lemma B.1. *Under the misspecified model $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$, for any bandwidth $u > 0$, we have*

$$\begin{aligned} \|\Sigma^{(t)} - \mathfrak{T}_d(\Sigma^{(t)}; u)\|_2 & \lesssim (M_{\max} u^{-\alpha_{\max}} + ((M_t u^{-\alpha_t}) \wedge \delta) + \eta)\mathbf{1}(u < d), \\ \|\Sigma - \mathfrak{T}_d(\Sigma; u)\|_2 & \lesssim (M_{\max} u^{-\alpha_{\max}} + ((M_{\max} u^{-\alpha_{\min}}) \wedge \delta) + \eta)\mathbf{1}(u < d), \end{aligned}$$

where we define $M_{\max} := M_t \vee M_s$ and $\alpha_{\min} := \alpha_t \wedge \alpha_s$.

It is now left to control the variance term. Recall the discussion about the blockwise tridiagonal operator $\mathfrak{T}(\cdot; u)$ and definition of the induced collection $\mathcal{T}(\cdot; u)$ in Figure 7a. Applying norm compression inequality (Cai and Yuan, 2012, Theorem 3.4), the variance term can be represented by the sub-matrices in collection $\mathcal{T}_d(\check{\Sigma}^{(t)}; u)$ as follows:

$$\begin{aligned} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2 & \leq 3 \max_{B \in \mathcal{T}_d(\check{\Sigma}^{(t)}; u)} \|B - \mathbb{E}B\|_2 + \|\mathfrak{T}_d(\mathbb{E}\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2 \\ & \leq 3 \max_{B \in \mathcal{T}_d(\check{\Sigma}^{(t)}; u)} \|B - \mathbb{E}B\|_2 + C\rho^2 n_t^{-1}. \end{aligned} \quad (40)$$

Indeed, $\mathbb{E}\check{\Sigma}^{(t)} = (1 - n_t^{-1})\Sigma^{(t)}$, while uniform sub-Gaussianity implies $\|\Sigma^{(t)}\|_2 \leq C\rho^2$. The block-compression inequality also gives $\|\mathfrak{T}_d(A; u)\|_2 \leq 3\|A\|_2$ for every symmetric A . Applying it to $A = -n_t^{-1}\Sigma^{(t)}$ proves the second line above, and this deterministic term is absorbed by the stochastic bound below. The subsequent lemma now provides a concentration inequality for those sub-matrices.

Lemma B.2. *Suppose B is a full coordinate rectangular sub-matrix of $\check{\Sigma}^{(t)}$, with both dimensions at most u . We obtain that for any given $\kappa > 0$,*

$$\mathbb{P} \left\{ \|B - \mathbb{E}B\|_2 \geq \lambda_\kappa \rho^2 \sqrt{\frac{u + \log d}{n_t}} \right\} \leq d^{-\kappa},$$

provided that $u \vee \log d \leq c_\kappa n_t$ holds. Here, $c_\kappa > 0$ is a small enough constant and $\lambda_\kappa > 0$ is a sufficiently large constant, where both only depend on the given $\kappa > 0$.

Since the collection $\mathcal{T}_d(\check{\Sigma}^{(t)}; u)$ consists of full coordinate rectangular blocks with both dimensions at most u , Lemma B.2 guarantees the existence of a small enough constant $c_t > 0$ and large enough constant $\lambda > 0$ such that the following holds: provided $u \vee \log d \leq c_t n_t$,

$$\mathbb{P} \left\{ \|B - \mathbb{E}B\|_2 \leq \frac{\lambda}{6} \rho^2 \sqrt{\frac{u + \log d}{n_t}} \right\} \geq 1 - d^{-3}, \quad \text{for every } B \in \mathcal{T}_d(\check{\Sigma}^{(t)}; u).$$

Now that the collection $\mathcal{T}_d(\check{\Sigma}^{(t)}; u)$ includes at most $3d$ many sub-matrices, the simple union bound yields that

$$\mathbb{P} \left\{ \|B - \mathbb{E}B\|_2 \leq \frac{\lambda}{6} \rho^2 \sqrt{\frac{u + \log d}{n_t}} \text{ for every } B \in \mathcal{T}_d(\check{\Sigma}^{(t)}; u) \right\} \geq 1 - 3d^{-2}.$$

Combining it with Equation (40), we obtain a concentration inequality for the variance term:

$$\begin{aligned} \mathbb{P}(E_1) &\geq 1 - 3d^{-2}, \\ \text{where } E_1 &:= \left\{ \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2 \leq \lambda \rho^2 \sqrt{\frac{u + \log d}{n_t}} \right\}. \end{aligned} \quad (41)$$

Considering the definition of the event E_1 , it is certain to transfer the high-probability result into the upper bound of the expected value:

$$\begin{aligned} \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^2 &= \mathbb{E} \left(\|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^2 \mathbb{1}_{E_1} \right) + \mathbb{E} \left(\|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^2 \mathbb{1}_{E_1^c} \right) \\ &\leq \lambda^2 \rho^4 \cdot \frac{u + \log d}{n_t} + \sqrt{\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^4 (1 - \mathbb{P}(E_1))}. \end{aligned} \quad (42)$$

Thanks to the uniformly sub-Gaussian assumption, the fourth moments of the centered entrywise fluctuations of the target sample covariance matrix can be controlled as the subsequent lemma.

Lemma B.3. *By letting $\Sigma^{(t)} = (\sigma_{jk}^{(t)})_{j,k \in [d]}$ and $\check{\Sigma}^{(t)} = (\check{\sigma}_{jk}^{(t)})_{j,k \in [d]}$, there exists an absolute constant $C > 0$ such that for any $j, k \in [d]$,*

$$\mathbb{E}(\check{\sigma}_{jk}^{(t)} - \sigma_{jk}^{(t)})^4 \leq \frac{C\rho^8}{n_t^2}.$$

Since the blockwise tridiagonal operator $\mathfrak{T}_d(\cdot; u)$ leaves at most $3ud$ non-zero entries, it follows from Lemma B.3 and Cauchy-Schwarz inequality that

$$\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^4 \leq \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_F^4 \leq 9C\rho^8 \frac{u^2 d^2}{n_t^2}.$$

Combining it with the high-probability result in Equation (41), the variance term in Equation (42) is further bounded from above by:

$$\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^2 \leq \lambda^2 \rho^4 \cdot \frac{u + \log d}{n_t} + \sqrt{27C} \rho^4 \frac{u}{n_t}.$$

As a result, Equation (39) yields the desired result immediately: $u \vee \log d \leq c_t n_t$,

$$\begin{aligned} \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)}; u) - \Sigma^{(t)}\|_2^2 &\lesssim \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}^{(t)} - \Sigma^{(t)}; u)\|_2^2 + (u^{-2\alpha_{\max}} + u^{-2\alpha_t} \wedge \delta^2 + \eta^2) \mathbb{1}(u < d) \\ &\lesssim \frac{u + \log d}{n_t} + (u^{-2\alpha_{\max}} + (u^{-2\alpha_t} \wedge \delta^2) + \eta^2) \mathbb{1}(u < d). \end{aligned}$$

□

B.2 Proof of Proposition A.7

For given any bandwidth $v > 0$, it is straightforward to decompose the ℓ_2 -loss into variance and bias terms. By Lemma B.1, we have

$$\begin{aligned} & \|\mathfrak{T}_d(\check{\Sigma}; v) - \Sigma\|_2 \\ & \leq \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2 + \|\Sigma - \mathfrak{T}_d(\Sigma; v)\|_2 \\ & \lesssim \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2 + (v^{-\alpha_{\max}} + (v^{-\alpha_{\min}} \wedge \delta) + \eta) \mathbf{1}(v < d), \end{aligned} \quad (43)$$

We are now left to control the variance term. Recall the discussion about the blockwise tridiagonal operator $\mathfrak{T}(\cdot; v)$ and definition of the induced collection $\mathcal{T}(\cdot; v)$ in Figure 7a. Applying norm compression inequality (Cai and Yuan, 2012, Theorem 3.4), the variance term can be represented by the sub-matrices in collection $\mathcal{T}_d(\check{\Sigma}; v)$ as follows:

$$\begin{aligned} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2 & \leq 3 \max_{B \in \mathcal{T}_d(\check{\Sigma}; v)} \|B - \mathbb{E}B\|_2 + \|\mathfrak{T}_d(\mathbb{E}\check{\Sigma} - \Sigma; v)\|_2 \\ & \leq 3 \max_{B \in \mathcal{T}_d(\check{\Sigma}; v)} \|B - \mathbb{E}B\|_2 + C\rho^2 n^{-1}. \end{aligned} \quad (44)$$

Here

$$\mathbb{E}\check{\Sigma} - \Sigma = -\frac{1}{n} \sum_{g: n_g > 0} \Sigma^{(g)},$$

and there are at most two groups. Applying $\|\mathfrak{T}_d(A; v)\|_2 \leq 3\|A\|_2$ to each covariance matrix shows that the deterministic bias is again absorbed by the stochastic term. The subsequent lemma now provides a concentration inequality for those sub-matrices.

Lemma B.4. *Suppose B is a full coordinate rectangular sub-matrix of $\check{\Sigma}$, with both dimensions at most v . We obtain that for any given $\kappa > 0$,*

$$\mathbb{P} \left\{ \|B - \mathbb{E}B\|_2 \geq \lambda_\kappa \rho^2 \sqrt{\frac{v + \log d}{n}} \right\} \leq d^{-\kappa},$$

provided that $v \vee \log d \leq c_\kappa n$ is true. Here, $c_\kappa > 0$ is a small enough constant and $\lambda_\kappa > 0$ is a sufficiently large constant, where both only depend on the given $\kappa > 0$.

Since the collection $\mathcal{T}_d(\check{\Sigma}; v)$ consists of full coordinate rectangular blocks with both dimensions at most v , Lemma B.4 guarantees the existence of a small enough constant $c_p > 0$ and large enough constant $\lambda > 0$ such that the following holds: provided $v \vee \log d \leq c_p n$,

$$\mathbb{P} \left\{ \|B - \mathbb{E}B\|_2 \leq \frac{\lambda}{6} \rho^2 \sqrt{\frac{v + \log d}{n}} \right\} \geq 1 - d^{-3}, \quad \text{for every } B \in \mathcal{T}_d(\check{\Sigma}; v).$$

Now that the collection $\mathcal{T}_d(\check{\Sigma}; v)$ includes at most $3d$ many sub-matrices, the simple union bound yields that

$$\mathbb{P} \left\{ \|B - \mathbb{E}B\|_2 \leq \frac{\lambda}{6} \rho^2 \sqrt{\frac{v + \log d}{n}} \text{ for every } B \in \mathcal{T}_d(\check{\Sigma}; v) \right\} \geq 1 - 3d^{-2}.$$

Combining it with Equation (44), we obtain a concentration inequality for the variance term:

$$\mathbb{P}(E_2) \geq 1 - 3d^{-2}, \quad (45)$$

$$\text{where } E_2 := \left\{ \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2 \leq \lambda \rho^2 \sqrt{\frac{v + \log d}{n}} \right\}.$$

Considering the definition of the event E_2 , it is apparent to transfer the high-probability result into the upper bound of the expected value.

$$\begin{aligned} & \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^2 \\ &= \mathbb{E} \left(\|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^2 \mathbb{1}_{E_2} \right) + \mathbb{E} \left(\|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^2 \mathbb{1}_{E_2^c} \right) \\ &\leq \lambda^2 \rho^4 \cdot \frac{v + \log d}{n} + \sqrt{\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^4 (1 - \mathbb{P}(E_2))}. \end{aligned} \quad (46)$$

Thanks to the uniformly sub-Gaussian assumption, the fourth moments of the centered entrywise fluctuations of the pooled sample covariance matrix can be controlled as the subsequent lemma.

Lemma B.5. *By letting $\Sigma = (\sigma_{jk})_{j,k \in [d]}$ and $\check{\Sigma} = (\check{\sigma}_{jk})_{j,k \in [d]}$, there exists an absolute constant $C > 0$ such that for any $j, k \in [d]$,*

$$\mathbb{E}(\check{\sigma}_{jk} - \sigma_{jk})^4 \leq \frac{C\rho^8}{n^2}.$$

Since the blockwise tridiagonal operator $\mathfrak{T}_d(\cdot; v)$ leaves at most $3vd$ non-zero entries, it follows from Lemma B.5 and Cauchy-Schwarz inequality that

$$\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^4 \leq \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_F^4 \leq 9C\rho^8 \frac{v^2 d^2}{n^2}.$$

Combining it with the high-probability result in Equation (45), the variance term in Equation (46) is further bounded from above by:

$$\mathbb{E} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^2 \leq \lambda^2 \rho^4 \cdot \frac{v + \log d}{n} + \sqrt{27C} \rho^4 \frac{v}{n}.$$

As a result, Equation (43) yields the desired result immediately: for any $v \vee \log d \leq c_p n$,

$$\begin{aligned} & \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma}; v) - \Sigma\|_2^2 \\ &\lesssim \mathbb{E} \|\mathfrak{T}_d(\check{\Sigma} - \Sigma; v)\|_2^2 + (v^{-2\alpha_{\max}} + v^{-2\alpha_{\min}} \wedge \delta^2 + \eta^2) \mathbb{1}(v < d) \\ &\lesssim \frac{v + \log d}{n} + (v^{-2\alpha_{\max}} + v^{-2\alpha_{\min}} \wedge \delta^2 + \eta^2) \mathbb{1}(v < d). \end{aligned}$$

□

B.3 Proof of Lemma A.9

Define $D := \Sigma_1 - \Sigma_2$ and $A := \Sigma_2^{-1/2} D \Sigma_2^{-1/2}$. It follows that

$$\begin{aligned} D_{\text{KL}}(N(\mathbf{0}, \Sigma_1) \parallel N(\mathbf{0}, \Sigma_2)) &= \frac{1}{2} \left(\text{tr}(\Sigma_2^{-1} \Sigma_1) - \log |\Sigma_2^{-1} \Sigma_1| - p \right) \\ &= \frac{1}{2} \left(\text{tr}(A) - \log |I_p + A| \right) \\ &= \frac{1}{2} \sum_{i=1}^p \left(\gamma_i(A) - \log(1 + \gamma_i(A)) \right). \end{aligned}$$

Since A is symmetric, the assumption of the lemma gives

$$|\gamma_i(A)| \leq \|A\|_2 \leq \frac{\|D\|_2}{\gamma_{\min}(\Sigma_2)} \leq \frac{1}{2}, \quad i = 1, \dots, p.$$

The inequality $x - \log(1 + x) \leq x^2$, valid for $x \in [-1/2, 1/2]$, now yields

$$\begin{aligned} D_{\text{KL}}(N(\mathbf{0}, \Sigma_1) \parallel N(\mathbf{0}, \Sigma_2)) &\leq \frac{1}{2} \sum_{i=1}^p \gamma_i^2(A) = \frac{1}{2} \|A\|_F^2 \\ &\leq \frac{\|D\|_F^2}{2\gamma_{\min}^2(\Sigma_2)}. \end{aligned}$$

□

B.4 Proof of Lemma B.1

Proof. It suffices to consider any bandwidth $u \in (0, d)$. Write $\mathcal{R}_d(B; u) := B - \mathfrak{T}_d(B; u)$. Every entry retained by $\mathcal{R}_d(B; u)$ lies outside the ordinary u -band. Hence, for $A \in \mathcal{B}_\alpha(M)$,

$$\|\mathcal{R}_d(A; u)\|_2 \leq \|\mathcal{R}_d(A; u)\|_1 \leq \|\mathfrak{A}_d(A; u)\|_1 \leq M u^{-\alpha}.$$

If H is symmetric and upper-tail stable at level r , then

$$\|\mathcal{R}_d(H; u)\|_2 \leq \|H\|_2 + \|\mathfrak{T}_d(H; u)\|_2 \leq 4r,$$

where the last inequality follows from the $u = 0$ case of upper-tail stability and the block-compression inequality.

The first part follows from the misspecified decomposition in Equation (11), the bandable assumption on $\Sigma_0^{(t)}$, and the upper-tail stability of $R^{(t)}$:

$$\|\mathcal{R}_d(\Sigma^{(t)}; u)\|_2 \leq \begin{cases} \|\mathcal{R}_d(\Sigma_0^{(t)}; u)\|_2 + \|\mathcal{R}_d(R^{(t)}; u)\|_2 \leq M_t u^{-\alpha_t} + 4\eta, \\ \|\mathcal{R}_d(\Sigma^{(s)}; u)\|_2 + \|\mathcal{R}_d(\Delta^{(s)}; u)\|_2 \leq M_s u^{-\alpha_s} + 4\eta + 4\delta. \end{cases}$$

Hence

$$\begin{aligned} \|\mathcal{R}_d(\Sigma^{(t)}; u)\|_2 &\leq (M_t u^{-\alpha_t} + 4\eta) \wedge (M_s u^{-\alpha_s} + 4\eta + 4\delta) \\ &\lesssim M_{\max} u^{-\alpha_{\max}} + ((M_t u^{-\alpha_t}) \wedge \delta) + \eta. \end{aligned}$$

On the other hand, let us decompose the pooled covariance matrix as

$$\Sigma = \frac{n_t}{n} \Sigma_0^{(t)} + \frac{n_s}{n} \Sigma_0^{(s)} + \frac{n_t}{n} R^{(t)} + \frac{n_s}{n} R^{(s)}.$$

For every bandwidth $u \geq 0$, the pooled remainder satisfies

$$\|\mathcal{R}_d\left(\frac{n_t}{n} R^{(t)} + \frac{n_s}{n} R^{(s)}; u\right)\|_2 \leq \frac{n_t}{n} \|\mathcal{R}_d(R^{(t)}; u)\|_2 + \frac{n_s}{n} \|\mathcal{R}_d(R^{(s)}; u)\|_2 \leq 4\eta.$$

Therefore

$$\|\mathcal{R}_d(\Sigma; u)\|_2 \leq \begin{cases} \frac{n_t}{n} \|\mathcal{R}_d(\Sigma_0^{(t)}; u)\|_2 + \frac{n_s}{n} \|\mathcal{R}_d(\Sigma_0^{(s)}; u)\|_2 \\ \quad + \|\mathcal{R}_d\left(\frac{n_t}{n} R^{(t)} + \frac{n_s}{n} R^{(s)}; u\right)\|_2 \leq M_{\max} u^{-\alpha_{\min}} + 4\eta, \\ \|\mathcal{R}_d(\Sigma^{(t)}; u)\|_2 + \|\mathcal{R}_d(\Delta; u)\|_2 \leq M_t u^{-\alpha_t} + 4\eta + 4\delta, \\ \|\mathcal{R}_d(\Sigma^{(s)}; u)\|_2 + \frac{n_t}{n} \|\mathcal{R}_d(\Delta^{(s)}; u)\|_2 \leq M_s u^{-\alpha_s} + 4\eta + 4\delta. \end{cases}$$

Hence

$$\begin{aligned} \|\mathcal{R}_d(\Sigma; u)\|_2 &\leq (M_{\max} u^{-\alpha_{\min}} + 4\eta) \wedge (M_t u^{-\alpha_t} + 4\eta + 4\delta) \wedge (M_s u^{-\alpha_s} + 4\eta + 4\delta) \\ &\lesssim M_{\max} u^{-\alpha_{\max}} + ((M_{\max} u^{-\alpha_{\min}}) \wedge \delta) + \eta. \end{aligned}$$

□

B.5 Proof of Lemma B.2

The following two standard auxiliary facts will also be used below.

Lemma B.6. *For every positive integer r and $\varepsilon > 0$, the unit sphere in $\mathbb{R}^r$ has an ε -net $N_r(\varepsilon)$ with*

$$|N_r(\varepsilon)| \leq (1 + 2/\varepsilon)^r.$$

Moreover, for every $r \times s$ matrix A and $\varepsilon \in (0, 1/2)$,

$$\|A\|_2 \leq \frac{1}{1 - 2\varepsilon} \sup_{\substack{a \in N_r(\varepsilon) \\ b \in N_s(\varepsilon)}} |a^\top A b|.$$

Lemma B.7. *Let $Z_1, \dots, Z_m$ be independent mean-zero random variables. There is a universal constant $C > 0$ such that*

$$\|Z_1 + \dots + Z_m\|_{\psi_r}^2 \leq C \sum_{i=1}^m \|Z_i\|_{\psi_r}^2, \quad r \in \{1, 2\}.$$

We now give a proof that uses only full coordinate rectangles. By translation invariance, assume that $\mathbb{E}X_1^{(t)} = 0$. There are coordinate sets $\mathcal{J}, \mathcal{K} \subset [d]$, with $|\mathcal{J}| \vee |\mathcal{K}| \leq u$, such that

$$B = \frac{1}{n_t} \sum_{i=1}^{n_t} X_{i,\mathcal{J}}^{(t)} (X_{i,\mathcal{K}}^{(t)})^\top - \bar{X}_{\mathcal{J}}^{(t)} (\bar{X}_{\mathcal{K}}^{(t)})^\top.$$

Fix unit vectors $a \in \mathbb{R}^{|\mathcal{J}|}$ and $b \in \mathbb{R}^{|\mathcal{K}|}$, and put $U_i = a^\top X_{i,\mathcal{J}}^{(t)}$ and $V_i = b^\top X_{i,\mathcal{K}}^{(t)}$. Uniform sub-Gaussianity and Lemmas B.8 and B.9 give

$$\|U_i V_i - \mathbb{E}(U_i V_i)\|_{\psi_1} \leq C\rho^2.$$

In addition, Lemma B.7 implies $\|\bar{U}\|_{\psi_2} \vee \|\bar{V}\|_{\psi_2} \leq C\rho/\sqrt{n_t}$, and hence

$$\|\bar{U}\bar{V} - \mathbb{E}(\bar{U}\bar{V})\|_{\psi_1} \leq C\rho^2/n_t.$$

Bernstein's inequality and a union bound therefore yield, for every $y \geq 1$,

$$\mathbb{P}\left\{|a^\top(B - \mathbb{E}B)b| > C\rho^2\left(\sqrt{\frac{y}{n_t}} + \frac{y}{n_t}\right)\right\} \leq 4e^{-y}. \quad (47)$$

Apply Lemma B.6 with 1/4-nets of the two unit spheres. Their product has cardinality at most 9^{2u} . Taking $y = C_0(u + \kappa \log d)$, applying a union bound, and using $u \vee \log d \leq c_\kappa n_t$ to absorb the linear term in Equation (47) proves Lemma B.2 after adjusting c_κ and λ_κ . $\square$

B.6 Proof of Lemma B.3

We shall assume $\mathbb{E}X_1^{(t)} = 0$ without loss of generality since the target sample covariance matrix $\check{\Sigma}^{(t)}$ is invariant under translation. Let us write the entry of each observation $X_i^{(t)}$ as

$$X_i^{(t)} = \left(X_{ij}^{(t)} : j \in [d]\right), \quad (i = 1, \dots, n_t).$$

For any given index $j, k \in [d]$, the (j, k) -th entry of target sample covariance matrix, $\check{\sigma}_{jk}^{(t)}$, is represented as

$$\check{\sigma}_{jk}^{(t)} = \frac{1}{n_t} \sum_{i=1}^{n_t} X_{ij}^{(t)} X_{ik}^{(t)} - \bar{X}_j^{(t)} \bar{X}_k^{(t)}, \quad \text{where } \bar{X}_j^{(t)} := \frac{1}{n_t} \sum_{i=1}^{n_t} X_{ij}^{(t)}. \quad (48)$$

We will argue that the centered fluctuation $\check{\sigma}_{jk}^{(t)} - \sigma_{jk}^{(t)}$ is uniformly sub-exponential over all indices $j, k \in [d]$. The following lemma states that the product of two sub-Gaussian variables is sub-exponential. See, e.g. Vershynin (2018, Lemma 2.7.7), for the detailed proof.

Lemma B.8. *For any given random variables Z_1 and Z_2 , we have*

$$\|Z_1 Z_2\|_{\psi_1} \leq \|Z_1\|_{\psi_2} \|Z_2\|_{\psi_2}.$$

Lemma B.9. *There is a universal constant $C_4 > 0$ such that for any random variable Z ,*

$$\|Z - \mathbb{E}Z\|_{\psi_1} \leq C_4 \|Z\|_{\psi_1}.$$

It is worth noting that the target observations $X_1^{(t)}, \dots, X_{n_t}^{(t)}$ consist of sub-Gaussian entries with the standard-deviation proxy ρ since each observation is uniformly sub-Gaussian in Assumption 2.1. For the centered empirical product term in Equation (48), let us define

$$Y_i := X_{ij}^{(t)} X_{ik}^{(t)} - \sigma_{jk}^{(t)}, \quad (i = 1, \dots, n_t).$$

Then $Y_1, \dots, Y_{n_t}$ are independent and mean-zero random variables. By Lemma B.8 and Lemma B.9,

$$\|Y_i\|_{\psi_1} \leq C_4 \|X_{ij}^{(t)} X_{ik}^{(t)}\|_{\psi_1} \leq C_4 \|X_{ij}^{(t)}\|_{\psi_2} \|X_{ik}^{(t)}\|_{\psi_2} \leq C_4 \rho^2, \quad (i = 1, \dots, n_t).$$

Consequently, Lemma B.7 yields that

$$\left\| \frac{1}{n_t} \sum_{i=1}^{n_t} Y_i \right\|_{\psi_1}^2 \leq \frac{C_2}{n_t^2} \sum_{i=1}^{n_t} \|Y_i\|_{\psi_1}^2 \leq \frac{C_2 C_4^2 \rho^4}{n_t}.$$

On the other hand, Lemma B.7 and Lemma B.8 imply that

$$\begin{aligned} \|\bar{X}_j^{(t)}\|_{\psi_2}^2 &\leq \frac{C_2}{n_t^2} \sum_{i=1}^{n_t} \|X_{ij}^{(t)}\|_{\psi_2}^2 \leq \frac{C_2 \rho^2}{n_t}, \\ \|\bar{X}_j^{(t)} \bar{X}_k^{(t)}\|_{\psi_1} &\leq \|\bar{X}_j^{(t)}\|_{\psi_2} \|\bar{X}_k^{(t)}\|_{\psi_2} \leq \frac{C_2 \rho^2}{n_t}. \end{aligned}$$

Combining those inequalities with Equation (48), we obtain

$$\begin{aligned} \|\check{\sigma}_{jk}^{(t)} - \sigma_{jk}^{(t)}\|_{\psi_1} &\leq \left\| \frac{1}{n_t} \sum_{i=1}^{n_t} Y_i \right\|_{\psi_1} + \|\bar{X}_j^{(t)} \bar{X}_k^{(t)}\|_{\psi_1} \\ &\leq \rho^2 C_4 \sqrt{\frac{C_2}{n_t}} + \frac{C_2 \rho^2}{n_t} \leq (C_4 \sqrt{C_2} + C_2) \frac{\rho^2}{\sqrt{n_t}}, \end{aligned} \tag{49}$$

where the last inequality follows from $n_t \geq 1$. The following lemma gives the relationship between the fourth moment and sub-exponential norm. See, e.g. Vershynin (2018) and Wainwright (2019), for the detailed proof.

Lemma B.10. *There is a universal constant $C_3 > 0$ such that for any random variable Z ,*

$$\mathbb{E} Z^4 \leq C_3 \|Z\|_{\psi_1}^4.$$

The desired result now follows from Lemma B.10 and Equation (49):

$$\mathbb{E} (\check{\sigma}_{jk}^{(t)} - \sigma_{jk}^{(t)})^4 \leq C \frac{\rho^8}{n_t^2}$$

for a universal constant $C > 0$. □

B.7 Proof of Lemma B.4

By translation invariance, assume that both group means vanish. Let B be the coordinate block indexed by $\mathcal{J} \times \mathcal{K}$, where $|\mathcal{J}| \vee |\mathcal{K}| \leq v$. For fixed unit vectors a, b , write

$$U_i^{(g)} = a^\top X_{i,\mathcal{J}}^{(g)}, \quad V_i^{(g)} = b^\top X_{i,\mathcal{K}}^{(g)}, \quad g \in \{t, s\},$$

omitting a group when its sample size is zero. Then

$$\begin{aligned} a^\top (B - \mathbb{E}B)b &= \frac{1}{n} \sum_g \sum_{i=1}^{n_g} \left(U_i^{(g)} V_i^{(g)} - \mathbb{E}(U_i^{(g)} V_i^{(g)}) \right) \\ &\quad - \sum_g \frac{n_g}{n} \left(\bar{U}^{(g)} \bar{V}^{(g)} - \mathbb{E}(\bar{U}^{(g)} \bar{V}^{(g)}) \right). \end{aligned}$$

The summands in the first line are independent, centered, and have ψ_1 -norm at most $C\rho^2$. By Lemmas B.7 to B.9, each centered term in the second line, after multiplication by n_g/n , has ψ_1 -norm at most $C\rho^2/n$. Since there are at most two groups, Bernstein's inequality and a union bound give, for every $y \geq 1$,

$$\mathbb{P} \left\{ |a^\top (B - \mathbb{E}B)b| > C\rho^2 \left(\sqrt{\frac{y}{n}} + \frac{y}{n} \right) \right\} \leq 6e^{-y}. \quad (50)$$

Applying Lemma B.6 with 1/4-nets, taking $y = C_0(v + \kappa \log d)$, and using $v \vee \log d \leq c_\kappa n$ to absorb the linear term proves Lemma B.4 after adjusting the constants. $\square$

B.8 Proof of Lemma B.5

Let $\mathcal{G}_+ := \{g \in \{t, s\} : n_g > 0\}$. By translation invariance of the within-group sample covariances, assume that every nonempty group has mean zero. Write

$$X_i^{(g)} = \left(X_{ij}^{(g)} : j \in [d] \right), \quad i = 1, \dots, n_g, \quad g \in \mathcal{G}_+.$$

For any given $j, k \in [d]$, the (j, k) -th entry of the pooled sample covariance matrix, $\check{\sigma}_{jk}$, is represented as

$$\check{\sigma}_{jk} = \frac{1}{n} \left(\sum_{g \in \mathcal{G}_+} \sum_{i=1}^{n_g} X_{ij}^{(g)} X_{ik}^{(g)} - \sum_{g \in \mathcal{G}_+} n_g \bar{X}_j^{(g)} \bar{X}_k^{(g)} \right), \quad (51)$$

where it is defined that

$$\bar{X}_j^{(g)} := \frac{1}{n_g} \sum_{i=1}^{n_g} X_{ij}^{(g)}, \quad j = 1, \dots, d, \quad g \in \mathcal{G}_+.$$

We will argue that each centered fluctuation $\check{\sigma}_{jk} - \sigma_{jk}$ is uniformly sub-exponential over all indices $j, k \in [d]$. It is worth noting that both target observations, $X_1^{(t)}, \dots, X_{n_t}^{(t)}$, and source observations, $X_1^{(s)}, \dots, X_{n_s}^{(s)}$, consist of sub-Gaussian entries with the standard-deviation proxy ρ since every observation is uniformly sub-Gaussian in Assumption 2.1. For each group $g \in \mathcal{G}_+$ and each observation index $i = 1, \dots, n_g$, let us define

$$Y_i^{(g)} := X_{ij}^{(g)} X_{ik}^{(g)} - \sigma_{jk}^{(g)}.$$

Then all variables $\{Y_i^{(g)} : i = 1, \dots, n_g, g \in \mathcal{G}_+\}$ are independent and mean-zero. By Lemma B.8 and Lemma B.9,

$$\|Y_i^{(g)}\|_{\psi_1} \leq C_4 \|X_{ij}^{(g)} X_{ik}^{(g)}\|_{\psi_1} \leq C_4 \rho^2, \quad i = 1, \dots, n_g, \quad g \in \mathcal{G}_+.$$

Consequently, Lemma B.7 yields

$$\begin{aligned} \left\| \frac{1}{n} \sum_{g \in \mathcal{G}_+} \sum_{i=1}^{n_g} Y_i^{(g)} \right\|_{\psi_1}^2 &\leq \frac{C_2}{n^2} \sum_{g \in \mathcal{G}_+} \sum_{i=1}^{n_g} \|Y_i^{(g)}\|_{\psi_1}^2 \\ &\leq \frac{C_2 C_4^2 \rho^4}{n}. \end{aligned}$$

On the other hand, Lemma B.7 and Lemma B.8 imply that

$$\begin{aligned} \|\bar{X}_j^{(g)}\|_{\psi_2}^2 &\leq \frac{C_2}{n_g^2} \sum_{i=1}^{n_g} \|X_{ij}^{(g)}\|_{\psi_2}^2 \leq \frac{C_2 \rho^2}{n_g}, \quad g \in \mathcal{G}_+, \\ \left\| \frac{1}{n} \sum_{g \in \mathcal{G}_+} n_g \bar{X}_j^{(g)} \bar{X}_k^{(g)} \right\|_{\psi_1} &\leq \frac{1}{n} \sum_{g \in \mathcal{G}_+} n_g \|\bar{X}_j^{(g)}\|_{\psi_2} \|\bar{X}_k^{(g)}\|_{\psi_2} \leq \frac{2C_2 \rho^2}{n}. \end{aligned}$$

Combining those inequalities with Equation (51), we obtain

$$M_{jk} := \frac{1}{n} \sum_{g \in \mathcal{G}_+} n_g \bar{X}_j^{(g)} \bar{X}_k^{(g)}.$$

$$\begin{aligned} \|\check{\sigma}_{jk} - \sigma_{jk}\|_{\psi_1} &\leq \left\| \frac{1}{n} \sum_{g \in \mathcal{G}_+} \sum_{i=1}^{n_g} Y_i^{(g)} \right\|_{\psi_1} \\ &\quad + \|M_{jk}\|_{\psi_1} \\ &\leq \rho^2 C_4 \sqrt{\frac{C_2}{n}} + \frac{2C_2 \rho^2}{n} \leq (C_4 \sqrt{C_2} + 2C_2) \frac{\rho^2}{\sqrt{n}}, \end{aligned} \tag{52}$$

where the last inequality follows from $n \geq 1$.

By Lemma B.10 and Equation (52),

$$\mathbb{E}(\check{\sigma}_{jk} - \sigma_{jk})^4 \leq C \frac{\rho^8}{n^2}$$

for a universal constant $C > 0$. □

B.9 Proof of Lemma B.6

The first result is immediate from a simple volume argument:

$$|N_u(\varepsilon)| \left(\frac{\varepsilon}{2}\right)^u \leq \left(1 + \frac{\varepsilon}{2}\right)^u.$$

To claim the next result, we can find two unit vectors $a_u^* \in \mathbb{R}^u$ and $a_v^* \in \mathbb{R}^v$ such that

$$\|B\|_2 = (a_u^*)^\top B a_v^*.$$

Thanks to the definition of an ε -covering set, there always exist $\tilde{a}_u \in N_u(\varepsilon)$ and $\tilde{a}_v \in N_v(\varepsilon)$ that approximate the previous vectors, $a_u^* \in \mathbb{R}^u$ and $a_v^* \in \mathbb{R}^v$, respectively:

$$\|a_u^* - \tilde{a}_u\|_2 \leq \varepsilon \quad \text{and} \quad \|a_v^* - \tilde{a}_v\|_2 \leq \varepsilon.$$

Notice that $\tilde{a}_u^\top B \tilde{a}_v$ could be a good approximate for $\|B\|_2$ or $(a_u^*)^\top B a_v^*$.

$$\begin{aligned} |(a_u^*)^\top B a_v^* - \tilde{a}_u^\top B \tilde{a}_v| &= \frac{1}{2} \left| (a_u^* - \tilde{a}_u)^\top B (a_v^* + \tilde{a}_v) + (a_u^* + \tilde{a}_u)^\top B (a_v^* - \tilde{a}_v) \right| \\ &\leq \frac{1}{2} \|a_u^* - \tilde{a}_u\|_2 \|B\|_2 \|a_v^* + \tilde{a}_v\|_2 + \frac{1}{2} \|a_u^* + \tilde{a}_u\|_2 \|B\|_2 \|a_v^* - \tilde{a}_v\|_2 \\ &\leq \left(\|a_u^* - \tilde{a}_u\|_2 + \|a_v^* - \tilde{a}_v\|_2 \right) \|B\|_2 \leq 2\varepsilon \|B\|_2. \end{aligned}$$

The desired result is now straightforward:

$$\|B\|_2 = (a_u^*)^\top B a_v^* \leq 2\varepsilon \|B\|_2 + |\tilde{a}_u^\top B \tilde{a}_v| \leq 2\varepsilon \|B\|_2 + \sup_{\substack{a_u \in N_u(\varepsilon) \\ a_v \in N_v(\varepsilon)}} |a_u^\top B a_v|.$$

□

B.10 Proof of Lemma B.7

When one of the random variables $Z_1, \dots, Z_m$ is not sub-Gaussian or sub-exponential, the desired result is immediate. Vershynin (2018, Proposition 2.6.1) has already shown the same statement when all of them are sub-Gaussian. We will focus on the case where $Z_1, \dots, Z_m$ are independent and mean-zero sub-exponential variables.

The sub-exponential properties (Vershynin, 2018, Proposition 2.7.1) give us a useful tool for analyzing the moment generating function of the sum: there exists a generic constant $C_{21} > 0$ satisfying

$$\mathbb{E} e^{\lambda(Z_1 + \dots + Z_m)} = \prod_{k=1}^m \mathbb{E} e^{\lambda Z_k} \leq \prod_{k=1}^m e^{\lambda^2 C_{21}^2 \|Z_k\|_{\psi_1}^2}, \quad \text{for any } |\lambda| \leq C_{21}^{-1} \min_{1 \leq k \leq m} \|Z_k\|_{\psi_1}^{-1}.$$

We may further refine this inequality so that

$$\mathbb{E} e^{\lambda(Z_1 + \dots + Z_m)} \leq \exp \left(\lambda^2 C_{21}^2 \sum_{k=1}^m \|Z_k\|_{\psi_1}^2 \right), \quad \text{for any } |\lambda| \leq C_{21}^{-1} \left(\sum_{k=1}^m \|Z_k\|_{\psi_1}^2 \right)^{-1/2}$$

Applying again the sub-exponential properties (Vershynin, 2018, Proposition 2.7.1), the desired result is immediate: there exists generic constants $C_{22} > 0$ such that

$$\|Z_1 + \dots + Z_m\|_{\psi_1}^2 \leq C_{22} C_{21}^2 \sum_{k=1}^m \|Z_k\|_{\psi_1}^2.$$

□

C Proof of Technical Results in Adaptivity

Throughout this section, $n = n_t + n_s$. The exact model is obtained by setting $\eta = 0$. We use the pairing convention introduced in the main text, so the observations denoted by $X_i^{(g)}$ below satisfy $\mathbb{E}X_i^{(g)} = 0$, $\text{Cov}(X_i^{(g)}) = \Sigma^{(g)}$, and $\sup_{\|u\|_2=1} \|u^\top X_i^{(g)}\|_{\psi_2} \leq C\rho$. Write

$$p_r(a) := r^{-2a/(2a+1)} \wedge \frac{d}{r}, \quad P_r(a) := p_r(a) + \frac{\log(ed)}{r}.$$

Using $\log(ed)$ adds at most r^{-1} to these rates, which is absorbed by $p_r(a)$.

C.1 Sample splitting and the adaptive estimator

Let m_f be the validation size and put $m_0 = n_t - m_f$. Choose deterministic disjoint target folds $\mathcal{I}_t^{(0)}$ and $\mathcal{I}_t^{(f)}$, with $|\mathcal{I}_t^{(0)}| = m_0$ and $|\mathcal{I}_t^{(f)}| = m_f$. Write $m = m_0 + n_s$, $\mathcal{D}_t = \{X_i^{(t)} : i \in \mathcal{I}_t^{(0)}\}$, and $\mathcal{D}_p = \mathcal{D}_t \cup \{X_i^{(s)} : 1 \leq i \leq n_s\}$. The pooled training second moment and its expectation are

$$\widehat{\Sigma}^{(p)} = \frac{1}{m} \left(\sum_{i \in \mathcal{I}_t^{(0)}} X_i^{(t)} (X_i^{(t)})^\top + \sum_{i=1}^{n_s} X_i^{(s)} (X_i^{(s)})^\top \right), \quad \Sigma^{(p)} = \frac{m_0 \Sigma^{(t)} + n_s \Sigma^{(s)}}{m}. \quad (53)$$

The fitting sizes satisfy $m_0 \asymp n_t$ and $m \asymp n$.

Choose the fixed envelope $L = C_L \rho^2$ so that $0 \preceq \Sigma^{(g)} \preceq L I_d$. Let Clip_L denote eigenvalue truncation to $[0, L]$. If $0 \preceq \Theta \preceq L I_d$, Weyl's inequality gives

$$\|\text{Clip}_L(A) - \Theta\|_2 \leq 2\|A - \Theta\|_2. \quad (54)$$

Set $A_{\text{tar}} = \text{FitAdaptive}(\mathcal{D}_t)$ and $A_{\text{pool}} = \text{FitAdaptive}(\mathcal{D}_p)$. Both are measurable with respect to data outside $\mathcal{I}_t^{(f)}$. The final estimator is obtained by applying the selector in Algorithm 2.

C.2 Block concentration

Lemma C.1 (Block concentration). *Let $Z_1, \dots, Z_r$, $r \geq 1$, be independent centered observations, not necessarily identically distributed, with $\mathbb{E}Z_i Z_i^\top = \Sigma_i$, and*

$$\sup_{\|u\|_2=1} \|u^\top Z_i\|_{\psi_2} \leq C\rho.$$

Define

$$\widehat{\Theta} = \frac{1}{r} \sum_{i=1}^r Z_i Z_i^\top, \quad \Theta = \frac{1}{r} \sum_{i=1}^r \Sigma_i.$$

For complete coordinate sets $J, K \subset [d]$, with $|J| \vee |K| \leq q$, and every $x \geq 0$,

$$\mathbb{P} \left\{ \|(\widehat{\Theta} - \Theta)_{J,K}\|_2 > C\rho^2 \left(\sqrt{\frac{q+x}{r}} + \frac{q+x}{r} \right) \right\} \leq 4e^{-x}. \quad (55)$$

Moreover, for every integer $1 \leq q \leq d$,

$$\mathbb{P} \left\{ \|\mathfrak{T}_d(\widehat{\Theta} - \Theta; q)\|_2 > C\rho^2 \left(\sqrt{\frac{q + \log(ed) + x}{r}} + \frac{q + \log(ed) + x}{r} \right) \right\} \leq 4e^{-x}, \quad (56)$$

$$\mathbb{E} \|\mathfrak{T}_d(\widehat{\Theta} - \Theta; q)\|_2^2 \leq C\rho^4 \left[\frac{q + \log(ed)}{r} + \frac{(q + \log(ed))^2}{r^2} \right]. \quad (57)$$

Proof. The zero-mean convention removes the additional sample-mean correction from the covariance decomposition.

Fix unit vectors $u \in \mathbb{R}^{|J|}$ and $v \in \mathbb{R}^{|K|}$, extended by zero outside J and K . The product

$$(u^\top Z_{i,J})(v^\top Z_{i,K}) - \mathbb{E}((u^\top Z_{i,J})(v^\top Z_{i,K}))$$

has ψ_1 -norm at most $C\rho^2$. Bernstein's inequality for the average of these independent variables gives variance and maximal-scale proxies bounded respectively by $C\rho^4/r$ and $C\rho^2/r$.

Consequently, Bernstein's inequality yields

$$\mathbb{P} \left\{ |u^\top (\widehat{\Theta} - \Theta)_{J,K} v| > C\rho^2 \left(\sqrt{\frac{x}{r}} + \frac{x}{r} \right) \right\} \leq 4e^{-x}.$$

Let N_J, N_K be $1/4$ -nets of the two unit spheres, with $|N_J| \leq 9^{|J|}$, $|N_K| \leq 9^{|K|}$. The rectangular net inequality

$$\|M\|_2 \leq 2 \max_{u \in N_J, v \in N_K} |u^\top M v|$$

and a union bound prove Equation (55).

Partition a matrix into consecutive q -blocks. If D is block tridiagonal and $M = \max_{|r-s| \leq 1} \|D_{rs}\|_2$, then

$$\|(Dz)_r\|_2 \leq M \sum_{|s-r| \leq 1} \|z_s\|_2, \quad \text{and consequently} \quad \|D\|_2 \leq 3M.$$

The retained blocks of $\mathfrak{T}_d(\widehat{\Theta} - \Theta; q)$ are complete rectangles, and there are at most $3\lceil d/q \rceil \leq 3d$ of them. A second union bound proves Equation (56). Integrating that tail and using $(\sqrt{y/r} + y/r)^2 \leq 2y/r + 2y^2/r^2$ proves Equation (57). $\square$

C.3 Fixed bandwidth target and pooled risks

Lemma C.2 (Deterministic tridiagonal approximation). *For every matrix D and every integer $q \geq 1$,*

$$\|\mathfrak{T}_d(D; q)\|_2 \leq 3\|D\|_2.$$

If $\Sigma_0 \in \mathcal{B}_\alpha(M)$, then

$$\|\Sigma_0 - \mathfrak{T}_d(\Sigma_0; q)\|_2 \leq Mq^{-\alpha} \mathbf{1}(q < d).$$

If a symmetric R is upper-tail stable at level η , then

$$\|R - \mathfrak{T}_d(R; q)\|_2 \leq 4\eta.$$

Proof. Every retained block of D has norm at most $\|D\|_2$, so the first assertion follows from the block-compression argument in the proof of Lemma C.1. Every entry removed by $\mathfrak{T}_d(\cdot; q)$ has $|j - k| > q$. Thus the removed part of Σ_0 is entrywise dominated by $\mathfrak{A}_d(\Sigma_0; q)$, and, by symmetry,

$$\|\Sigma_0 - \mathfrak{T}_d(\Sigma_0; q)\|_2 \leq \|\Sigma_0 - \mathfrak{T}_d(\Sigma_0; q)\|_1 \leq \|\mathfrak{A}_d(\Sigma_0; q)\|_1 \leq Mq^{-\alpha}.$$

Finally, the $u = 0$ convention in Equation (9) gives $\|R\|_2 \leq \eta$. Hence

$$\|R - \mathfrak{T}_d(R; q)\|_2 \leq \|R\|_2 + \|\mathfrak{T}_d(R; q)\|_2 \leq 4\eta.$$

□

Proposition C.3 (Fixed bandwidth target and pooled risks). *Uniformly over $\tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}$, for every $1 \leq q \leq d$,*

$$\begin{aligned} \mathbb{E} \|\text{Clip}_L \left(\mathfrak{T}_d(\hat{\Sigma}_{\mathcal{I}_t^{(0)}}^{(t)}; q) \right) - \Sigma^{(t)}\|_2^2 &\lesssim \rho^4 \left(\frac{q + \log(ed)}{m_0} + \frac{(q + \log(ed))^2}{m_0^2} \right) \\ &\quad + M_t^2 q^{-2\alpha_t} \mathbf{1}(q < d) + \eta^2. \end{aligned} \quad (58)$$

For every $1 \leq q \leq d$, the pooled candidate satisfies both

$$\begin{aligned} \mathbb{E} \|\text{Clip}_L \left(\mathfrak{T}_d(\hat{\Sigma}^{(p)}; q) \right) - \Theta\|_2^2 &\lesssim \rho^4 \left(\frac{q + \log(ed)}{m} + \frac{(q + \log(ed))^2}{m^2} \right) \\ &\quad + M_{\max}^2 q^{-2\alpha_{\max}} \mathbf{1}(q < d) + \delta^2 + \eta^2, \end{aligned} \quad (59)$$

for $\Theta = \Sigma^{(p)}$ and for $\Theta = \Sigma^{(t)}$.

Proof. Apply Lemma C.1 to the target training fold. By Equation (11) and Lemma C.2,

$$\|\mathfrak{T}_d(\Sigma^{(t)}; q) - \Sigma^{(t)}\|_2 \leq M_t q^{-\alpha_t} \mathbf{1}(q < d) + 4\eta.$$

Squaring the stochastic and bias decomposition and using Equation (54) proves Equation (58).

For the pooled estimator, apply Lemma C.1 directly to the m independent observations in the combined training fold. Let $g_* \in \{t, s\}$ be a group with smoothness $\alpha_{\max}$. The $u = 0$ disparity condition gives

$$\|\Sigma^{(t)} - \Sigma^{(s)}\|_2 \leq \delta, \quad \|\Sigma^{(p)} - \Sigma^{(g_*)}\|_2 \leq \delta, \quad \|\Sigma^{(p)} - \Sigma^{(t)}\|_2 \leq \delta.$$

Consequently, for $\Theta = \Sigma^{(p)}$ or $\Sigma^{(t)}$,

$$\begin{aligned} \|\mathfrak{T}_d(\Sigma^{(p)}; q) - \Theta\|_2 &\leq \|\mathfrak{T}_d(\Sigma^{(p)} - \Sigma^{(g_*)}; q)\|_2 \\ &\quad + \|\mathfrak{T}_d(\Sigma^{(g_*)}; q) - \Sigma^{(g_*)}\|_2 + \|\Sigma^{(g_*)} - \Theta\|_2 \\ &\lesssim M_{\max} q^{-\alpha_{\max}} \mathbf{1}(q < d) + \delta + \eta. \end{aligned}$$

The concentration bound, clipping, and $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$ prove Equation (59). □

C.4 Uniform superband selection

For a sample collection $\mathcal{D} = \{Z_i : 1 \leq i \leq r\}$, write $\Theta = r^{-1} \sum_i \mathbb{E} Z_i Z_i^\top$. The following oracle bound applies to any sample collection.

Proposition C.4 (Superbanding oracle for FitAdaptive). *Suppose that $\Theta = \Theta_0 + R$, where $\Theta_0 \in \mathcal{B}_a(M)$ and symmetric R is upper-tail stable at level ζ . For $r \geq 8$, uniformly over independent centered sub-Gaussian observations with average covariance Θ ,*

$$\mathbb{E} \|\text{FitAdaptive}(\mathcal{D}) - \Theta\|_2^2 \lesssim P_r(a) + \zeta^2.$$

Proof. Write $\ell_d = \log(ed)$, $\hat{\Theta} = r^{-1} \sum_{Z \in \mathcal{D}} Z Z^\top$, and

$$D_u(B) = \mathfrak{T}_d(B; 2u) - \mathfrak{T}_d(B; u), \quad p(u) = \lambda \rho^2 \sqrt{\frac{u + \ell_d}{r}}.$$

If $\mathcal{U}_r$ is a singleton, the fixed bandwidth decomposition in Lemmas C.1 and C.2 proves the assertion directly. Indeed, either its sole bandwidth is at least $d/2$, in which case the estimator is the full sample second moment, or it is a fixed positive fraction of $r/\log(er)$, in which case the variance is absorbed by ℓ_d/r and the smoothness term at that index is absorbed by $p_r(a)$. We henceforth suppose that the grid has at least two points.

We use two ingredients. Whenever $u, 2u \in \mathcal{U}_r$, the two partitions are nested and the nonzero pieces of D_u are complete coordinate rectangles. Thus, Lemma C.1 gives

$$\|D_u(\hat{\Theta} - \Theta)\|_2 \leq C \rho^2 \sqrt{\frac{u + \ell_d}{r}}. \quad (\text{C.1})$$

Simultaneously for all such u , (C.1) holds on an event $\mathcal{E}$ with $\mathbb{P}(\mathcal{E}^c) \leq C(ed)^{-6}$. Indeed, apply Equation (56) to the two complete block tridiagonal matrices in D_u , with confidence parameter $x = C_0 \ell_d$, and take a union bound over the dyadic grid. For a non-singleton grid, $u_- = \lceil \ell_d \rceil$, and the definition of u_+ gives

$$|\mathcal{U}_r| \leq 1 + \log_2(u_+/u_-) \lesssim \ell_d.$$

If $u_+ = d$, this uses $u_+ \leq d$; if $u_+ < d$, non-singletonness forces $u_+ = \lfloor r/\log(er) \rfloor < d$. Thus C_0 can be chosen so that the union bound is at most $C(ed)^{-6}$, while $u + \ell_d + x \lesssim u + \ell_d$. If the grid has more than one point, its largest bandwidth is at most $r/\log(er)$; the same observation applies to ℓ_d , and hence the linear term in Equation (56) is absorbed by its square-root term. If the grid is a singleton, no simultaneous assertion is needed.

The deterministic increment has the corresponding bound at one scale

$$\|D_u(\Theta)\|_2 \leq C(Mu^{-a} + \zeta). \quad (\text{C.2})$$

To see this, every entry retained by D_u lies outside the u -band. Therefore $D_u(\Theta_0)$ is an entrywise restriction of $\mathfrak{A}_d(\Theta_0; u)$, and symmetry gives

$$\|D_u(\Theta_0)\|_2 \leq \|D_u(\Theta_0)\|_1 \leq \|\mathfrak{A}_d(\Theta_0; u)\|_1 \leq Mu^{-a}.$$

For the remainder, partition the coordinates into consecutive u -blocks. In the upper triangular part of $D_u(R)$, the nonzero block diagonals have block offsets two and three: the

offset-one blocks are retained by both $\mathfrak{T}_d(R; u)$ and $\mathfrak{T}_d(R; 2u)$. Each such block is a compression of $\mathfrak{A}_d^+(R; u)$, and therefore has operator norm at most ζ . Since there are at most two such blocks in every block row and column, block compression gives an upper-part norm at most 2ζ . The lower part is its transpose, so $\|D_u(R)\|_2 \leq 4\zeta$. This proves (C.2).

Choose C_0 large enough, depending only on the fixed constants, and set

$$q_0 = C_0 \left(\left(\frac{M^2 r}{\rho^4} \right)^{1/(2a+1)} \vee \frac{r\zeta^2}{\rho^4} \vee \ell_d \right).$$

Every nontrivial grid starts at ℓ_d , so $p(2u) - p(u) \geq c\lambda\rho^2\sqrt{(u + \ell_d)/r}$. Thus (C.1), (C.2), and a sufficiently large fixed λ imply, on $\mathcal{E}$,

$$F_{\mathcal{D}}(u) \leq F_{\mathcal{D}}(2u) \quad \text{for every } u \geq q_0 \text{ with } u, 2u \in \mathcal{U}_r. \quad (\text{C.3})$$

Here we used the exact adjacent-scale identity

$$F_{\mathcal{D}}(u) - F_{\mathcal{D}}(2u) = \|D_u(\widehat{\Theta})\|_2 + p(u) - p(2u).$$

Let $q_+ = \max \mathcal{U}_r$. First suppose that $q_0 \leq q_+$, and let q_* be the first grid point not smaller than q_0 . By (C.3), $\widehat{u} \leq q_*$ on $\mathcal{E}$. Cancelling the common tail of the two objective functions yields

$$\|\mathfrak{T}_d(\widehat{\Theta}; \widehat{u}) - \mathfrak{T}_d(\widehat{\Theta}; q_*)\|_2 \leq p(q_*). \quad (\text{C.5})$$

The fixed bandwidth decomposition in Lemmas C.1 and C.2 gives

$$\mathbb{E}\|\mathfrak{T}_d(\widehat{\Theta}; q_*) - \Theta\|_2^2 \lesssim \rho^4 \left(\frac{q_* + \ell_d}{r} + \frac{(q_* + \ell_d)^2}{r^2} \right) + M^2 q_*^{-2a} + \zeta^2 \lesssim P_r(a) + \zeta^2. \quad (\text{C.6})$$

Since $q_* \leq 2q_0 \leq 2q_+$, and a non-singleton grid has $q_+ \leq r/\log(er)$, the quadratic concentration term is absorbed by the first one. The three terms in the definition of q_0 respectively give

$$\rho^4 q_*/r \lesssim P_r(a) + \zeta^2, \quad M^2 q_*^{-2a} \lesssim p_r(a), \quad \zeta^2 \leq P_r(a) + \zeta^2.$$

Combining (C.5) and (C.6) proves the desired bound in the interior regime.

Finally, suppose that $q_0 > q_+$. If $u_+ = d$, then $q_+ \geq d/2$ and $\mathfrak{T}_d(\widehat{\Theta}; q_+) = \widehat{\Theta}$. The same deterministic comparison as in (C.5) gives

$$\|\mathfrak{T}_d(\widehat{\Theta}; \widehat{u}) - \widehat{\Theta}\|_2 \leq p(q_+).$$

Since $q_0 = C_0(q_{\text{sm}} \vee r\zeta^2/\rho^4 \vee \ell_d)$ and $q_+ \geq d/2$, one of the three terms in parentheses is at least $d/(2C_0)$. Consequently,

$$\frac{d}{r} \lesssim \begin{cases} p_r(a), & q_{\text{sm}} \geq d/(2C_0), \\ \zeta^2, & r\zeta^2/\rho^4 \geq d/(2C_0), \\ \ell_d/r, & \ell_d \geq d/(2C_0), \end{cases}$$

and hence

$$\frac{d + \ell_d}{r} \lesssim P_r(a) + \zeta^2,$$

and

$$p(q_+)^2 \lesssim \rho^4(d + \ell_d)/r.$$

The full-sample second-moment bound therefore gives

$$\mathbb{E}\|\text{FitAdaptive}(\mathcal{D}) - \Theta\|_2^2 \lesssim P_r(a) + \zeta^2$$

in this endpoint case. If $u_+ < d$, the same comparison as in (C.5), now with q_+ in place of the full bandwidth, gives $\|\mathfrak{T}_d(\hat{\Theta}; \hat{u}) - \mathfrak{T}_d(\hat{\Theta}; q_+)\|_2 \leq p(q_+)$. Since $q_+ \asymp r/\log(er)$, the endpoint fixed bandwidth bound is

$$\rho^4 \left(\frac{q_+ + \ell_d}{r} + \frac{(q_+ + \ell_d)^2}{r^2} \right) + M^2 q_+^{-2a} + \zeta^2 \lesssim P_r(a) + \zeta^2. \quad (\text{C.7})$$

Indeed, the quadratic term is absorbed by the linear term, while

$$\rho^4 q_+/r \lesssim \begin{cases} \zeta^2, & q_0 \text{ is driven by } r\zeta^2/\rho^4, \\ \ell_d/r, & q_0 \text{ is driven by } \ell_d. \end{cases}$$

If $q_{\text{sm}} := (M^2 r / \rho^4)^{1/(2a+1)}$ is the largest term in q_0 , then $q_+ < C_0 q_{\text{sm}}$. Since $q_+ \asymp r/\log(er)$ whereas $q_{\text{sm}} \asymp r^{1/(2a+1)}$, this case can occur only for bounded r , whose contribution is absorbed into the implicit constant. Otherwise, q_+ is above the smoothness scale for all sufficiently large r , and $M^2 q_+^{-2a} \lesssim p_r(a)$; the remaining bounded range is absorbed in the same way. This proves (C.7).

Combining the two cases on $\mathcal{E}$, together with Equation (54) control the returned estimator. On $\mathcal{E}^c$, both the estimator and Θ have spectra in $[0, L]$, so its contribution is at most $L^2 \mathbb{P}(\mathcal{E}^c)$, which is absorbed by $P_r(a)$. This proves Proposition C.4. $\square$

Proposition C.5. *The target fit satisfies*

$$\sup_{\mathbb{P} \in \tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E}\|A_{\text{tar}} - \Sigma^{(t)}\|_2^2 \lesssim P_{n_t}(\alpha_t) + \eta^2.$$

Proof. Apply Proposition C.4 to $\mathcal{D}_t$, with $\Theta_0 = \Sigma_0^{(t)}$, $R = R^{(t)}$, and $r = m_0 \asymp n_t$. $\square$

Proposition C.6. *The pooled fit satisfies*

$$\sup_{\mathbb{P} \in \tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E}\|A_{\text{pool}} - \Sigma^{(t)}\|_2^2 \lesssim P_n(\alpha_{\max}) + \delta^2 + \eta^2. \quad (60)$$

Proof. Let $g_* \in \{t, s\}$ have smoothness $\alpha_{\max}$. The weighted difference $\Sigma^{(p)} - \Sigma^{(g_*)}$ is upper-tail stable at level at most $C\delta$. To use the decomposition required by Proposition C.4, set

$$R_p := \Sigma^{(p)} - \Sigma_0^{(g_*)} = (\Sigma^{(p)} - \Sigma^{(g_*)}) + R^{(g_*)}.$$

By the triangle inequality for the upper-tail operator, R_p is upper-tail stable at level $C(\delta + \eta)$. The oracle proposition therefore gives

$$\mathbb{E}\|A_{\text{pool}} - \Sigma^{(p)}\|_2^2 \lesssim P_m(\alpha_{\max}) + (\delta + \eta)^2.$$

Finally, $\|\Sigma^{(p)} - \Sigma^{(t)}\|_2 \leq \delta$ by the $u = 0$ disparity condition, and hence

$$\mathbb{E}\|A_{\text{pool}} - \Sigma^{(t)}\|_2^2 \leq 2\mathbb{E}\|A_{\text{pool}} - \Sigma^{(p)}\|_2^2 + 2\|\Sigma^{(p)} - \Sigma^{(t)}\|_2^2 \lesssim P_n(\alpha_{\max}) + \delta^2 + \eta^2,$$

where $m \asymp n$ is used in the last step. $\square$

C.5 Validation and assembly

Lemma C.7 (One-sided directional selector). *Let $\mathcal{H}$ be a sigma-field and let A, B be $\mathcal{H}$ -measurable matrices satisfying $0 \preceq A, B \preceq LI_d$. Let $0 \preceq \Sigma \preceq LI_d$, and let $Y_1, \dots, Y_m$, independent of $\mathcal{H}$, be independent centered validation vectors satisfying $\mathbb{E}Y_i = 0$, $\mathbb{E}Y_i Y_i^\top = \Sigma$, and $\sup_{\|u\|_2=1} \|u^\top Y_i\|_{\psi_2} \leq C\rho$. Choose measurably a unit vector v such that*

$$|v^\top (A - B)v| = \|A - B\|_2,$$

put $a = v^\top Av$, $b = v^\top Bv$, and

$$\hat{q} = \Pi_{[0,L]} \left(\frac{1}{m} \sum_{i=1}^m (v^\top Y_i)^2 \right).$$

For $x \geq 1$, set

$$\tau_x = C_1 \rho^2 \left(\sqrt{\frac{x}{m}} + \frac{x}{m} \right),$$

and select A if $|a - \hat{q}| + 4\tau_x \leq |b - \hat{q}|$, otherwise select B . For C_1 sufficiently large,

$$\mathbb{E} \left[\|\hat{\Sigma} - \Sigma\|_2^2 \mid \mathcal{H} \right] \leq 18 \min(\|A - \Sigma\|_2^2, \|B - \Sigma\|_2^2) + C\tau_x^2 + 2L^2 e^{-x}, \quad (61)$$

$$\mathbb{E} \left[\|\hat{\Sigma} - \Sigma\|_2^2 \mid \mathcal{H} \right] \leq 9\|B - \Sigma\|_2^2 + 2L^2 e^{-x}. \quad (62)$$

Proof. Let $q = v^\top \Sigma v$ and $G = \{|\hat{q} - q| \leq \tau_x\}$. Conditional scalar Bernstein and clipping give $\mathbb{P}(G^c \mid \mathcal{H}) \leq 2e^{-x}$.

Suppose first that the rule selects A . On G ,

$$|a - q| \leq |a - \hat{q}| + \tau_x \leq |b - \hat{q}| - 3\tau_x \leq |b - q| - 2\tau_x.$$

Hence

$$\|A - B\|_2 = |a - b| \leq |a - q| + |b - q| \leq 2\|B - \Sigma\|_2,$$

and therefore $\|A - \Sigma\|_2 \leq 3\|B - \Sigma\|_2$. If the rule selects B , the reverse strict inequality and G give

$$|b - q| < |a - q| + 6\tau_x.$$

When A is the better candidate,

$$\|B - \Sigma\|_2 \leq \|B - A\|_2 + \|A - \Sigma\|_2 \leq 3\|A - \Sigma\|_2 + 6\tau_x.$$

Combining the cases shows on G that

$$\|\hat{\Sigma} - \Sigma\|_2 \leq 3 \min(\|A - \Sigma\|_2, \|B - \Sigma\|_2) + 6\tau_x, \quad \|\hat{\Sigma} - \Sigma\|_2 \leq 3\|B - \Sigma\|_2.$$

On G^c , spectral clipping gives the deterministic bound $\|\hat{\Sigma} - \Sigma\|_2^2 \leq L^2$. Hence, writing $r_A = \|A - \Sigma\|_2^2$ and $r_B = \|B - \Sigma\|_2^2$,

$$\mathbb{E} \left[\|\hat{\Sigma} - \Sigma\|_2^2 \mid \mathcal{H} \right] \leq 18 \min(r_A, r_B) + 72\tau_x^2 + 2L^2 e^{-x},$$

and

$$\mathbb{E} \left[\|\hat{\Sigma} - \Sigma\|_2^2 \mid \mathcal{H} \right] \leq 9r_B + 2L^2 e^{-x}.$$

Enlarging the numerical constant in front of τ_x^2 proves Equations (61) and (62). $\square$

Proposition C.8 (Adaptive estimator upper bound). *The estimator in Algorithm 2 satisfies*

$$\begin{aligned} & \sup_{\mathbb{P} \in \tilde{\mathcal{P}}_{\alpha_t, \alpha_s, \delta, \eta}} \mathbb{E} \|\hat{\Sigma}_{\text{ad}}^{(t)} - \Sigma^{(t)}\|_2^2 \\ & \lesssim \Phi_{\alpha_t, \alpha_s, \delta}(n_t, n_s, d) + \eta^2 + \left[\delta^2 \wedge \rho^4 \left(\frac{\log(en)}{m_f} + \frac{\log^2(en)}{m_f^2} \right) \right]. \end{aligned} \quad (63)$$

Proof. Apply Lemma C.7 conditionally on all data outside $\mathcal{I}_t^{(f)}$, with $A = A_{\text{tar}}$, $B = A_{\text{pool}}$, $m = |\mathcal{I}_t^{(f)}|$, and $x = 6 \log(en)$. Since $|\mathcal{I}_t^{(f)}| = m_f$,

$$\tau_x^2 \lesssim \rho^4 \left(\frac{\log(en)}{m_f} + \frac{\log^2(en)}{m_f^2} \right) = \Gamma_{m_f, n}.$$

The first selector bound and Proposition C.5 give

$$R_{\text{ad}} \lesssim P_{n_t}(\alpha_t) + \eta^2 + \Gamma_{m_f, n},$$

and the second selector bound and Proposition C.6 give

$$R_{\text{ad}} \lesssim P_n(\alpha_{\max}) + \delta^2 + \eta^2.$$

Here $r_A = \|A_{\text{tar}} - \Sigma^{(t)}\|_2^2$ and $r_B = \|A_{\text{pool}} - \Sigma^{(t)}\|_2^2$ denote the conditional candidate risks. For the first inequality, we used $\mathbb{E} \min(r_A, r_B) \leq \mathbb{E} r_A$, and for the default inequality we used the second candidate bound directly. More generally, $\mathbb{E} \min(r_A, r_B) \leq \min(\mathbb{E} r_A, \mathbb{E} r_B)$, so the conditional selector bounds can be combined with the two candidate risk propositions. In both displays the bad-event contribution is at most $C\rho^4(en)^{-6}$, which is absorbed by $P_n(\alpha_{\max}) \geq n^{-1}$. Write

$$r = p_{n_t}(\alpha_t), \quad \ell = \frac{\log d}{n_t}, \quad u = P_n(\alpha_{\max}), \quad z = \delta^2.$$

Taking the smaller of the two bounds and using

$$\min(r + \ell + C\Gamma_{m_f, n}, u + z) \leq u + \min(r + \ell + C\Gamma_{m_f, n}, z) \leq u + (r \wedge z) + (\ell \wedge z) + C(\Gamma_{m_f, n} \wedge z)$$

proves Equation (63). $\square$

C.6 Proof of Corollary 3.3

Proof. Set

$$r_t := p_{n_t}(\alpha_t), \quad \ell_t := \frac{\log d}{n_t}, \quad z := \delta^2.$$

By Equation (63) with $\eta = 0$, it is enough to show

$$z \wedge \Gamma_{m_f, n} \lesssim (r_t \wedge z) + (\ell_t \wedge z). \quad (\text{C.8})$$

Under the first condition of the corollary, $z \lesssim r_t \vee \ell_t$. Thus either $z \lesssim r_t$ or $z \lesssim \ell_t$, and consequently

$$z \wedge \Gamma_{m_f, n} \leq z \lesssim (r_t \wedge z) + (\ell_t \wedge z).$$

For the second condition, $n \asymp n_t^{\max\{1, \kappa\}}$ and $m_f \asymp n_t / \log n_t$ for sufficiently large n_t . Hence

$$\Gamma_{m_f, n} \lesssim \frac{\log^2 n_t}{n_t}.$$

Let $\beta = 2\alpha_t / (2\alpha_t + 1) < 1$. Since $d \asymp n_t^\xi$ with $\xi > 0$,

$$r_t \asymp n_t^{-\gamma}, \quad \gamma = \begin{cases} \max\{\beta, 1 - \xi\}, & 0 < \xi < 1, \\ \beta, & \xi \geq 1, \end{cases}$$

where in all cases $\gamma < 1$. Therefore

$$\Gamma_{m_f, n} = o(r_t), \quad z \wedge \Gamma_{m_f, n} \lesssim r_t \wedge z \leq (r_t \wedge z) + (\ell_t \wedge z).$$

This proves (C.8), and the corollary follows. □

D Additional Numerical Experiments

This section presents additional simulation studies that examine the robustness of the adaptive transfer block-tridiagonal estimator beyond the baseline setting in the main text. Throughout, we compare its transfer version with the target-only counterpart. Unless otherwise stated, we fix $n_t = 50$, $n_s = 200$, and $d = 50$, and report squared operator-norm loss over 1,000 repetitions.

D.1 General Disparity Under Weaker Similarity Conditions

We begin with a source-target discrepancy model motivated by Proposition A.1, which shows that the upper-tail stability condition is implied by several norm-based notions of similarity, including Schatten- p control for $1 \leq p \leq 2$. Starting from the same bandable target covariance matrix as in the main text, we generate the disparity matrix as a mixture of two components: a decaying component and a random dense symmetric component. Specifically, we form

$$\Delta^{(s)} = (1 - \rho)\Delta_{\text{decay}} + \rho\Delta_{\text{dense}},$$

where each component is first normalized in operator norm, and the final mixture is then rescaled so that $\|\Delta^{(s)}\|_1 = \delta$ for $\delta \in \{0.1, 0.7, 1.3\}$. The mixing weight $\rho \in \{0, 0.5, 1\}$ therefore controls how diffuse the source-target discrepancy is while keeping the overall matrix ℓ_1 -disparity level aligned with the baseline simulations.

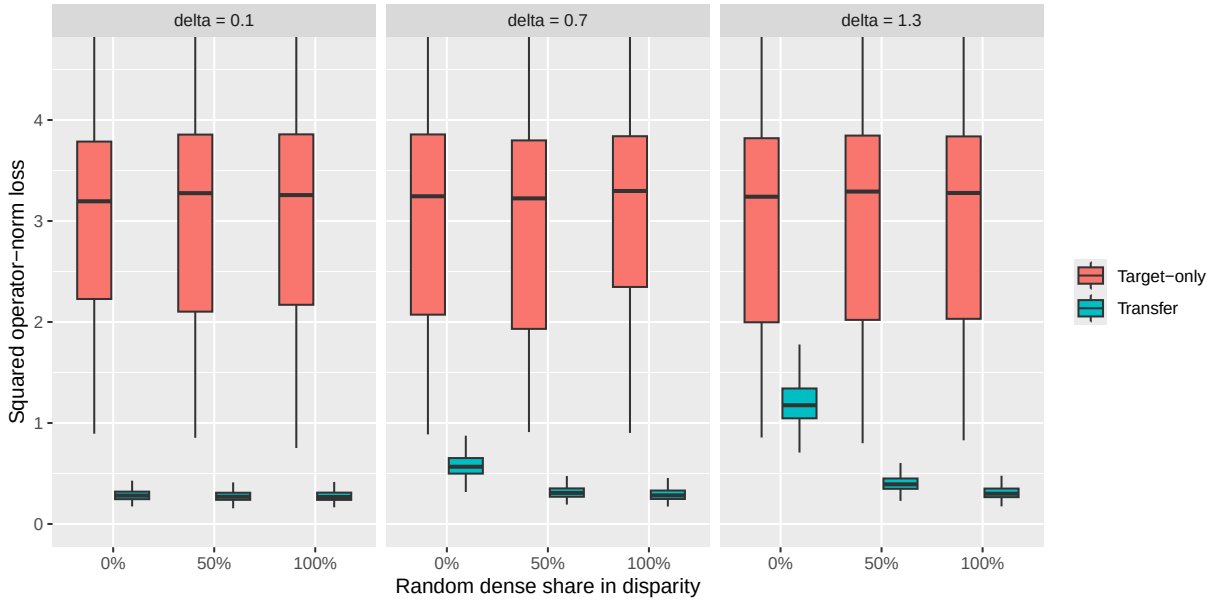


Figure 8: Squared operator-norm loss for the target-only and transfer estimators under a more general disparity model. The horizontal axis records the proportion of the random dense component in the source-target disparity matrix.

Figure 8 shows that, after aligning the disparity calibration, the purely decaying case $\rho = 0$ becomes substantially harder when δ is large. In particular, at $\delta = 1.3$, the transfer loss under $\rho = 0$ rises to a level comparable to that in the misspecification experiment. By

contrast, when the same matrix ℓ_1 -disparity budget is spread over a more diffuse component with $\rho = 0.5$ or 1, the transfer estimator remains much more accurate than the target-only benchmark. This suggests that the discrepancy shape matters in addition to its overall size: a concentrated decaying disparity is more harmful for transfer than a diffuse disparity with the same matrix ℓ_1 magnitude.

D.2 Misspecification of the Bandable Structure

We next turn to misspecification of exact bandability. The misspecified-model setup in Theorems A.3 and A.4 allows the target and source covariance matrices to deviate from the ideal bandable model through a tail-stable remainder term. Motivated by this setup, we consider a misspecified target covariance matrix of the form

$$\Sigma^{(t)} = \Sigma_0^{(t)} + \eta H,$$

where $\Sigma_0^{(t)}$ is generated from the same bandable model as in the main simulation study, H is a random symmetric perturbation normalized in Frobenius norm, and $\eta \in \{0, 0.3, 0.6\}$ controls the misspecification level. Given $\Sigma^{(t)}$, we then generate the source covariance matrix under the same matrix ℓ_1 -disparity calibration as in the main text, with $\delta \in \{0.1, 0.7, 1.3\}$.

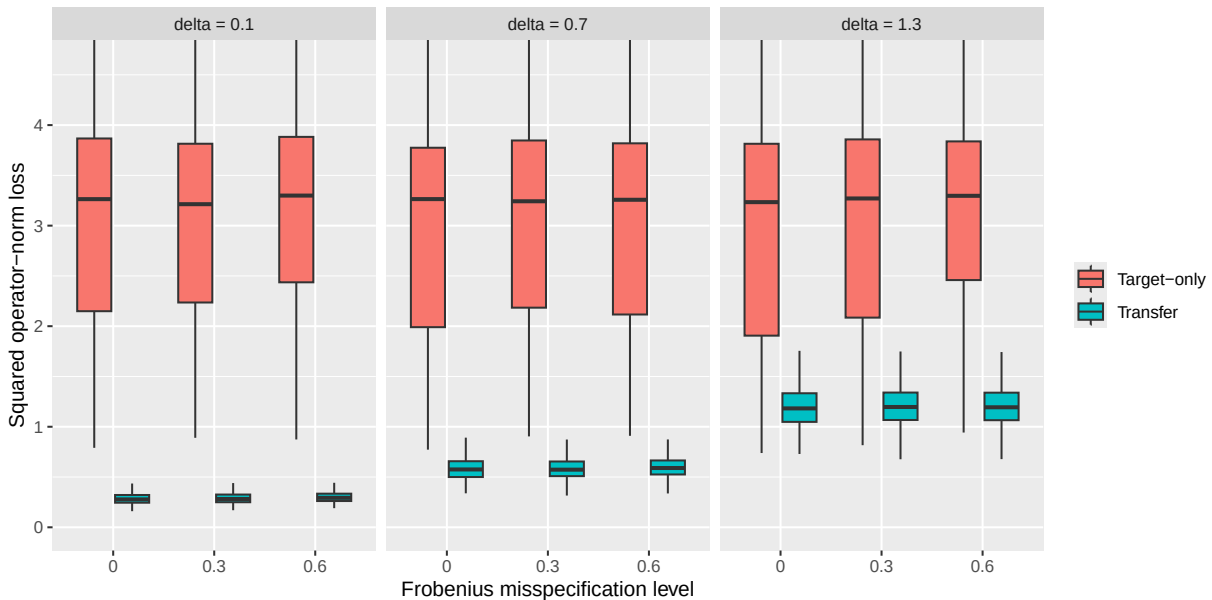


Figure 9: Squared operator-norm loss for the target-only and transfer estimators under misspecification of exact bandability. The horizontal axis records the Frobenius misspecification level η .

Figure 9 shows that the transfer estimator has lower loss than its target-only counterpart at every displayed disparity and misspecification level. When $\delta = 0.1$ or 0.7 , the improvement from transfer learning remains substantial even after the perturbation is added. When $\delta = 1.3$, the gain becomes smaller, but transfer still does not collapse relative to the target-only benchmark in this experiment. Moreover, increasing η produces only a

mild deterioration in empirical risk for the adaptive estimator over the range considered here.

To make this degradation pattern more explicit, we also rerun the misspecification study on a finer grid

$$\eta \in \{0, 0.25, 0.5, 0.75, 1, 1.25, 1.5\},$$

while keeping the same three disparity levels $\delta \in \{0.1, 0.7, 1.3\}$. Figure 10 reports the mean loss together with interquartile bands for each method.

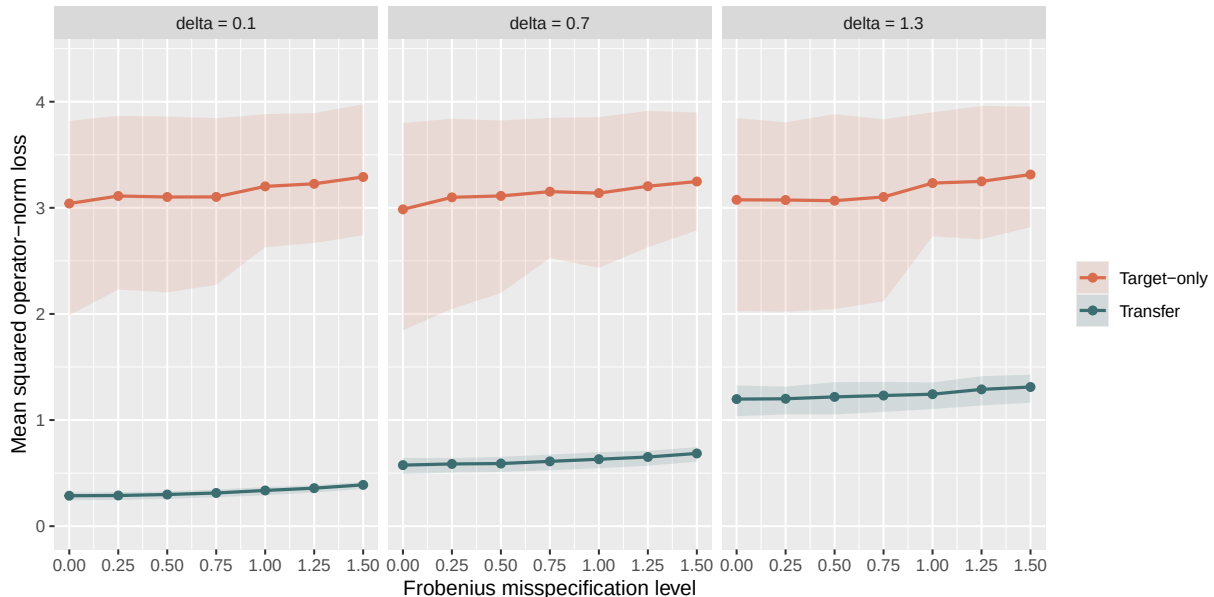


Figure 10: Mean squared operator-norm loss and interquartile bands for the target-only and transfer estimators under a finer misspecification grid. The horizontal axis records the Frobenius misspecification level η .

Figure 10 shows a mild, gradual deterioration as misspecification increases. For all three disparity levels, the transfer curve drifts upward with η rather than exhibiting an abrupt breakdown, while the target-only benchmark changes much less over the same range. When $\delta = 0.1$ or 0.7 , the transfer estimator stays clearly below the target-only estimator throughout the grid. When $\delta = 1.3$, the transfer advantage narrows steadily as η increases, but it still remains positive over the range considered here. This finer-grid experiment shows a gradual rather than abrupt weakening of the transfer benefit over the displayed range.

D.3 Robustness Beyond Gaussian Sampling

Finally, we investigate robustness to the sampling distribution itself. Our theoretical analysis assumes uniformly sub-Gaussian observations. To evaluate how sensitive the method is to this assumption, we repeat the same simulation design under three sampling distributions with the same covariance matrix: Gaussian, Uniform, and Student- $t(5)$. The target and source covariance matrices are generated from the baseline bandable model, and we again consider $\delta \in \{0.1, 0.7, 1.3\}$.

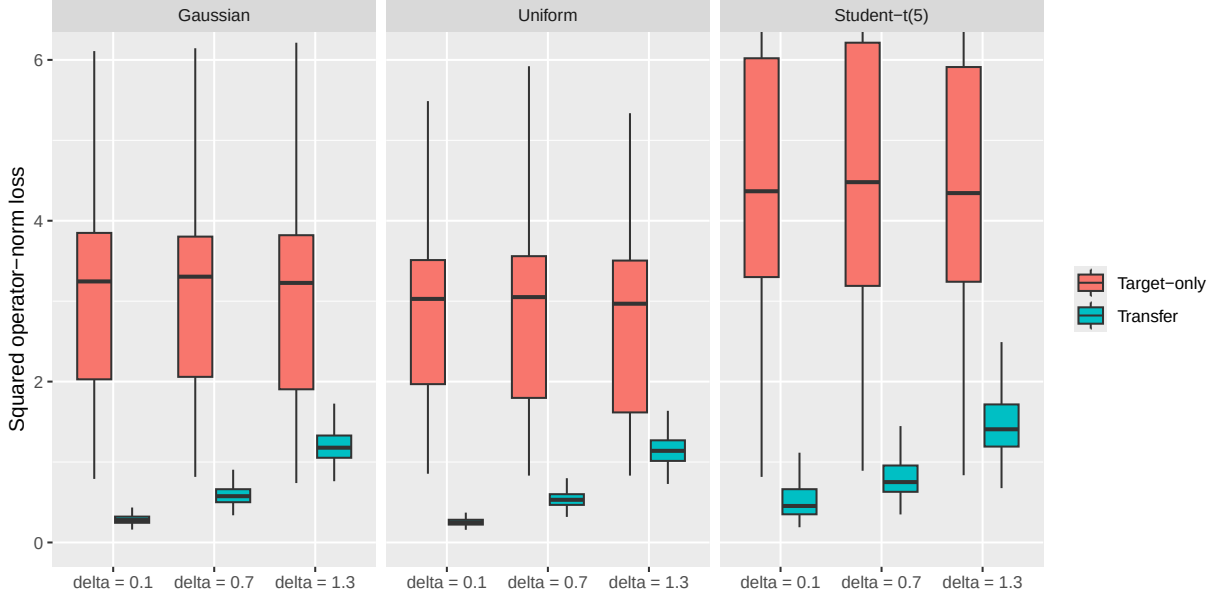


Figure 11: Squared operator-norm loss for the target-only and transfer estimators under Gaussian, Uniform, and Student- $t(5)$ sampling distributions.

As shown in Figure 11, the qualitative transfer-learning benefit persists across all three distributions. Under Gaussian and Uniform sampling, the patterns are very similar to those in the baseline experiment: the transfer estimator yields a clear reduction in loss when δ is small or moderate, and the gain shrinks as δ becomes large. Under Student- $t(5)$ sampling, both estimators become more variable, reflecting the heavier tails, but the transfer estimator still exhibits a clear downward shift relative to the target-only estimator. This experiment is only empirical and does not extend the current theory to heavy-tailed observations, but it suggests that the practical benefit of the adaptive estimator is not confined to the exact Gaussian setting.

D.4 Sensitivity to the Validation-Sample Proportion

We next examine how the validation-sample proportion affects the adaptive transfer rule. To isolate the role of validation, we use the known zero mean directly and do not pair the observations, while retaining the target sample split and the validation selector. We use the same bandable covariance model as in the main simulation study, fix $n_t = 50$, $n_s = 200$, and $d = 50$, and consider $\delta \in \{0.1, 0.7, 1.3\}$. For each validation proportion

$$p_f \in \{0.1, 0.2, 0.3, 0.4, 0.5\},$$

we define

$$m_f(p_f) = \lfloor p_f n_t \rfloor, \quad m_0(p_f) = n_t - m_f(p_f),$$

where $m_f(p_f)$ target observations are used for validation and $m_0(p_f)$ for fitting. The target-only estimator is computed once from all n_t target observations for each replicate, while the transfer estimator is recomputed at each p_f using the corresponding fitting and validation sizes. The same target and source draws are reused across the five validation proportions. For each setting, we record squared operator-norm loss over 1,000 repetitions.

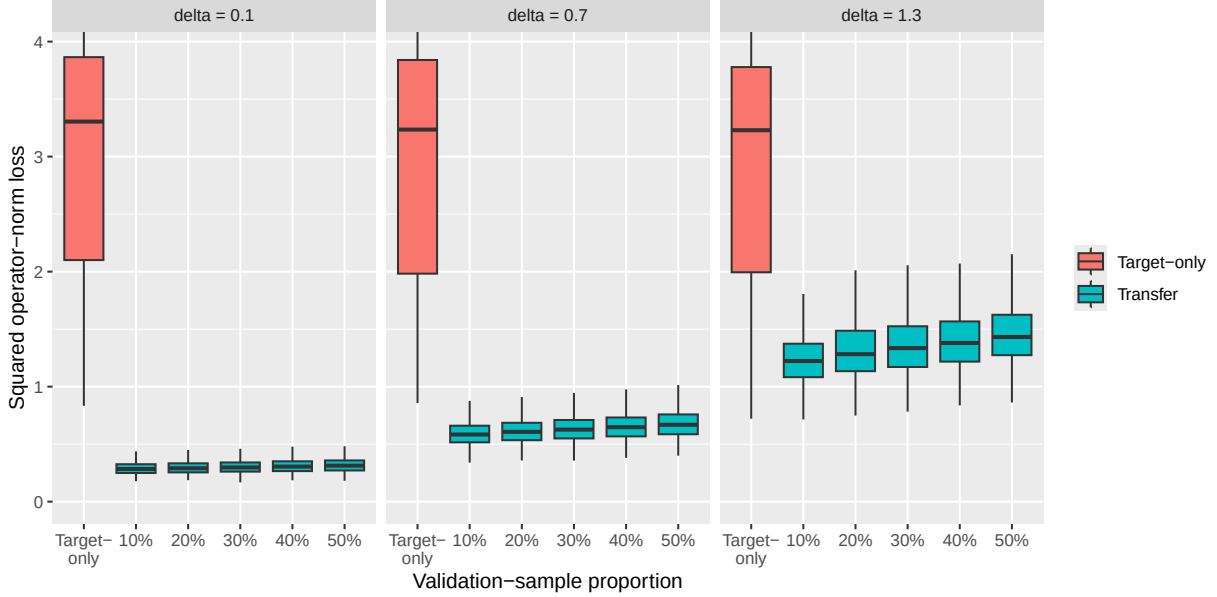


Figure 12: Squared operator-norm loss for the target-only baseline and the transfer estimator as a function of the validation-sample proportion. The target-only baseline is shown once using all n_t target observations; the transfer estimates use the five displayed proportions. The experiments use $n_t = 50$, $n_s = 200$, $d = 50$, and the three displayed disparity levels.

As shown in Figure 12, the target-only baseline is common to all five comparisons, whereas the transfer estimator is evaluated separately at each validation proportion. At $\delta = 0.1$ and 0.7 , the validation selector almost always retains the transfer estimate, and the transfer loss changes only gradually as p_f increases. At $\delta = 1.3$, the selector occasionally falls back to the target-only estimate, but the transfer estimator continues to have lower mean loss throughout the displayed range. The gradual increase in transfer loss for larger p_f reflects the smaller fitting sample available to the pooled estimate. Thus, in this experiment, a smaller validation proportion provides the most favorable estimation tradeoff while retaining a validation sample for detecting substantial source-target discrepancy.

D.5 Empirical Negative-transfer Diagnostics

To diagnose negative transfer, we consider the same bandable model as in the main text with $(\alpha_t, \alpha_s) = (2, 1)$, $n_t = 200$, and $d = 50$. We vary n_s over $\{200, 400, 600, 800, 1000\}$ and δ over $\{0.4, 0.7, 1, 1.3\}$. For each replicate, we record the paired excess loss

$$L_{\text{method}} - L_{\text{target}},$$

where positive values indicate negative transfer for the transfer estimator.

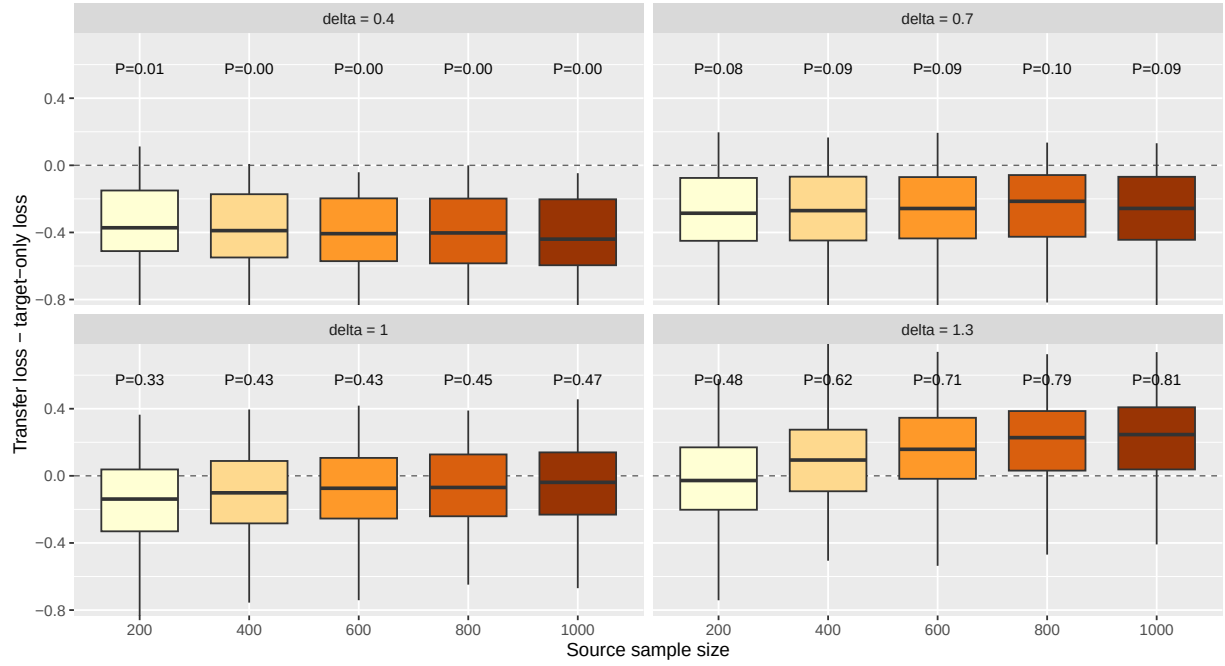


Figure 13: Paired negative-transfer diagnostic for the transfer estimator under the bandable model with $(\alpha_t, \alpha_s) = (2, 1)$, $n_t = 200$, and $d = 50$. Each boxplot shows the replicate-wise excess loss relative to the target-only estimator; values above zero indicate negative transfer. The labels report the empirical probability of negative transfer.

Figure 13 shows a clear transition across disparity levels. At $\delta = 0.4$, the excess loss is typically below zero and becomes more negative as n_s increases, whereas at $\delta = 1$ and 1.3 it is more often positive and becomes more severe. Thus, in this least-favorable regime, negative transfer can still occur in individual realizations. This does not contradict the theoretical guarantee in Theorem 3.2: although individual realizations may exhibit negative transfer, its risk-level effect is controlled through the additive remainder. The figure illustrates this controlled limitation under a large source-target disparity.

D.6 Additional Details for the Phoneme Experiment

We briefly supplement the real-data analysis in Subsection 4.2. For each random split, QDA estimates separate covariance matrices for the AA and AO classes, while LDA estimates a pooled covariance matrix. When $n_{\text{IY}} > 0$, we compute the sample covariance of an auxiliary IY subsample and use it as the source input to the adaptive transfer estimator; the class means and class priors are still estimated from the AA/AO training data only.

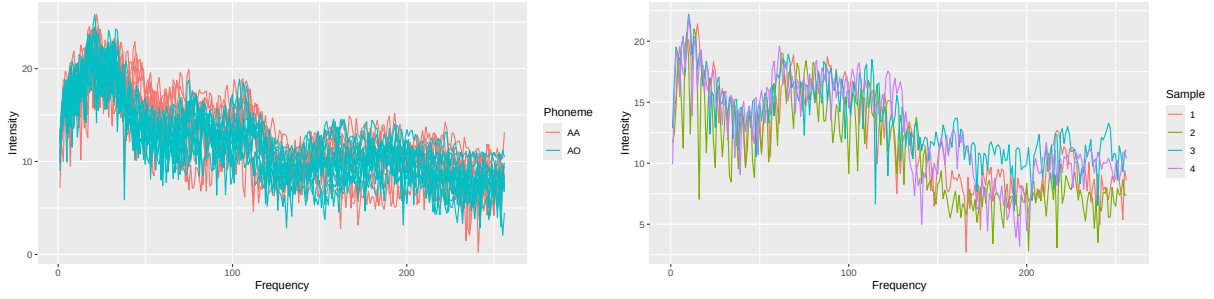


Figure 14: Representative log-periodograms for the target phonemes AA/AO (left) and the auxiliary phoneme IY (right).

Figure 14 provides a complementary view of source-target similarity. Although IY is distinct from AA and AO, all three phonemes exhibit smooth spectral trajectories with similar large-scale shapes, which helps explain why IY can serve as a useful auxiliary phoneme.

The mean accuracies over the 100 random splits, together with the gain relative to the target-only adaptive fit, are summarized below.

Table 2: Mean phoneme-classification accuracy and gain over the target-only adaptive fit. Values are averaged over 100 random train/test splits and rounded to three decimals.

n_{IY}	QDA		LDA	
	Accuracy	Gain	Accuracy	Gain
0	0.562	—	0.677	—
200	0.683	0.121	0.743	0.067
600	0.741	0.179	0.752	0.076
1100	0.801	0.239	0.801	0.124

To quantify the downstream effect of transfer more directly, we also record the paired accuracy gain

$$\text{Acc}_{\text{transfer}} - \text{Acc}_{\text{target}}$$

for each of the 100 random train/test splits.

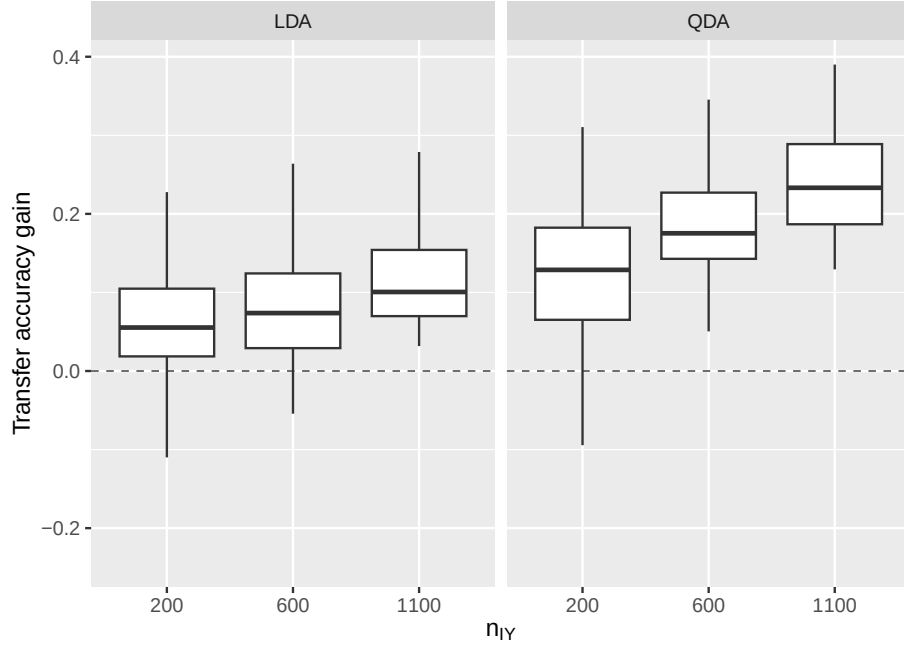


Figure 15: Paired accuracy gain in the phoneme experiment. Each boxplot shows the split-wise difference $\text{Acc}_{\text{transfer}} - \text{Acc}_{\text{target}}$, where the target benchmark is the target-only classifier; positive values indicate an improvement from transfer learning.

Figure 15 shows that the gain is heterogeneous when $n_{\text{IY}} = 200$, especially for QDA, where a moderate auxiliary sample does not help every split. At $n_{\text{IY}} = 600$, the gain is positive in 96% of QDA splits and 82% of LDA splits; at $n_{\text{IY}} = 1100$, it is positive in all splits for both methods. This complements the main-text accuracy boxplots by showing that the practical benefit of transfer becomes both larger and more stable as the auxiliary sample grows.

E Preliminary Classification Guarantee

Motivated by the reviewers' suggestion during this revision, in this section, we include a preliminary theoretical guarantee for the downstream classification task in a stylized binary Gaussian classification model. The goal is not to provide a full analysis of the data-driven QDA/LDA procedures used in the phoneme experiment, where means, priors, and covariance matrices are all estimated, but rather to isolate the covariance-stability component needed by plug-in discriminant rules.

Consider binary Gaussian LDA with equal priors and known means:

$$Y \in \{0, 1\}, \quad \mathbb{P}\{Y = 0\} = \mathbb{P}\{Y = 1\} = \frac{1}{2},$$

and, conditional on $Y = k$,

$$X | (Y = k) \sim N(\mu_k, \Sigma), \quad k \in \{0, 1\},$$

for some positive-definite covariance matrix Σ . Write

$$\Delta_\mu := \mu_1 - \mu_0, \quad \bar{\mu} := \frac{\mu_0 + \mu_1}{2}.$$

For any positive-definite matrix A , let

$$g_A(x) := \mathbf{1}\{\Delta_\mu^\top A^{-1}(x - \bar{\mu}) \geq 0\}$$

denote the plug-in LDA rule obtained by replacing Σ with A , and let

$$R(A) := \mathbb{P}\{g_A(X) \neq Y\}$$

be its misclassification risk. The Bayes rule in this model is g_Σ , and we write $R^* := R(\Sigma)$.

Proposition E.1. *Assume that*

$$\lambda_{\min}(\Sigma) \geq \underline{\lambda} > 0, \quad \lambda_{\max}(\Sigma) \leq \bar{\lambda}, \quad \|\Delta_\mu\|_2 \leq M,$$

and that the Mahalanobis separation satisfies

$$\Delta_\mu^\top \Sigma^{-1} \Delta_\mu \geq \gamma^2$$

for some $\gamma > 0$. Then there exist constants $c_0, C > 0$, depending only on $\underline{\lambda}$, $\bar{\lambda}$, M , and γ , such that the following holds.

For any symmetric matrix $\hat{\Sigma}$ satisfying

$$\|\hat{\Sigma} - \Sigma\|_2 \leq c_0,$$

the matrix $\hat{\Sigma}$ is positive definite and the corresponding plug-in LDA rule obeys

$$R(\hat{\Sigma}) - R^* \leq C\|\hat{\Sigma} - \Sigma\|_2.$$

Proof. For any positive-definite matrix A , a direct Gaussian calculation gives

$$R(A) = \Phi_N \left(-\frac{\Delta_\mu^\top A^{-1} \Delta_\mu}{2\sqrt{\Delta_\mu^\top A^{-1} \Sigma A^{-1} \Delta_\mu}} \right),$$

where Φ_N is the standard normal distribution function. Write

$$D := \Delta_\mu^\top \Sigma^{-1} \Delta_\mu, \quad a(A) := \Delta_\mu^\top A^{-1} \Delta_\mu, \quad b(A) := \Delta_\mu^\top A^{-1} \Sigma A^{-1} \Delta_\mu.$$

Then $D \geq \gamma^2$, $a(\Sigma) = b(\Sigma) = D$, and

$$R(A) = \Phi_N \left(-\frac{a(A)}{2\sqrt{b(A)}} \right), \quad R^* = \Phi_N \left(-\frac{\sqrt{D}}{2} \right).$$

Set $A = \widehat{\Sigma}$ and let $\varepsilon := \|\widehat{\Sigma} - \Sigma\|_2$. If $\varepsilon \leq \underline{\lambda}/2$, then Weyl's inequality gives $\lambda_{\min}(\widehat{\Sigma}) \geq \underline{\lambda}/2$, so $\widehat{\Sigma}$ is positive definite and $\|\widehat{\Sigma}^{-1}\|_2 \leq 2/\underline{\lambda}$. The resolvent identity implies

$$\|\widehat{\Sigma}^{-1} - \Sigma^{-1}\|_2 \leq \|\Sigma^{-1}\|_2 \|\widehat{\Sigma} - \Sigma\|_2 \|\widehat{\Sigma}^{-1}\|_2 \leq \frac{2}{\underline{\lambda}^2} \varepsilon.$$

Hence

$$|a(\widehat{\Sigma}) - D| \leq \|\Delta_\mu\|_2^2 \|\widehat{\Sigma}^{-1} - \Sigma^{-1}\|_2 \lesssim \varepsilon,$$

with the implicit constant depending only on M and $\underline{\lambda}$. Also,

$$\widehat{\Sigma}^{-1} \Sigma \widehat{\Sigma}^{-1} - \Sigma^{-1} = (\widehat{\Sigma}^{-1} - \Sigma^{-1}) \Sigma \widehat{\Sigma}^{-1} + (\widehat{\Sigma}^{-1} - \Sigma^{-1}),$$

so $\|\widehat{\Sigma}^{-1} \Sigma \widehat{\Sigma}^{-1} - \Sigma^{-1}\|_2 \lesssim \varepsilon$, and therefore

$$|b(\widehat{\Sigma}) - D| \leq \|\Delta_\mu\|_2^2 \|\widehat{\Sigma}^{-1} \Sigma \widehat{\Sigma}^{-1} - \Sigma^{-1}\|_2 \lesssim \varepsilon,$$

with the implicit constant depending only on $\underline{\lambda}$, $\bar{\lambda}$, and M .

Since $D \in [\gamma^2, M^2/\underline{\lambda}]$, after shrinking c_0 if necessary we may ensure that $a(\widehat{\Sigma})$ and $b(\widehat{\Sigma})$ both stay in a compact interval bounded away from zero, say

$$\left[\frac{\gamma^2}{2}, \frac{2M^2}{\underline{\lambda}} \right].$$

On this set, $(a, b) \mapsto a/(2\sqrt{b})$ is Lipschitz, with a constant depending only on $\underline{\lambda}$, M , and γ . Combining this with the $1/\sqrt{2\pi}$ -Lipschitz continuity of Φ_N , we obtain

$$|R(\widehat{\Sigma}) - R^*| \lesssim |a(\widehat{\Sigma}) - D| + |b(\widehat{\Sigma}) - D| \lesssim \varepsilon.$$

This proves the deterministic bound after adjusting the constant C . $\square$

Remark E.2. The proposition is intentionally modular. It shows that, in a simple binary Gaussian LDA model, improved operator-norm covariance estimation translates directly into improved excess classification risk up to constant factors. Hence, whenever the transfer estimator attains a smaller operator-norm covariance error than the target-only estimator, it also yields a smaller upper bound for the corresponding plug-in classification risk. A parallel perturbation argument applies to binary Gaussian QDA under analogous conditioning and margin assumptions when the two class-specific covariance matrices are both estimated in operator norm.

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